

Technical Exhibit A
Landfill Permit

US ARMY Corps of Engineers

COE POC and phone

Contract name and number

Contract completion date or end of authorization date

Building or areas affected i.e., Soldier's Development Center

Prime Contractor Name

Contractor POC and phone - someone on site that can get immediate action

NOTE: The following Items are available in hard copy only.

Request hard copy from DPW.

Part 4.IV TNRCC Liner Construction and Testing Handbook

Part 4.V Compliance Demonstration for Airport Safety

Part 4.VI Compliance Demonstration for Floodplains

Part 4.VII Compliance Demonstration for Unstable Areas

Appendix I Site Safety and Health Plan

Appendix II Standard Laboratory Operating Procedure

Appendix III Letter from Bell County W.C.I.D.

Appendix IV Field Permeability Tests, Soil Data, Core Borings

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- IV. TNRCC Liner Construction and Testing Handbook - Available in hard copy only from DPW
- V. Compliance Demonstration for Airport Safety - Available in hard copy only from DPW
- VI. Compliance Demonstration for Floodplains - Available in hard copy only from DPW
- VII. Compliance Demonstration for Unstable Areas - Available in hard copy only from DPW

APPENDICES

- I. Site Safety and Health Plan - Available in hard copy only from DPW
- II. Standard Laboratory Operating Procedure - Available in hard copy only from DPW
- III. Letter from Bell County W.C.I.D. - Available in hard copy only from DPW
- IV. Field Permeability Tests, Soil Data, Core Borings - Available in hard copy only from DPW
- V. ISCP - Available in hard copy only from DPW
- VI. Cost Estimate- Available in hard copy only from DPW

****NOTE**** Fort Hood changed name from Fort Cavazos on 28 July 2025.

PART 1, SECTION I

Written Acceptance Letters
From Texas Department of Health



Texas Department of Health

Robert Bernstein, M.D., F.A.C.P.
Commissioner

1100 West 49th Street
Austin, Texas 78756-3199
(512) 458-7111

Robert A. MacLean, M.D.
Deputy Commissioner

MAR 25 1991

Colonel Robert K. Jayne
Engineering Director, Engineering and Housing
U. S. Army, Fort Hood
ATTN: AFZF - DE - ED
Fort Hood, Texas 76544

Subject: Solid Waste - Coryell County
U. S. Army, Fort Hood - Permit No. 1866
Within the premises of the U.S. Military Reservation at Fort Hood,
approximately 3.0 miles NE of Copperas Cove, 2.6 miles N of U.S.
190, adjacent to NE side of junction of west Fort Hood entrance
road (Base Road) with Turkey Run Road.
Coordinates: N 31° 09.30' W 97° 50.00'

Dear Colonel Jayne:

A permit for your solid waste disposal facility at the above location is enclosed. We appreciate your cooperation in our evaluation and approval procedures.

Acceptance of this permit constitutes an acknowledgement that the permittee will comply with all of the terms, provisions, conditions, limitations, and restrictions embodied in this permit; with the "Municipal Solid Waste Management Regulations" of the Texas Department of Health; and with the pertinent laws of the State of Texas.

We previously sent you a copy of this Department's "Municipal Solid Waste Management Regulations" with all changes in effect as of February 18, 1988. Special attention is invited to Subchapter F which prescribes procedures to be followed with respect to ground and surface water protection; the disposal of industrial and special wastes; and disposal of Class I and hazardous wastes.

Please note that the regulations require the submission of a liner evaluation prior to disposal of waste in a new area or excavation. We are enclosing copies of the report forms which are to be used for that purpose.

Your solid waste disposal facility is subject to inspection by State and local authorities and public health officials at any reasonable time. If this Department can provide any further information or should you have any questions regarding the requirements of this Department's "Municipal Solid Waste Management Regulations", please feel free to contact us at any time.

Sincerely,

L. D. Thurman, P.E.
Associate Commissioner for Environmental
and Consumer Health Protection

TAO:aam
Enclosures

cc: Region 1, TDH
Coryell County Health Authority
Senator Temple Dickson, Dist. 24
Representative Bob Melton, Dist. 57



Texas Department of Health

Robert Bernstein, M.D., F.A.C.P.
Commissioner

Robert A. MacLean, M.D.
Deputy Commissioner

1100 West 49th Street
Austin, Texas 78756-3199
(512) 458-7111

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Barbara T. Slover, R.Ph.
Sr. Marian Strohmeyer, R.N., M.P.H.
Edward H. Zunker, O.D.

FAX#
(512) 458-7477

Permit No. 1866
Coordinates: N 31° 09.30' W 97° 50.00'

PERMIT FOR A MUNICIPAL SOLID WASTE FACILITY
issued under provisions of Texas Solid Waste Disposal Act,
Chapter 361, Health and Safety Code and the Texas Department
of Health's "Municipal Solid Waste Management Regulations"

Permittee

U. S. Army, Fort Hood
ATTN: AFZF - DE - ED
Fort Hood, Texas 76544

Site Owner

U. S. Army, Fort Hood
ATTN: AFZF - DE - ED
Fort Hood, Texas 76544

Legal Description of Site: The legal description as submitted with the application is hereby made a part of this permit.

Size and Location of Site: The site consists of approximately 154.27 acres of land and is located within the premises of the U. S. Military Reservation at Fort Hood, approximately 3.0 miles northeast of Copperas Cove, 2.6 miles north of U.S. 190, adjacent to the northeast side of the junction of West Fort Hood entrance road (Base Road) with Turkey Run Road in Coryell County.

Operational Classification of Site: Type I

Waste Disposal Methods Used at Site: Trench and area fill method of disposal with compaction of solid waste and cover with a minimum of six (6) inches of compacted earth not less often than once on each day of operation.

Description of Waste Materials Processed at Site: Solid waste under the regulatory jurisdiction of the Texas Department of Health, when disposed of or processed in accordance with the Department's "Municipal Solid Waste Management Regulations".

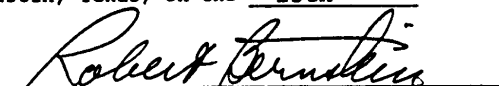
Findings of Fact and Conclusions of Law: Attached hereto and made a part hereof.

Standard Provision: Acceptance of this permit constitutes an acknowledgment that the permittee will comply with all of the terms, provisions, conditions, limitations and other restrictions embodied in this permit; with the "Municipal Solid Waste Management Regulations" of the Texas Department of Health; and with the pertinent laws of the State of Texas.

Special Provisions: None

This permit will be valid until canceled or revoked by the Commissioner of Health of the Texas Department of Health or until the site is completely filled and rendered unusable, whichever occurs first.

Given under my hand and Seal of Office at Austin, Texas, on the 25th
of March, 1991.


Robert Bernstein, M.D., F.A.C.P.
Commissioner of Health



Texas Department of Health

Robert Bernstein, M.D., F.A.C.P.
Commissioner

1100 West 49th Street
Austin, Texas 78756-3199
(512) 458-7111

Robert A. MacLean, M.D.
Deputy Commissioner

MAY 24 1991

CERTIFIED MAIL # 501-395
RETURN RECEIPT REQUESTED

Colonel Robert F. Harris
Director of Engineering and Housing
U.S. Army, Fort Hood
ATTN: AFZF-DE-ED
Fort Hood, Texas 76544

Subject: Solid Waste - Coryell County
U.S. Army, Fort Hood - Permit No. 1866
0.01 Mile N of Turkey Run Road and
0.15 Mile W of Clear Creek Road in Fort Hood

Dear Colonel Harris:

A permit for the subject solid waste disposal facility was issued to the United States Army, Fort Hood, Texas, on March 25, 1991.

All pre-opening requirements of Sections 325.112 (a) and (b) of the Department's "Municipal Solid Waste Management Regulations" (MSWMR) have been met and you are hereby authorized to open your facility to the receipt of waste.

Subchapter P of the MSWMR requires that you submit fees and reports on a periodic basis. It is important that you keep your billing address current with us so that our notices will receive your proper attention.

If you have any questions concerning this letter or if we may be of any assistance to you regarding solid waste management, you may contact Mr. Philip A. Spry, P.E., of my staff here in Austin at telephone number (512) 458-7271 or you may prefer to contact Mr. Oran Buckner, P.E., Regional Director of Environmental and Consumer Health Protection at 2408 South 37th Street, Temple, Texas 76504-7168; telephone number (817) 778-6744.

Sincerely,

Hans J. Mueller, P.E., Director
Surveillance and Enforcement Division
Bureau of Solid Waste Management

PAS:sm

cc: Region 1, TDH, Temple

PAGE 1 OF 2

PART 1, SECTION II

Permit Modification Form
For Type I Landfills

TEXAS WATER COMMISSION

Permit Modification Form for Type I - III Landfills

PART A - GENERAL DATA

Submit three copies of this form and the supporting documentation unless otherwise instructed by the Commission. Failure to complete all entries and provide all necessary attachments will delay processing the application. Please type or print.

GENERAL PERMIT INFORMATION

Permit Application No.: 1866

Date permit was originally issued: March 25, 1991

Is the site a "grandfathered" site? Yes X No

Has the permit been amended? Yes X No

If the permit has been amended, please give the date of the amendment(s) and the type of amendment(s):

Date of amendment(s):

Type of amendment(s):

Is the facility currently in receipt of waste for disposal? X Yes No

If the facility has not been receiving waste for disposal, has the final cover been completed? Yes No

Date of final cover completion:

Applicant is a (check one): ☐ City ☐ County ☒ Other

Applicant Name: U.S. Army

Business Address: Headquarters III Corps and Fort Hood

Fort Hood

Texas

76544

(city)

(state)

(zip)

Business Contact:

Clyde W.

Reynolds, P.E.

Phone No. (817) 287-9813

(title)

(first name)

(last name)

(area code)

SITE INFORMATION

Site Name: Fort Hood Landfill

Physical location: Intersection of Clarke & Turkey Run Rds County: Coryell

Geographic Coordinates (if known): N31°09.30' W97°50.00'

Site located in: County of Coryell City of Fort Hood

Nearest town: (applicable only if site is outside the city limits or ETJ of any city).

PERMITS AND LICENSES

List any other existing permits or licenses issued by this or any government agency, whether local, state, or federal, which pertain to this facility (attach additional sheet if necessary):

1) TNRCC Solid Waste 1283, 1284

2) TWC Wastewater Permits 12096-001, 02233, 02230

3) TWC Hazardous Waste Permit 50323

4) EPA Storm Water Permit TXR10H927

Has facility received an Arid Exemption (31 TAC 330.3(e)(f)) from the TWC? Yes No X

SITE OPERATIONS AND SPECIFICATIONS

The following information should be taken from the latest (most recent) permit application:

- Facility is: 154.27 acres in size.
- Approximate percentage of total permitted landfill area remaining open after October 9, 1993: 95%.
- Approximate percentage of total permitted landfill area remaining unused after October 9, 1993: 91%.
- Approximate percentage of total permitted landfill area which has final cover in place as of October 9, 1993: 0%.
- Facility will service approximately: 45,000 persons.
- Will facility plan on extending service area? Yes X No.

If yes, please describe the expansion:

- Facility will receive an average of approximately: 200 tons per day of municipal waste.
- With permit modification facility's estimated life is approximately: 20 years.



DEPARTMENT OF THE ARMY
HEADQUARTERS III CORPS AND FORT HOOD
FORT HOOD, TEXAS 76544-50 57

REPLY TO
ATTENTION OF

November 17, 1986

Directorate of Engineering and Housing

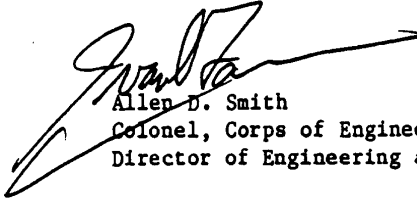
Robert Bernstein, M.D.
Commissioner of Health
Texas Department of Health
1100 West 49th Street
Austin, Texas 78752

Dear Doctor Bernstein:

This is to advise you that the officials of the U.S. Army, Fort Hood, Texas, have duly appointed the U.S. Army Engineer District, Fort Worth, as consulting and designing engineer for the purpose of submitting engineering reports, planning material, plans, and specifications of the new Fort Hood sanitary landfill. The engineers who work on this project are registered professional engineers in good standing in accordance with State statutes and have had experience in the design of similar facilities.

We herewith authorize you to review and comment on such reports, planning material, data, and specifications on this proposed project as he may submit to you.

Sincerely,



Allen D. Smith
Colonel, Corps of Engineers
Director of Engineering and Housing

PART 1, SECTION IV
Engineers' Assignment

ORDER FOR SUPPLIES OR SERVICES (Contractor must submit four copies of invoice.)		Form Approved OMB No. 0704-0187 1997		PAGE 1 OF 1
DUPLICATE ORIGINAL				
Public reporting burden for this collection of information is estimated to average 1 hour per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0187), Washington, DC 20503. Please DO NOT RETURN your form to either of these addresses. Send your completed form to the procurement official identified in item 6.				
1. CONTRACT / PURCH ORDER NO. DACA63-94-D-0009		2. DELIVERY ORDER NO. 0003		3. DATE OF ORDER See Block 24
4. ISSUED BY U.S. Army Engineer District, Fort Worth P.O. Box 17300 Fort Worth, TX 76102-0300		5. ADMINISTERED BY (If other than 6) See Block 6		6. CERTIFIED FOR NATIONAL DEFENSE UNDER DMS REG 1 DO
7. CONTRACTOR NAME AND ADDRESS H. Platt Thompson Engineering Co, Inc. 5249 Langfield Road Houston, TX 77040-6618		8. FACILITY CODE		9. DELIVERY FOR ☒ DEST ☐ OTHER (See Schedule if other)
10. SHIP TO See Detailed Statement of Work		11. PAYMENT WILL BE MADE BY Disbursing Officer U.S. Army Engineer District, Fort Worth P.O. Box 17300 Fort Worth, TX 76102-0300		12. MARK IF BUSINESS IS ☒ SMALL ☐ SMALL DISADVANTAGED ☐ WOMEN-OWNED
13. DELIVERY ☒ DELIVERY ☐ PURCHASE		14. This delivery order is issued on another Government agency or in accordance with and subject to terms and conditions of above numbered contract. Reference your ACCEPTANCE. THE CONTRACTOR HEREBY ACCEPTS THE OFFER REPRESENTED BY THE NUMBERED PURCHASE ORDER AS IT MAY PREVIOUSLY HAVE BEEN OR IS NOW MODIFIED, SUBJECT TO ALL OF THE TERMS AND CONDITIONS SET FORTH, AND AGREES TO PERFORM THE SAME.		
15. NAME OF CONTRACTOR ACCOUNTING AND APPROPRIATION DATA/LOCAL USE 2142020 08 8073 P1170566 26CZ 284 S41443 (QK) RP48614350A009A		16. SIGNATURE JOHN H. BODGES BY: John H. Bodges CONTRACTING / ORDERING OFFICER		
17. QUANTITY IN COLUMN 20 HAS BEEN ☐ INSPECTED ☐ RECEIVED ☐ ACCEPTED, AND CONFORMS TO THE CONTRACT EXCEPT AS NOTED		18. UNITED STATES OF AMERICA 3/7/94		
19. DATE DATE		20. SIGNATURE OF AUTHORIZED GOVERNMENT REPRESENTATIVE SIGNATURE AND TITLE OF CERTIFYING OFFICER		
21. I certify this account is correct and proper for payment.		22. 31. PAYMENT ☐ COMPLETE ☐ PARTIAL ☐ FINAL		
23. DATE		24. 40. TOTAL CONTAINERS		
25. RECEIVED AT		26. RECEIVED BY		
27. DATE RECEIVED		28. 41. S/R ACCOUNT NUMBER		
29. 37. RECEIVED AT		30. 42. S/R VOUCHER NO.		

DD Form 1155, MAY 90

Previous editions are obsolete.

459-722

Legal Description

METES AND BOUNDS DESCRIPTION
OF THE SOUTH FORT HOOD SANITARY LANDFILL

Described herein is a tract of land in the State of Texas, County of Coryell. Said strip being a part of the original 157,588,023 acre Camp Acquisition and more particularly a portion of the Fort Hood Military Reservation released from state authority to exclusive Federal jurisdiction indeed cessation agreement, approved by the State legislature and signed by the Governor elect, A. Shivers, 6 September 1950.

Beginning from centerline intersection of Base Road and Turkey Run Road, thence southeasterly along centerline of Turkey Run Road approximately 933 linear feet, to a chiseled cross mark in center of Turkey Run Road, thence N 20° 53'00" E approximately, 65.08 linear feet approximately to concrete monument in ground with embedded railroad spike, designated point "A", point of beginning.

Direction of survey is clockwise.

Thence N 69° 07'00" W approximately, 2100 linear feet approximate, to concrete monument in ground with embedded railroad spike, designated point "B".

Thence N20 °07'00" E approximate, 3200 linear feet approximate, to concrete monument in ground with embedded railroad spike, designated point "C".

Thence S 69 °07'00" E approximate, 2100 linear feet approximate, to concrete monument in ground with embedded railroad spike, designated point "D".

Thence S 20 °53'00" w approximate, 3200 linear feet approximate, to concrete monument in ground with embedded railroad spike, designated point "A", point of beginning.

Above survey contains 154.27 acres more or less.

PART 1, SECTION VI
Evidence of Financial Responsibility

EVIDENCE OF FINANCIAL RESPONSIBILITY

Fort Hood will contract the operation of the site on a yearly basis. The applicant makes a firm commitment to provide backup equipment by lease, purchase, or diversion from other activities. As a Federal Government installation, the applicant requests a waiver from the requirements for evidence of financial responsibility.

PART 1, SECTION VII

Written Concurrence from Texas
Department of Health for the use
Of High Density Polyethylene Liner



Texas Department of Health

Robert Bernstein, M.D., F.A.C.P.
Commissioner

1100 West 49th Street
Austin, Texas 78756-3199
(512) 458-7111

Robert A. MacLean, M.D.
Deputy Commissioner
Professional Services
Hermas L. Miller
Deputy Commissioner
Management and Administration

JUL 12 1988

Colonel Robert K. Jayne
Director, Engineering and Housing
Fort Hood, Texas 76544

Attention: AFZF - DE - ED

Subject: Solid Waste - Coryell County
Fort Hood - Permit Application No. 1866

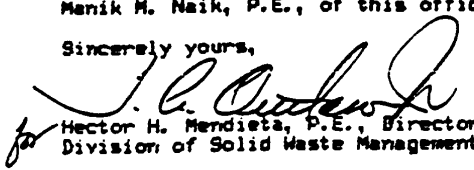
Dear Colonel Jayne:

This is in response to the recent telephone conversation between Mr. William C. Bodkin of your staff and Mr. Manik M. Naik, P.E., our Project Engineer, and the letter from Mr. Edwin E. Davis, P.E., of Parkhill, Smith and Cooper, Inc., requesting approval of a synthetic liner for lining waste disposal excavations of the above-captioned application site.

Considering the proposal indicated in Mr. Davis's letter that a 60-mil thick High Density Polyethylene (HDPE) liner will be used for lining disposal excavations at the proposed site and that a standard warranty of 20 years on the liner can be obtained from the liner manufacturer against its failure, the Division of Solid Waste Management deems that the proposed liner for the proposed site will be acceptable. However, the details of the liner placement design and quality control plan has to be submitted with the Site Development Plan for the Department's review purposes.

If you have any questions concerning any of the above comments, or if this Department can assist you in any way, please do not hesitate to contact Mr. Manik M. Naik, P.E., of this office at (512) 458-7271.

Sincerely yours,


for Hector H. Mendietta, P.E., Director
Division of Solid Waste Management

MMN:rg

cc: Region 1, TDH
Coryell County Health Authority
Mr. Edwin E. Davis, P.E.

PART 1, SECTION VIII
Written Acceptance from the
TNRCC for the use of High

Density Polyethylene Liner

John Hall, *Chairman*
Pam Reed, *Commissioner*
Peggy Garner, *Commissioner*
Anthony Grigsby, *Executive Director*



TEXAS NATURAL RESOURCE CONSERVATION COMMISSION

Protecting Texas by Reducing and Preventing Pollution

March 16, 1994

Colonel Albert G. Bungard
Director of Engineering and Housing
U.S. Army, Fort Hood
ATTN: AFZF-DE-ED
Fort Hood, TX 76544

Re: Solid Waste - Coryell County
U.S. Army, Fort Hood - MSW Permit No. 1866
Soil and Liner Evaluation Report Acceptance

Dear Colonel Bungard:

Based on a site visit on March 14th by Mr. Chet Safe, Ms. Maryann Ryan, and myself, we have concluded that the SLER portion of the composite liner for Phase 1 Cell 4 is acceptable. Waste cannot be placed in this area until the FMLER and a permit modification that includes the final version of SLQCP are approved.

No future SLERs or FMLERs will be accepted for this permit until a permit modification addressing Subtitle D requirements and a SLQCP finalized as a stand-alone document has received written acceptance by the TNRCC showing that the SLQCP complies with the ground-water protection requirements contained in the State of Texas Municipal Solid Waste Rules.

All submittals, including the cover letter, are to be in triplicate, as in the past. Please be sure to include an original and two copies. The permit number for the site and the name of the county are necessary on all submittals and correspondence to allow proper tracking and filing. Your cooperation in this matter will greatly be appreciated.

If you have any questions concerning this matter, please contact Ms. Maryann Ryan, Geologist, Ground-Water Protection Team, at (512) 239-6264.

Sincerely,

A handwritten signature in cursive script, appearing to read "Ronald J. Purviance".

Ronald J. Purviance, C.P.G., Team Leader
Ground-Water Protection Team
Compliance and Enforcement Section
Municipal Solid Waste Division

RJP/MAR/mr

cc: TNRCC Region 9 Office
Mr. Richard E. Weathers, P.E., TETCO

SITE DEVELOPMENT PLAN

I. AREA MASTER PLAN

The estimated population served by the United States Army, Fort Hood, Coryell County, Texas Type I municipal solid waste landfill is 71,000 based on troop strength and family members. A location map including areas to be served, road net, major drainage systems, and nearby communities is shown on the Highway Map in Part 3. Located approximately 4 miles northeast of Copperas Cove, 2.5 miles north of U.S. 190, and 5.5 miles west of Killeen, Bell County, Texas. This landfill serves only U.S. Government properties and facilities.

II. SITE PLAN

A. The Fort Hood municipal solid waste landfill is located within the boundaries of the military installations, property of the United States Government, and is shown on the Area Master Plan.

B. This landfill was initially permitted in March 1991 with wastes deposited in the southeast corner of the Phase I, into Area 1, Cell 1. Since that time, development has proceeded in a westerly direction into Cells 2 and 3. Design of Cell 4 was developed according to the Texas Administrative Code (TAC) Chapter 330, Municipal Solid Waste standards. The Soil and Liner Evaluation Report (SLER) was accepted on March 16, 1994 by the Texas Natural Resource Conservation Commission (TNRCC) (see TNRCC's SLER letter of Acceptance, Part 1). According to this letter, wastes can be placed in Cell 4 once the following items are approved: (1) the final version of the Soil and Liner Quality Control Plan (SLQCP); (2) the Flexible Membrane Liner Evaluation Report (FMLER); and (3) submittal of this permit modification according to the U.S. Environmental Protection Agency (EPA) and Texas Municipal Solid Waste (MSW) final rules implementing 40 Code of Federal Regulation Parts 257 and 258 of Subtitle D of the Federal Resource Conservation and Recovery Act (Subtitle D) requirements.

The landfill site is divided into three phases with two areas in each phase so as to maintain the integrity of surface runoff and reduce possibilities of surface erosion. The landfill phases are identified as Phase I, Areas 1 and 2; Phase II, Areas 3 and 4; and Phase III, Areas 5 and 6. Operation of the landfill is based on an area fill method, whereas the cover and fill soils are developed by excavation and stockpiled for subsequent usage. Development of the landfill will continue in the form of lateral expansions which are typically 420-feet in length by 330-feet wide. The width of expansions is not critical and may vary depending on the geology, control of the surface water runoff, and the leachate collection and removal system.

The entire 154 acre site has been developed according to the Subtitle D standards with this modification submittal. Final Closure and Post-Closure Care of the entire site will be according to these standards. Cells 1,2, and 3 will be closed according to Subtitle D standards. Development is continuing in a westerly direction in Area 1, which contains an additional 8.46 acres. Development will continue in a northerly direction into Area 2, which contains 16.69 acres; followed by Area 3, with 20.21 acres; Area 4, with 20.18 acres; Area 5, with 20.19 acres; and Area 6, with 20.19 acres to give a total acreage for the landfill of 105.92 acres. Solid waste placement rate averages 252 cubic yards per day at gross in-place density rates of approximately 700 pounds per cubic yard. The total landfill life of this facility is estimated to be 30 years.

C. Design characteristics of the various systems of the landfill will be addressed in the sections that follow. Protection of the groundwater is an integral part of the design with monitoring systems installed to indicate problems. An important aspect of this program is the leachate collection system. At each lateral expansion an 18-inch cleanout pipe will be provided at each sump for removal of the leachate. Each header pipe shall have a clean out and a cap at the end. As the area of fill expands, the cap will be removed and additional pipe shall be added to expand the header either east or west, depending on the direction of expansion. This procedure shall be utilized until the expansion reaches the highest point of the area as shown on drawings C8 to C11 (See Part 3). At this point the area shall be closed to receipt of refuse and the pipe plugged.

D. A minimum 7-foot-high chain link fence or an equivalent metal fence currently surrounds the present working area of the landfill site. As the landfill areas are developed fencing will be erected which will provide security for the site and identify the boundary. This boundary will be identified by appropriate signage. Fence locations are shown in Part 3. The access points utilized for ingress and egress can be secured by lock and key. A access control building is located at the landfill entrance.

E. Truck scales are installed at the control building. All vehicles admitted utilizing the landfill are weighed, and those are recorded and maintained by the scale operator. Inspections of vehicles shall be accomplished at the truck scale area, recorded, and maintained by the scale operator also.

F. Fire stations are located at 4 and 7 road miles from the landfill. A stockpile of earth shall be maintained within 2,500-feet of the working face for fire suppression also.

G. Robert Gray Army Airfield is the nearest airport at 5.2 miles south of the landfill. Fort Hood Airport is 6.8 miles east of the landfill. Killeen Municipal Airport, the nearest public airport, is 9.7 miles

southeast of the site. These airports and the roads and highways leading to Fort Hood can be seen on the Highway Map, Part 3.

III. SOLID WASTE DATA

A. Nature of Solid Waste

Solid waste disposed at the Fort Hood landfill consists of municipal solid wastes, construction and demolition wastes, all generated within the Base limits. Grease trap waste is not being accepted at this time due to the elimination from being permissible to be landfilled unless it has been solidified. A proposed design of a solidification process for the grease trap waste will be submitted for approval. If approved by the Executive Director the solidified grease trap waste would be allowed to be disposed in the landfill.

A design of a Special Waste Trench is included with this submittal that will be constructed in Area 2 upon approval by the TNRCC. Renovation/demolition debris with lead-based paint identified as Special Waste will require evaluation and approval by the TNRCC before acceptance at the landfill. Debris waste shipments shall be analyzed according to the Toxicity Characteristic Leaching Procedure (TCLP) and found to contain levels of lead to be in the range of non-hazardous waste. Specifically the waste must be in the range between 5 and 1.5 milligrams/Liter [(mg/L), approximately equal to parts per million], noninclusive (i.e., <5.0 mg/L and >1.5 mg/L) to be included in the Special Waste Trench. All waste below and including 1.5 mg/L (1.5 mg/L) may be placed with the municipal solid waste. Hazardous waste (5.0 mg/L) shall not be accepted at this facility. Other special wastes that would be allowed in the Special Waste Trench would be regulated and non-regulated asbestos-containing material that has lead-based paint applied on it. This would be contingent on the TCLP analysis results described above. Other design characteristics are discussed below. Solid waste shall be placed on top of the Special Waste Trench after it has been covered with 4-feet of compacted clay.

Solid waste will be collected by contracted private firms and hauled to the landfill site. The site will also be operated by contracted private firms. Regulations for the site will prohibit access of the general public to the site; however, military units will be permitted to haul to the site. No scavenging operations are allowed.

IV. DESIGN DATA

A. Land Use

1. General

The proposed sanitary landfill is sited in the southwest portion of Fort Hood Military Reservation within an area utilized for tactical training and maneuvers. Fort Hood also leases the area to ranchers as range land. The soils are not suited to cultivate crops or improve pasture. The land closest to the site utilized for purposes other than military training and range land is a housing development approximately 2,400-feet southeast of the site. The housing development is owned and controlled by Fort Hood. Immediately west of this housing area is a storage area comprised of six bunkers used for storage of furniture, mattresses, redeployment boxes, and communication supplies (wire, etc.). The map provided as Attachment 3 in Part 3, indicates land use within 1-mile radius consists of military training, range land, residential, and storage of general housing items.

2. Zoning

Fort Hood has no zoning restrictions. Long-range planning has reserved the area south of Turkey Run Road for future expansion of the cantonment area should such an expansion be required. However, there are presently no plans for development in this area.

3. Growth Trends

The population of Fort Hood is projected to remain essentially the same for the design life of the landfill. However, growth trends as indicated by 1978 photo revisions of the U.S. Geological Survey quadrangle Fort Hood, Texas, dated 1957, are towards the east and west. For the 21-year period from 1957 to 1978, residential construction has extended approximately 2.5 miles towards the west. Within the same time period, residential construction in Fort Hood has extended to its eastern boundary where it borders Killeen, a distance of approximately 1 mile. Growth trends of Copperas Cove, Texas, approximately 3.4 miles southwest of the landfill site, are predominantly towards the southeast, south and southwest. These trends were determined from 1978 photo revisions on the U.S. Geological Survey quadrangle Copperas Cove, Texas. Northerly growth is restricted by the Fort Hood Military Reservation boundary.

Growth has occurred at Fort Hood since 1978, primarily from base closures. New construction consists of a westerly expansion of Fort Hood 's industrial complex approximately 2 miles south-southwest of the landfill site. New family housing units are planned in sections of Fort Hood and the West Fort Hood area. The growth trend of Copperas Cove since 1978 has been to the south along FM 116.

4. Proximity to Sensitive Facilities

The boundary of a housing area is situated approximately 2,400-feet southeast of the proposed landfill site. Additionally, one school is situated approximately 4,100-feet southeast of the site. The closest business is approximately 3.5 miles southwest of the site in Copperas Cove. The nearest chapel is approximately 1.9 miles east of the site. Small surface archaeological sites are scattered throughout the installation and are under the protection of the Fort Hood Staff Archaeologist. The nearest archaeological site to the proposed sanitary landfill is approximately 900-feet to the west. There are no archaeological sites on the proposed site, based on results of surveys which were contracted in Fiscal Year 1984 and Fiscal Year 1985. Within a 1-mile radius of the site there are approximately 230 duplexes and no business establishments.

5. Availability of Roads

The landfill facility will serve the installation of Fort Hood. The government maintains the network of access roads, including Clarke Road and Turkey Run Road (the two all-weather access roads to the landfill).

6. Proximity to Airports

There are no airports within 5 miles of the landfill site. The nearest airport, Robert Gray Army Airfield, is 5.2 miles south of the site. Yoakum Defrenn Army Heliport is 6.8 miles east of the site. Killeen Municipal Airport, the nearest public airport, is 9.7 miles southeast of the site. A demonstration of compliance is provided in Part 4.

7. Buffer Zone

A minimum of 50-feet will be maintained between the disposal operation and the site boundary. There are no pipelines, underground utilities, or electrical transmission line easements within 25-feet of disposal operations.

B. Access

1. Vehicular Survey and Traffic Projection

An estimate of the number, size, and maximum weight of vehicles expected to use the site daily has been calculated over a 6-month period. This estimate is presented as follows:

Number	Size	Maximum Weight (lbs.)
12	Front Load Truck	16,450
5	2 1/2-Ton Truck	17,250
5	Pickup	4,000
10	5-Ton Truck	28,000

The estimated number of vehicles to use the site daily is 32.

2. Provisions for All-Weather Access

The primary access roadway to be used in the disposal operation is Turkey Run Road, classified as an all-weather, hard-surface roadway. Located at the southern boundary of the proposed landfill site, the two-lane roadway consists of strip paving with no significant roadway restrictions. An estimated total vehicles per day (VPD) on Turkey Run Road is between 5,000 to 6,000 VPD.

There is no projected increase for the next 25 years. Turkey Run Road is maintained by Fort Hood.

Interior all-weather roads and the access route to the entrance of the site will consist of a 6-inch aggregate base course compacted subgrade to 100 percent standard density.

Temporary roads will not be used to deliver wastes to the working face from the permanent road system. Permanent roads will be constructed around each Area. Temporary roads will be used until the completion of the construction of the Area and for access to monitoring wells in the areas that have not been completed. Temporary roads will be constructed by compacting the natural soil present, and by controlling drainage, or by topping them with a layer of a tractive material such as gravel or crushed stone. The use of gravel or crushed stone will help minimize mud tracking and aid in dust control. For additional dust control, water trucks will be used for sprinkling roads.

The intersection of Turkey Run Road and the access road are shown in Part 3. The actual gates for admittance into the site are located approximately 50-feet north of this intersection at the entrance to the landfill site. The radius of the access road at the entrance to the landfill site will be 50-feet. The maximum length of vehicles traveling on this entrance road will be 60-feet.

3. Access Control and Fencing

The long-range operation of the proposed sanitary landfill will be divided into three phases. Each phase will include one-third of the entire permitted facility. The entire landfill site is presently enclosed by a minimum 7-foot-high chain link fence or an equivalent metal fence. Once the landfill

is in operation, the fence will deter the entry of wild animals, protect the public from exposure to potential health and safety hazards, and discourage unauthorized entry or uncontrolled disposal of materials. Access to the site during times when the operators are not present will be prevented by means of a new 22-foot wide, 6-foot high, double swing gate with lock.

As the Phase I area becomes filled to capacity, additional fencing as previously described, will be constructed to encompass the Phase II disposal area. Phase III will be fenced in a similar manner at the appropriate time.

In the future, the internal fences between the Phases may be taken down as the area is returned to its native conditions. All boundary line fences will be maintained at all times.

C. Engineering Considerations

1. Landfill Method

The proposed method of landfill operation will be a combination of trench and area fill method.

2. Wet Weather Provisions

All-weather access to the site will be provided by means of all-weather roads. During the wet weather period, the site operator will designate the site location to deposit wastes. The site operator is also responsible for providing the exact location and the surface design of access road to be used during the wet weather period.

3. Special Handling Wastes

The landfill will not have specified areas for the disposal of special waste, except for the proposed special waste trench for depositing construction debris that has been tested to be non-hazardous according to the toxicity characteristic leaching procedure. The categories of special wastes to be allowed are given in the Solid Waste Data section above.

Special Waste Trench Design

The Special Waste Trench was designed according to guidelines obtained from the TNRCC Municipal Solid Waste Division and 30 Texas Administrative Code §§ 330.2-330.256, and in particular §330.137. The trench must have a 3-foot compacted clay liner directly overlain by a flexible membrane liner (FML), a leachate collection system, and finally covered by 4-feet of clay.

The composite liner system consists of a 3-foot clay layer overlain with a 60-mil high density polyethylene (HDPE) FML component, overlain with a 12-inch drainage layer (1×10^{-2} cm/sec coefficient of permeability), overlain by a 12-inch soil layer. The liner system must comply with the Soils and Liner Quality Control Plan in conjunction with the TNRCC's Liner Construction and Testing Handbook in Part 4.

The Special Waste Trench is designed with a leachate collection and removal system designed and constructed to maintain less than 30-cm depth of leachate over the liner. To predict the amount of leachate that will be produced in the Special Waste Trench, a U.S. EPA computer program, Hydrologic Evaluation and Liner Performance (HELP), Model 2.05, was used. Parameters pertaining to the geology/soils, climatological data, and solid waste placement, and liner data were used to predict the height of leachate on the liner to verify the slopes were sufficient and to avoid leachate ponding on the liner. It was determined that the leachate would not accumulate in levels greater than 30 cm (12-inches) after using the HELP model. See HELP Runs, Part 3.

At each lateral expansion of the Special Waste Trench is an 18-inch cleanout pipe for each sump for removal of the leachate. Each header pipe shall have a clean out and a cap at the end. As the area of fill expands, the cap will be removed and additional pipe shall be added to expand the header either east or west depending on the direction of expansion. This procedure shall be utilized until the expansion reaches the highest point of the area as shown on drawings C8 to C11 (See Part 3). At this point the area shall be closed to receipt of waste and the pipe plugged.

The special waste shall only be placed to the existing surrounding natural ground surface elevation, with the exception of asbestos-containing waste. Solid waste shall be placed on top of the 4-foot compacted clay layer to reach the final contours as shown on the closure for Area 2 (See Sheet 13 of 16). Inquiries to the TNRCC, Municipal Solid Waste Division, Permits Section resulted in the requirements for the liner system and cover. Mr. Philip A. Spry, P.E. advised that 4-feet of clay cover would be required; however, the 18-inches of topsoil would not be required since the municipal solid waste will be placed on top of the 4-feet of clay, which will ultimately receive a Subtitle D cover system.

Operations of the municipal solid waste section above the Special Waste Trench shall be as described in this permit as directed by the Executive Director.

Placement of the special waste in Area 2 shall begin on the east side and proceed in a westerly direction until the highest point in Area 2 is reached. Placement shall continue upward until the specified elevations are reached. Placement rates will be determined after the demolition debris is accepted for disposal, but as the demolition projects are started, rates of waste placement shall increase accordingly. Rates of waste placement shall be kept separate from the municipal solid waste. Rates of compaction shall be approximately 700 to 800 pounds per cubic yard. The daily cover material shall be placed on the special waste to protect personnel and environment from exposures, to prevent disease vectors, fires, odors, windblown waste, and scavenging. Shipments containing asbestos-containing materials shall be covered immediately with the location noted in the operators log. Leachate collection and removal, groundwater monitoring, closure and post-closure care shall be conducted according to sections given in this permit or as directed by the Executive Director. Personnel shall be trained in the safe operations of the Special Waste Trench and in personal protective measures to be taken while working in the Special Waste Trench, and the contingency and cleanup of any accidental spills occurring during the delivery and disposal operations. Personnel shall also be trained in the personal protective equipment to be worn. These items shall become a part of the Site Operating Plan.

A manifest system shall be established to accommodate the special waste disposal shipments. Requests for shipments shall be evaluated and must not be allowed without the authorization by the Executive Director to accept the waste, or not authorized by permit provisions. Upon approval, shipments shall be accounted for and waste shipment control tickets maintained for three years. A written report shall be submitted to the Executive Director no later than the 25th day of the month following the month in which the special waste was received. Reports shall be submitted on forms provided by the TNRCC.

When unexploded ammunition is found on-site, the location will be marked and the Explosive Ordinance personnel will be called at (254) 287-2309 to take charge of the round.

4. Control of Wind-Blown Wastes

Blowing litter will be kept at a minimum by maintaining a small-size working face and covering portions of the cell as they are constructed. Cover material is essential for maintaining a proper appearance of the sanitary landfill. Windblown material which is not contained by the cover material should be contained by a minimum 7-foot high security perimeter fencing. The landfill operator's personnel will cleanup litter periodically during the working day, especially near the close of the day. The litter will be placed on the working face before it is covered.

5. Rate of Waste Deposition.

Design Population 45,000

Solid Waste Generation 22,910 tons/year (average)

Solid Waste Weight 700 lbs/yd³ (compacted)

Yearly Volume 65,457 yd³/year

Daily Volume 252 yd³ day

The remaining available volume at the proposed site for compacted solid waste is estimated to be 2,703,402 yd³ (see Calculations, Part 3) as of August 10th 2023. Daily cover will occupy some of this volume. The ratio of solid wastes to daily cover material volume usually ranges between 4:1 and 3:1. Using the 4:1 ratio, the volume available for solid waste is 2,027,551 yd³. The yearly volume of compacted solid waste will occupy 65,457 yd³. Therefore, the design life of the proposed site is:

$$\frac{2,027,551 \text{ yd}^3}{65,457 \text{ yd}^3/\text{year}} = 31 \text{ years}$$

Therefore, the design life will be approximately 30 years depending on the volume of solid waste delivered to the site.

6. Subsequent Use of Site

It is anticipated that the disposal site will be returned to native conditions. The site will be seeded with native grasses upon completion of the disposal operations. The selection of such grasses and vegetation will be coordinated with the Land Management personnel located in the Directorate of Public Works.

7. On-Site Cover Material

Both daily cover and final cover material will be obtained from the excavations made at the proposed site. Final cover material will be a clayey soil and have a Unified Soil Classification of SC, CL or CH. The soil borings will be used to determine the depth and extent of the excavations with these soil classifications. The materials will be stockpiled for use as final cover. An estimated

444,214 yd³ of material are required for the final cover. From the soil borings, the volume of material available which meets the requirements for the final cover averages 11/2-feet over the entire area of the site. The volume of soil available that meets final cover requirements is estimated to be 384,000 yd³. This will require approximately 60,214 yd³ of the material which will meet final cover requirements to be provided from an off-site source. When the operation requires, select off-site material may be imported from just south of the proposed site across Turkey Run Road. The USDA-SCS soil survey indicates that the soil source is mapping unit Brackett. The top 3-feet are classified as CL or SC. Before this off-site source is utilized, the operator will provide the necessary laboratory analysis to indicate its suitability for final cover.

The material not stockpiled for final cover will be used for daily cover. The estimated volume of daily cover for the proposed site is 1,137,154 yd³. From the anticipated excavations, the volume of material available for daily cover is approximately 4,164,616 yd³ (volume of excavation less volume suitable for final cover). A surplus of daily cover material is expected and will be disposed of off-site of the landfill area, but on Fort Hood property as directed by the applicant.

A constructed 60-mil high density polyethylene (HDPE) liner system is proposed for this site. The construction is detailed in the Soil and Liner Quality Control Plan (SLQCP) in conjunction with the TNRCC's Liner Construction and Testing Handbook in Part 4 of this application. The site operator shall notify the regional office of the TNRCC, of their intent to begin construction of any liner prior to commencement of construction. The completed liner system will consist of a 2-foot compacted soil layer, overlain by a 60-mil HDPE liner, a 16 ounce geotextile fabric, and a 2-foot protective cover layer of soil and sand material above the 60-mil liner. The 2-foot protective cover layer will consist of 1-foot of drainage layer (1×10^{-2} cm/sec coefficient of permeability), overlain by a 1-foot layer of uncompacted soil. The material excavated from the site will be used, when acceptable. An estimated 444,214 yd³ are required for the 2-foot protective layer (volume approximately equal to final cover volume).

All stockpiles of soil material will be located along ridges or high points within the site, to avoid conflicts with existing drainage paths. The stockpile for final cover will be kept separate from the remainder of excavations. Adequate volumes of the cover materials will be stockpiled to provide the anticipated cover volumes.

8. Fire Control Facilities

No burning of wastes will be permitted at the sanitary landfill. The use of daily cover will keep any accidental fire confined to a cell that is under construction, from spreading laterally to other cells. All equipment operators will keep a fire extinguisher on their machines at all times. A stockpile of earth shall be maintained within 2,500-feet of the working face for fire suppression also. Should the need arise, the Fort Hood Fire Department will be called to control large fires.

9. Provisions for Compaction

The proposed site will use equipment common to earthmoving operations and equipment designed for the special task of compacting and otherwise handling solid waste. Solid waste will be compacted by the compressive forces developed by the overall massive loading of a landfill machine. The waste will be spread in layers approximately 2-feet deep and then compacted by tracked, rubber-tired, or steel-wheeled vehicles that pass over it 2 to 5 times. A compaction equivalent of 700 to 900 pounds per yd³ is anticipated.

10. Provisions for Subsidence

The sanitary landfill may settle as a result of waste decomposition, filtering of fines, superimposed loads, and its own weight. If settling produces wide cracks in the cover material, the FML will be exposed which will require additional cover material to be added. The surface will be inspected monthly and soil will be added and graded when necessary. Additional material will be available from the stockpile of excavated material located on-site.

A subsidence equal to 10 percent or less of the depth of solid waste is anticipated. All areas affected by subsidence will be restored to final grades by adding additional cover material and grading.

11. Provisions for Site Inspection and Maintenance

The disposal site will be readily available for inspection of erosion and ponding for 30 years following closure of the site. The inspection will check for erosion, ponding, leachate migration and methane migration. Corrective actions will be taken as necessary. Leachate migration will be monitored by water samples from the monitoring wells on the site. Erosion will be checked by grading the surface, filling depressions, and by planting vegetation on the final surface cover.

Ponded water will be controlled to avoid becoming a nuisance. Measures will be taken to prevent accumulation of water, not only in the waste receipt areas, but also the surrounding areas.

Methane migration is not anticipated to be a problem. The characteristic nature of the soils used in the landfill will allow methane to migrate freely to the surface and disperse to the atmosphere without concentration pockets. A methane gas recovery and venting system has been designed for final closure of each Area, to prevent gas buildup within the covered waste areas. A synthetic membrane will be used in conjunction with a soil cover to cap the waste storage area with the vents located along the top elevations laterally across the Area. Methane gas monitoring will be done quarterly according to the Methane Monitoring Plan.

Corrective actions will be completed as needed throughout the life of the facility and for 30 years following final closure. A Post Closure Care Plan is provided herein.

D. Groundwater Protection

1. Protection of Groundwater

A contour map of the potentiometric surface developed from water level readings obtained on December 7, 1987, was presented as Sheet 1 in the initial permit application. Groundwater data has been obtained from temporarily cased exploratory borings serving as observation wells. Each boring was cored using drilling fluid. Therefore, the top of the zone of saturation cannot be determined while drilling (i.e., level first encountered). To obtain the depth of groundwater, each boring was bailed of all drilling fluids and allowed to stabilize. Water levels were measured periodically and recorded. Table 3, Part 3 is this record. The applicant maintains this tabulation of water levels reflects water table conditions. As such, the pressure at the surface of this saturated zone is equal to atmospheric pressure. No additional hydrostatic head needs to be compensated for at or above this level. In situ materials overlying the water table were examined extensively to determine their adequacy as protection separating solid waste from groundwater. Thirty-two field permeability tests were conducted as discussed in Section III.1.2.e(2). Results are presented in Appendix IV. Additionally, 30 laboratory permeability tests, both vertical and horizontal, were conducted. Results are presented in Appendix IV. Although some laboratory coefficient of permeability values is lower than 1.0×10^{-7} cm/sec, values obtained from field tests were consistently higher. It is believed the results from the field permeability tests more closely represent conditions in the field than do the laboratory tests. Accordingly, the applicant concludes that the naturally occurring barrier of in situ soils does not provide the protection of 3-feet of soil with a coefficient of permeability of no more than 1.0×10^{-7} cm/sec. The applicant proposes construction

of a man-made liner, constructed of synthetic material, to provide compliance with the groundwater protection requirements specified in 30 TAC §§330.200 - 330.206.

2. Soil and Liner Quality Control Plan (SLQCP)

The proposed site will be operated to avoid contamination of the groundwater. A 60-mil high-density polyethylene liner (HDPE) system, or alternate design if approved by the Executive Director, will be used as shown by the enclosed drawings in Part 3. The revised SLQCP in conjunction with the TNRCC's Liner Construction and Testing Handbook is detailed in the Site Operating Plan (Part 4). They include specifications and construction methods.

3. Soil and Liner Quality Control Testing

The applicant will be required to furnish all liner quality control testing. The procedures and sampling frequency are included in the revised SLQCP and Liner Construction and Testing Handbook. Reports and results from the liner quality control testing will be submitted to the TNRCC. A Soil Liner Evaluation Report (SLER) shall be used to report the 2-foot, or the 3-foot soil liner for the Special Waste Trench, and a Flexible Membrane Liner Evaluation Report (FMLER) shall be used to report the installation and Quality Assurance/Quality Control process of the FML. Quality control sampling and testing will be performed by an experienced quality control technician and certified by a registered professional engineer, or an experienced engineering geologist as specified (see the TNRCC's Liner Construction and Testing Handbook - Part 4).

4. Liner Used In Conjunction With Compacted Clay

The applicant proposes to use a 60-mil high-density polyethylene (HDPE) liner system as detailed in this application. The lining will be placed in strict accordance with the revised SLQCP and Liner Construction and Testing Handbook as described in the Site Operating Plan (Part 4). Alternate liner designs, if approved by the Executive Director, shall be installed according to the Alternate Liner Construction Quality Control (CQC) and Construction Quality Assurance (CQA) Plan submitted with the permit modification or permit amendment.

5. Water Balance

To predict the amount of leachate that is produced in this landfill, a U.S. EPA computer program, Hydrologic Evaluation and Liner Performance (HELP), Model 2.05, was used. Parameters pertaining to the geology/soils, climatological data, solid waste placement, and liner data were used

to predict the height of leachate on the liner to verify the slopes were sufficient and to avoid leachate ponding on the liner. It was determined that the leachate would not accumulate in levels greater than 30 cm (12-inches) after using the HELP model. See HELP Runs, Part 3.

Volumes of leachate removed from the leachate collection system along with the measured depths in the leachate collection system, shall be recorded and inserted in the operating record. A long, wooden measuring pole shall be used to measure the level of the leachate on the liner. It shall be lowered down the 18-inch leachate cleanout pipe and obtain a reading on the pole. The amount of leachate will be calculated by using the angle of the side slope and the distance on the pole.

6. Groundwater Monitoring Wells

Siting of groundwater monitoring wells is dictated by the direction of groundwater flow. Sheet 1, submitted in the initial permit application, presents the proposed locations of one upgradient well and seven downgradient wells. The well on the south perimeter is designated the upgradient well. The wells located on the west, north, and east perimeters are sited to intercept contamination at the earliest possible occurrence and are accordingly designated the downgradient wells. Typical details of groundwater monitoring well construction are shown on Figure 1. Texas Water Commission's Form SE 67, Monitor Well Data Sheets, for the 13 monitoring wells are included in Part 3. The depth of each well is designated to supply a sufficient reservoir for sampling during periods of low water table elevations. Five additional groundwater monitoring wells were developed in September 1992 after the initial permit submittal. Well No. 9 was drilled in the middle of the proposed Area 6 of Phase 3. This well will be plugged and abandoned according to the Groundwater Monitoring Well Plugging Procedures section below, or as directed by the Executive Director, to avoid providing a pathway from the refuse to the groundwater. Another monitoring well, No. 14, will be drilled in the southeast corner of the landfill to replace No. 9 but primarily to provide an additional up-gradient well. All existing and proposed groundwater monitoring wells are indicated on the enclosed drawings in Part 3. The Groundwater Sampling and Analysis Plan below describes the sampling and analysis of these wells.

E. Groundwater Sampling and Analysis Plan

Groundwater sampling will be conducted according to 30 TAC §330.233, and the TNRCC's Municipal Solid Waste Division, *Groundwater Sampling and Analysis Plan (GWSAP)*, Publication No. RG-74, May 1994. The U.S. Army at Fort Hood, Texas shall determine who shall perform the sampling and analysis. A baseline data of groundwater quality is presently available for the landfill based on the

indicator parameters as listed in the original permit submittal, but not for the Detection Monitoring Parameters listed in Table 1, Part 3. To establish a baseline database on all the parameters, a minimum of four independent samples from each well (upgrading and down-gradient) must be collected and analyzed during the first sampling event. Background groundwater quality shall be established for the landfill in each well independently for each groundwater detection monitoring parameter. Analyses for the Detection Monitoring Parameters will be accomplished during the first semi-annual sampling event. These samples shall be statistically independent, with additional samples having to be collected from each well should it be necessary to satisfy the requirements of the statistical methods chosen to evaluate the data. The duration of the first semi-annual sampling event may take up to one year to complete, to assure that statistically independent samples are obtained. Seasonality and temporal and spatial variability should be taken in to account to determine the statistical independence of each sample. Samples collected during both low and high rainfall periods would be useful to determine the effect of climatic changes on the groundwater chemistry. Enough time should be allowed to pass for the wells to recover and to be from different water. This is the amount of time necessary for water in each monitoring well to completely change and provide temporal independence of samples from each well. The time required for a change in the groundwater can be calculated using Darcy's equation with the diameter of the well.

Detection monitoring will begin six months after the completion of background sampling for the detection monitoring parameters and sampled every six months thereafter. The groundwater sampling and analysis program must be conducted for the life of the facility and for the duration of the post-closure care period of the facility which is specified as 30 years in the Texas Administrative Code Title 30. Groundwater monitoring may be extended beyond the period specified if deemed necessary by the Executive Director.

Preparation for the sampling event should include obtaining all the necessary equipment and supplies for the sampling and decontamination procedures. A detailed equipment list should be compiled and used by the samplers for preparation of sampling. A review of the Site Safety and Health Plan (SSHP) for responsibilities and precautions should be done. A copy of the SSHP, included in Appendix I, should be included in items to accompany the samplers. The categories of materials to consider are:

- Personnel protective equipment such as boots, gloves, goggles, etc.;
- Safety equipment such as first-aid kit, portable eyewash station, fire extinguisher;
- Field parameter monitoring equipment such as pH, conductivity, and turbidity meters, thermometer, water-level measurement device;

- Sample bottles for sampling wells plus additional bottles;
- Decontamination equipment and supplies for sampling equipment, field monitoring equipment, samplers, tools, etc.;
- Sampling equipment for problem correction such as tool kits; and
- Shipping and sampling labels such as tape, labels, shipping forms, chain-of-custody forms and seals, field notebooks, sampling forms, permanent pens, iced storage chests.

The groundwater sampling equipment should be inspected prior to sampling to determine its condition. Check valves should be inspected to ensure that fouling problems do not reduce the delivery capabilities or cause aeration of the samples. Other sampling practices to observe include the following:

- Sampling should proceed from upgradient or background wells to down-gradient wells;
- Before each well, samplers should disassemble and decontaminate the sampling equipment;
- Decontaminated sampling equipment should not be allowed to come into contact with the ground or other contaminated surfaces prior to lowering into the well;
- Bailers and other sampling equipment should never be dropped into the well, but lowered slowly to avoid degassing of the water upon impact;
- Groundwater samples should be collected as soon as possible after the well is purged;
- The contents of the sampling device should be transferred to sample containers in a controlled manner that will minimize sample agitation and aeration, and to avoid spillage;
- Groundwater samples collected for analysis for organic parameters should not be filtered in the field;

Preparation for sampling includes preparing field notebooks, sampling forms, labels, etc.

Well Inspection

Upon arrival at each monitoring well, the samplers should inspect the area around the well and make observations for anything unusual or of obvious concern. These would include broken piping or casing, broken concrete pads, well maintenance activities, spillage of water or other liquids, and/or odors. Notations should be made on the Groundwater Monitoring Well Sampling form (sampling form - Enclosed in Part 3) to request corrective action be taken. If the situation warrants (i.e., broken casing, spillage, etc.), the sampler should notify the Directorate of Public Works personnel to provide for immediate corrective action.

The samplers would prepare the sampling form for the sampling event and record all the necessary information. After the inspection, a sampling and decontamination area would be set up around the well to accomplish the sampling and decontamination procedures. Six mil polyethylene or plastic sheeting shall be placed around the well casing and pad to provide containment for accidental spillage of the groundwater, and to provide a clean area to prevent cross contamination of groundwater and sample equipment.

Decontamination

Decontamination of sampling equipment (i.e., water-level measurement device, well purging device, sample collection device) that is not dedicated to the well, and portable instrumentation for field parameter monitoring shall be accomplished according to the following procedure:

- Rinse with water (tap, distilled, or deionized);
- Wash and scrubbing with laboratory-grade detergent; and
- Two to three rinses with distilled water.

The introduction of other materials (i.e., acids and solvents) that may be incorporated into the decontamination procedure, must be evaluated with respect to its potential to impact sample integrity by imparting potential contaminants to the equipment being cleaned. It is recommended that this procedure be reviewed periodically by reviewing the QA/QC sample results with respect to the Detection Monitoring Parameters. Sampling equipment should be disassembled as much as possible to decontaminate. Several important issues of the decontamination procedures that the samplers must consider include the following:

- The location and construction of the decontamination area;
- Movement of clean and contaminated equipment in and out of the decontamination area;
- Preliminary cleaning of all equipment prior to sampling event;
- Segregation of clean and contaminated equipment;
- Controls to ensure that cleaned equipment does not become contaminated prior to sampling
- Chemical verification of the suitability of the portable water supply for cleaning; and
- Use of rinse (field) blanks and/or wipes samples in conjunction with trip blanks to verify the effectiveness of the decontamination procedures.

Field blanks should be used to determine the effectiveness of this and other decontamination practices. The use of latex gloves during well sampling and field analysis is required to protect

samplers and to prevent cross-contamination. Gloves shall be discarded between wells or whenever torn.

Water-level Measurement

After preparing the well and decontamination has been accomplished, the water-level measurement must be taken. A wetted-tape or a electrical probe can be used to measure the groundwater levels in each well. For the wetted-tape method, a steel tape with markings divided into tenths of feet, is coated with carpenter's chalk. A water-level measurement is made by slowly lowering the tape into the well about 1 or 2-feet into the water. It may be convenient to lower the tape in to the water a sufficient distance to allow the tape to read and even foot mark at the top of the well casing or surface reference point. Interpolation to the nearest 0.01-foot is possible. The water-level measurement is calculated by subtracting the submerged distance, as indicated by the absence of change in chalk color, from the reference point at the top of the well.

The electrical probe method of measuring the water level operates on the principle that a circuit is completed when two electrodes come in contact with the water surface in the well. Other instruments rely on physical characteristics as resistance, capacitance, or self-potential to produce a signal. These instruments use a two-wire conductor which is marked every foot, every 5-feet, or at 10-foot intervals. Some instruments use vinyl-, epoxy-, or Teflon-clad steel tapes as an insulated electrode and the well casing or grounding wire as the other. Measurements made with well maintained electrical probes are reproducible to within ± 0.02 -foot to 0.1-foot. Prior to sampling a well, the water level will be measured and used to calculate the water in the well which will be recorded on the sampling form. Decontamination of the water-level measuring instrument would follow the procedures described above.

Purging of the Wells

Using the calculated water in the well, the samplers would next calculate the volume of water to be purged. All wells will be purged prior to sampling to insure that samples are representative of the groundwater and do not contain water which has been standing in the casing. At least three to five volumes of water in the well casing will be removed unless low yield wells which do not quickly recharge after pumping are present. In such an event, the low yield well will be pumped dry, and the sample then obtained as soon as the well recharges. It is recommended that once a low-yield well has been purged dry, that the well be allowed to recover for up to seven days, collecting groundwater samples as it becomes available. It is anticipated that all monitoring wells installed at the landfill site

will be the low-yielding type, but typically have sufficient water for sampling after 24 hours. A dedicated fluorocarbon resin or stainless-steel bailer with double check valve and bottom emptying device and inert cable/chain (fluorocarbon resin or stainless steel) will be utilized. Since the bailer used for purging is the same for sampling, the bailer shall be lowered to the level of the well screen for purging. The bailer shall be removed, and the water emptied into a measuring device to determine volume removed. When the purged volume that was calculated is obtained or the well goes dry, purging shall stop. Purged water combined with the soap and water from the decontamination procedures shall be stored in a drum or barrel and retained until the analyses has been performed. The stored fluids should be sampled and analyzed according to the Toxicity Characteristic Leaching Procedure (TCLP), EPA Method 1311. The groundwater analysis should provide a basis to determine which parameters to analyze for. Disposal of the purged water and decontamination fluids would be by applicable federal, state, and local laws and regulations. Should the analyses indicate elevated levels of constituents in the water which have been identified or listed as a hazardous waste by the Administrator of the U.S. Environmental Protection Agency, then the waste must be disposed of as a hazardous waste. If the TCLP analysis indicates there are no listed toxic or hazardous wastes, then the fluids may be disposed at the Bell County W.C.I.D. No. 1 Wastewater Treatment Plant, pending approval.

Depending on the time elapsed and the equipment used, the decontamination of all the purging equipment may be necessary to ensure sample integrity and prevent contamination. When purging must be accomplished over a timed interval, the bailer should be decontaminated after each use. The bailer does not require decontamination before sampling, if the sampling follows immediately after the purged volume is obtained. Decontamination should always follow the completion of a sampling event.

Measurement of Field Parameters

Analyses of field parameters are conducted at each well upon completion of the purging procedures and there is sufficient groundwater to analyze. Those parameters measured include temperature, pH, specific conductance, turbidity, and physical/chemical characteristics.

1. Temperature:

Temperature as a screening parameter can be useful to determine possible contamination. Heat can ultimately enter the groundwater system since heat is transferred to the leachate from the anaerobic decomposition process of the putrescibles in landfills. Aquifers are good insulators,

therefore, the warmed water may extend for some distance downgradient. A zone of warmer water may be indicative of a leachate plume. The water temperature will be measured with a calibrated thermometer upon removal of the water from the well. Record the temperature to the nearest °C.

2. pH:

pH is another screening field parameter that will be indicative of changes in the ambient water quality. Wastes can cause groundwater to become alkaline or acidic if contacted with wastes. Leachate may influence an acidic or alkaline groundwater toward neutrality also. Samples should be analyzed as soon as possible at the time of sampling. The pH is measured with a portable pH meter. The pH meter is calibrated with the pH 7.0 and 4.0 buffers. The pH 10.0 buffer is used as a reference check.

Procedure:

A sample is placed in a clean disposable beaker using sufficient volume to cover the sensing elements of the electrodes and to give adequate clearance for the magnetic stirring bar. Rinse and wipe the electrodes before measuring the pH of samples. Immerse the electrodes into the sample beaker and gently stir at a constant rate to provide homogeneity and suspension of solids. Note and record the pH and temperature of the sample. If the sample temperature varies by more than 2°C from the buffer solution, the measured pH values must be corrected. Instruments are equipped with automatic or manual compensators that electronically adjust for temperature differences. Refer to the manufacturer's instructions. Repeat measurements on successive volumes of sample water until values differ by less than 0.1 pH units. Two or three volume changes are usually sufficient.

pH meters read directly in pH units. Report pH to the nearest 0.1 unit and temperature to the nearest °C.

3. Turbidity:

This method is based upon a comparison of the intensity of light scattered by the sample under defined conditions, with the intensity of the light scattered by a standard reference suspension. The higher the intensity of scattered light, the higher the turbidity. Readings are made in a nephelometer designed according to specifications, with a range of turbidity from 0 to 40 nephelometric turbidity units (NTU). A standard suspension of Formazin, prepared under closely defined conditions, is used to calibrate the instrument. Preservation of the sample is not practical,

analysis should begin as soon as possible. The presence of floating debris and coarse sediments will give low readings. Finely divided air bubbles will affect the results in a positive manner. The sensitivity of the instrument would permit detection of a turbidity difference of 0.02 unit or less in waters having turbidities less than 1 unit. The sample tubes to be used with the available instrument must be clear, colorless glass. They should be kept scrupulously clean, both inside and out, and discarded when they become scratched or etched.

a. Procedure:

- (1) Calibration of the turbidimeter should follow manufacturer's operating instructions. Measure standards on the turbidimeter covering the range of interest. At least one standard should be run in each instrument range to be used. Some instruments allow adjustments of sensitivity so that scale values will correspond to turbidities. If a pre-calibrated scale is not supplied, then calibration curves should be prepared for each range of the instrument.
- (2) For turbidities less than 40 units; Shake the sample to thoroughly disperse the solids. Wait until air bubbles disappear then pour the sample into the turbidimeter tube. Read the turbidity directly from the instrument scale or from the appropriate calibration curve.
- (3) For turbidities more than 40 units: Dilute the sample with one or more volumes of turbidity-free water, until the turbidity falls below 40 units. The turbidity of the original sample is then computed from the turbidity of the diluted sample and the dilution factor. For example, if 5 volumes of turbidity-free water were added to 1 volume of sample, and the diluted sample showed a turbidity of 30 units, then the turbidity of the original sample was 180 units.

b. Calculation:

- (1) Multiply sample readings by the appropriate dilution to obtain the final reading.
- (2) Report results as follows:

<u>NTU</u>	<u>Record to the Nearest</u>
0.0 - 1.0	0.05
1-10	0.1
10-40	1
40-100	5

100-400	10
400-1000	50
>1000	100

4. Specific Conductance:

The specific conductance of a sample is measured by the use of a self-contained conductivity meter. Samples are preferably analyzed at 25°C. If not, temperature corrections are made and results reported at 25°C. Instruments are standardized with KCl (Potassium Chloride, 0.01 M) solution before use. The conductivity cell must be kept clean. The sampler should follow the direction of the manufacturer for the operation of the instrument. Determine the temperature of the sample within 0.5°C. If the temperature of the sample is not 25°C, make the temperature correction in accordance with these instructions:

Temperature corrections are based on the standard KCl solution.

- a. If the temperature of the sample is below 25°C, add 2% of the reading per degree.
- b. If the temperature of the sample is above 25°C, subtract 2% of the reading per degree.

Results are to be reported as Specific Conductance, $\mu\text{mhos/cm}$ at 25°C.

5. Physical/Chemical Characteristics:

Several physical and chemical characteristics may be evident by observation upon removal of the groundwater from the wells. Color and odors may be present in groundwater samples, which may indicate the presence of contaminants. Odors may indicate the presence of certain contaminants such as volatile organic compounds. Another physical evidence of possible problems would be a two-phase (aqueous and nonaqueous) sample. This could indicate an introduction of an organic liquid. The organic phase may be either lighter or heavier than water, depending upon its composition.

Notes describing these observations should be placed in the "Comments" column on the sampling form. The Directorate of Public Works should be notified of these observations.

Sampling and Preservation

Sample containers will be rinsed with water pumped or bailed from the well prior to filling unless sample containers contain preservation agent(s), in which case, the container will be filled so that no

headspace remains. Sample containers will be filled in a manner which does not agitate or aerate the sample in order to prevent loss of volatile contaminants and to prevent chemistry changes due to either CO₂ loss with pH increase or oxygenation of the samples. All sampling equipment will be thoroughly decontaminated as described above.

Samples will be collected and containerized in the order of the volatilization sensitivity of the parameters being collected. The preferred order of collection should be as follows: volatile organics; dissolved gases; total organic carbon; large volume samples for organic compound analysis; total metals; inorganic species such as nitrate, ammonia, iron; cations and anions; QA/QC. Groundwater samples require varying sampling amounts, preservation methods, and holding times before analysis must be performed. Samples to be analyzed for volatile organic compounds require 2- 40 ml glass jars or bottles with Teflon-lined septa. The preservation of these samples required the addition of hydrochloric acid (HCl) to the groundwater to make the pH less than 2 (<2). Samples must be cooled to 4°C after sampling and maintained at that temperature until analysis. Maximum sample holding time by the analytical laboratory is 14 days. Groundwater to be analyzed for metals is sampled in 1-liter plastic or glass, jars or bottles. These samples are to be acidified with nitric acid (HNO₃) to a pH<2 after field filtering and must be maintained at 4°C after sampling. Maximum holding time for metals is 6 months except for mercury which is 28 days. Inorganic parameters can be sampled in plastic or glass, bottles or jars, with varying sampling amounts required, sample preservation methods, and holding times. Sample containers, (size and material of construction), preservative methods, and holding times should be obtained from Attachment A entitled "Recommended Sampling, Preservation, and Storage Procedures for Groundwater Monitoring" from the TNRCC Municipal Solid Waste Division's Technical Guidance Document, *Guidelines For Preparing A Groundwater Sampling And Analysis Plan (GWSAP)*, Publication No. RG-74, May, 1994, or Table 11-1, "Sampling and Preservation Procedures for Detection Monitoring, " from the *Test Methods for Evaluating Solid Waste Physical/Chemical Methods*, 3rd EPA Publication No. SW-846, 1986, with revisions.

Groundwater samples to be analyzed for metals will be filtered in the field. The groundwater sample is passed through a porous filter, made of glass microfiber or cellulose membrane, typically with a pore size of 0.45 μ m. The groundwater sample is pulled through the filter under a vacuum and flows into the sample container, or into a transfer vessel and then into the sample container. The potential for cross-contamination of samples exists; therefore, the vacuum device must be disassembled for thorough decontamination between samples. For this reason, disposable filters are recommended. There is also a potential to alter the chemistry of the groundwater by sample aeration/oxygenation,

degassing of volatile constituents, and imparting partial-pressure changes; therefore, only the samples for metal analysis will be filtered. Reevaluation of this practice should be done upon the determination of the analytical laboratory to perform the sampling and analysis. When samples are filtered the acid preservatives must be added after filtration. A space is provided on the Analysis Request and Chain of Custody Record form to indicate whether the sample was field filtered or not, by marking a check or "X" for yes or no.

QA/QC samples to be included in the shipment to the laboratory include field duplicates, trip blanks, field blanks, equipment blanks, and split samples. These samples shall be prepared for each groundwater sampling event. Field duplicates consist of two samples from the same well but labeled differently (i.e., different sample number). It is recommended that two field duplicates be prepared for each sampling event. Trip blanks are prepared before the sampling event by pouring laboratory-grade distilled water, into clean sample containers and then transported to and from the site with the other sample containers. Trip blanks are to determine if any of the sample bottles or samples collected have been contaminated before, or during sampling, or if sample shipment, handling, or storage have impacted the integrity of the samples. One trip blank for each sampling event will be used. Field blanks are prepared in the field by pouring laboratory-grade distilled water, brought from the laboratory, into sample containers opened in the field, then returned to the laboratory with the samples to be analyzed. Field blanks shall be collected when sampling the down-gradient wells at the site. One field blank for each sampling day or two field blanks, whichever is greater, will be used. Equipment blanks are a sample of laboratory-grade distilled water, processed through the sampling equipment in the same manner as the groundwater samples to determine the effectiveness of the decontamination procedures. One equipment blank per sampling event will be used. Split samples shall be taken from the same bailer and divided as evenly as possible between the sample containers. Sample pre-treatment, handling methods, and as much as a possible, the methods of analysis shall be the same for both sets of samples. This is to validate the analytical results and laboratory documentation. One split sample per sampling event will be taken.

Chain of Custody Records:

A water sample chain-of-custody record will be maintained to document sample possession and the handling of individual samples from the time of collection through the completion of laboratory analysis. The chain-of-custody program should include the following:

Sample labels to prevent misidentification of samples;

Sample custody seals to preserve the integrity of the samples from the time they are collected until they are opened in the laboratory;

Field notes to record information about each sample collected during the groundwater monitoring program;

Chain-of-custody record to document sample possession from the time of collection to analysis; and

Laboratory storage and analysis records, which are maintained at the laboratory and which record pertinent information about each sample.

The various components of this chain-of-custody program are described below:

Sample Labels:

Each sample's identification must be marked clearly on the sample container labels in waterproof ink.

The information should be printed on each label and placed on the sample container prior to filling.

The labels should be durable and legible even when wet and contain the following information:

- Sample identification number;
- Name and signature of the sampler;
- Date and time of sample collection;
- Well number; and
- Analyses requested.

Sample Custody Seal:

Sample custody seals are placed on the shipping container and/or each sample bottle in a manner that will break the seal if the container or sample is tampered with.

Field Logbook:

The field logbook is to provide documentation of all activities involved in sampling activities. Items to be noted in the logbook are measurements, observations, information about samples including color, turbidity, degassing odor, and surface film or layered sample. Weather conditions should be noted including temperature, humidity, wind, and precipitation. Well purging activities should also be noted.

The field logbook should document the following information:

- Well number and depth;
- Static water level depth and measurement technique;

- Presence and thickness of immiscible layer, if present, and the detection method;
- Well purging procedure, equipment and time;
- Well yield (high, same, low) and well recovery after purging (slow, same, fast);
- Collection method for immiscible layers, if needed;
- Sample withdrawal procedure, equipment, date and time, and results;
- Field observations of sampling event, including weather condition;
- Name of samplers; and
- Internal temperature of field and shipping containers.

Sample Analysis Request and Chain of Custody Record:

The form to be used for this record is included in Part 3, as Analysis Request and Chain of Custody Record (field chain-of-custody). The samplers shall manage this from the time of the sampling event until delivery to the analytical laboratory. The samplers shall record all the necessary information on the form and sign their name. Items of information to be recorded include the following:

- Sample numbers;
- date and time of sample
- whether or not the sample was filtered in the field;
- size and material of sample container;
- sample type, liquid or if immiscible material present;
- type and quantity of preservatives added, unusual conditions; and
- analysis to be performed.

Once the samples are delivered to the laboratory, the samplers must sign in the "Relinquished by:" space and record the date and time relinquished on the form. The person receiving the samples must sign their name adjacent to the person relinquishing the samples, in the "Received by:" space, record the date and time received and whether or not the samples were intact. This procedure is followed each time someone else handles the samples, until delivery to the analytical laboratory. The laboratory personnel will sign the field chain of custody and take receipt of the sample bottles. An internal laboratory chain-of-custody is maintained at the laboratory. As with the field chain-of-custody, the laboratory custody and transfer of samples must be documented so that the identity and integrity of samples can be established and maintained.

Laboratory Analysis:

Groundwater samples will be analyzed for the Detection Monitoring Parameters listed in Table 1, Part 3, and the indicator parameters listed in Table 2, Part 3 by an analytical laboratory selected by the U.S. Army, Fort Hood. Analyses will be conducted in accordance with the standard methods from "Test Methods for Evaluating Solid Waste Physical/Chemical Methods," 3rd EPA Publication No. SW-846, 1986, as revised December 1987 and November 1990, or those in *Standard Methods for the Examination of Water and Wastewater*, 18th edition, 1992. The test methods to be used for all the volatile organic compounds (VOC) from the Detection Monitoring Parameters in Table 1 will be EPA solid waste Method 8260, revised November 1990. Method 8260 is based upon a purge and trap, gas chromatographic with capillary column and mass spectrometric (GC/MS) procedure. An alternate method, EPA Method 8240, would be used if it is determined that there is a constituent that has questionable results and there is a need for a comparative analysis. EPA Method 8240 is based upon a purge and trap, gas chromatographic with packed column and mass spectrometric. U.S. Army, Fort Hood in consultation with the analytical laboratory would make this determination. Groundwater samples to be analyzed for VOC's should be extracted by EPA Method 5030, purge and trap method prior to analysis by these methods. Heavy metals will be analyzed by inductively coupled plasma emission spectroscopy (ICP) EPA Method 6010. Metals such as antimony, beryllium, cadmium, lead, thallium, arsenic, and selenium may require a 7000 series method which would involve the use of a graphite furnace to achieve reporting limits that are at or below the maximum contaminant level of those metals. The following metals with the corresponding EPA methods for analysis are given:

<u>Metals</u>	<u>Suggested Method</u>
Antimony	7040 or 7041
Beryllium	7090 or 7091
Cadmium	7130 or 7131
Lead	7420 or 7421
Thallium	7840 or 7841
Arsenic	7060 or 7061
Selenium	7740 or 7741

The inorganic analyses can be performed using ICP Method 6010. These methods have a range of Practical Quantitation Limits (PQLs) from 12 to 50 $\mu\text{g/L}$ (micrograms per liter, approximately equal to parts per billion). For any of the analytical methods used, the PQL used in subsequent statistical analysis, is the lowest concentration level that can be reliably achieved within specified limits of precision and accuracy during routine laboratory operating conditions that are available to the facility. Procedures shall be in place for demonstrating proficiency with each analytical procedure used in the

laboratory. These shall include procedures for demonstrating the precision and bias of the method as performed by the laboratory and procedures for determining the method detection limit (MDL). Documented precision, bias, and MDL information shall be maintained for all methods performed in the laboratory. Procedure shall also be in place for demonstrating that the laboratory is in control during each data collection activity. Analytical data generated with laboratory control samples that fall within prescribed limits, are judged to be generated while the laboratory was in control. Conversely, data generated that fall outside the control limits are judged to be "out of control" situations. These data are considered suspect and must be repeated or reported with qualifiers. Laboratory control samples shall be analyzed with each sampling event to verify that the precision and bias of the analytical process are within control limits. A method blank shall be analyzed with each sampling event to assess contamination levels in the laboratory. Guidelines shall be in place for accepting or rejecting data based on the level of contamination in the blank. Procedures shall be in place for documenting the effect of the matrix on method performance. When appropriate for the method, there shall be at least one matrix spike and either one matrix duplicate or one matrix spike duplicate per analytical batch. Any activity not performed in accordance with laboratory procedures, is considered a deviation from plan. Errors, deficiencies, deviations, or laboratory events or data that fall outside of established acceptance criteria require investigation. In some instances, corrective action may be needed to resolve the problem and restore proper functioning to the analytical system. The investigation of the problem and any subsequent corrective action taken, shall be documented. The analytical laboratory must submit its Standard Laboratory Operating Procedure (SLOP) for review and approval by the TNRCC, before any analyses is conducted by the laboratory. The SLOP should demonstrate that the laboratory meets or exceeds the QA/QC standards described in Appendix II. The Approved SLOP will be substituted for the SLOP in Appendix II.

Several flowcharts are provided on the following pages to provide a general overview of the selection of methods for the three types of statistical comparisons: interwell comparisons, comparisons with the Maximum Contaminant Level (MCL) and intra-well comparisons. Selection of an appropriate statistical test must be based on the quality of the data available, the hydrogeology of the site, and the theoretical properties of the test. Figure 2 is a general flowchart indicating the various statistical comparison techniques utilized for the three types of groundwater monitoring comparisons. Figures 3 through 7 are provided to aid U.S. Army, Fort Hood in selecting and applying the statistical methods. It is important to note that the selection of a particular statistical procedure may be based also on the quantity of data available. This is of special concern when selecting a statistical procedure or group of procedures for this landfill since existing data is extremely limited.

Reporting:

The groundwater data collected during the detection monitoring phase of characterization shall be restructured, simplified, and finally presented in a manner to allow interpretation and verification. The physical and chemical analytical data shall be reported to Fort Hood by way of laboratory analysis sheets. The data will be compiled into tables and display formats that will facilitate the understanding and correlation of the information. A list of all monitoring data should be provided for each sampling event and updated as new data becomes available. The data should include the following: well identification, date of analysis, laboratory, measurement unit, limit of detection, and chemical concentration. Results for a given analyte are presented as a quantified value or as ND (not detected). Results are reported either as a quantified concentration or as less than (<) the method detection limit value.

U.S. Army, Fort Hood will submit a report to the Executive Director with the results of each sampling event and analysis for each of the monitoring wells sampled within 45 days after each sampling event. Triplicate copies of the results will be submitted with the TNRCC's form, No. TNRCC-0312 with all the heading information completed. All four pages should be completed with the results of the testing; analytical methods for each of the 62 parameters (47 VOCs and 15 metals) from the detection monitoring parameters of Table 1, §330.241; and the PQLs. The analyses should include the laboratory report, the laboratory QA/QC data, and copies of the Analysis Request and Chain of Custody Record forms. Each submittal must include a cover letter with a discussion of the monitoring results with an interpretation of the data. One original and two copies are submitted to the TNRCC Municipal Solid Waste Division.

In determining whether a statistically significant change has occurred, the U.S. Army, Fort Hood will compare the groundwater quality of each parameter at each monitoring well to the background value for that parameter, according to the statistical procedures and performance standards specified under this section. If there is a statistically significant change from background of any tested constituent at any monitoring well, the U.S. Army, Fort Hood shall notify the Executive Director in writing no later than 60 days after the sampling event. The report should include all the background information leading to the conclusion that a statistically significant change had occurred, what measures were being taken to determine the source, and possible corrective measures. Other information to include in the report would be laboratory analysis summary sheets, QA/QC results, and verification of data performance by analytical laboratory.

If a statistically significant change over background concentrations is determined for one or more parameters required to be monitored, the U.S. Army, Fort Hood shall immediately place a notice in the operating record, describing the increase and shall establish an assessment monitoring program within 90 days of the date of the written notice to the Executive Director as described above. The assessment monitoring program or corrective action program shall be followed as described in 30 TAC §§330.235-330.238 and as specified by the Executive Director. An exception to this would involve the U.S. Army having reasonable cause to think that a source other than the landfill unit has caused the contamination that resulted in a statistically significant change from background levels of any of the tested constituents, at any of the monitoring wells or that the statistically significant change resulted from error in sampling, analysis, statistical evaluation, or natural variation in groundwater quality. If so, then a report shall be prepared and certified by a qualified groundwater scientist providing documentation that the reason for the statistically significant change was a source other than the municipal solid waste landfill facility and submitted to the Executive Director for review and approval. If a demonstration to the Executive Director has not been satisfactory made within 90 days after the sampling event, the U.S. Army, Fort Hood shall initiate an assessment monitoring program as described in 30 TAC §§330.235-330.238 and the specification of the Executive Director.

F. Groundwater Monitoring Well Plugging Procedures

The following is provided, as required, for each of the groundwater monitoring wells on the landfill:

1. Monitoring well-surface elevations at the top of the casing with respect to mean sea level, as well as depth of wells, depth to groundwater, and elevation of groundwater are listed in Table 6 of Groundwater Monitoring Wells, Part 3; and
2. As-built diagrams showing all pertinent features are shown on Monitoring-Well Records in Part 3.
3. Plugging and abandonment of monitoring wells and geotechnical borings shall be done according to the rules of the State of Texas.
 - a. In addition to the standards in 31 TAC §330.48 and §330.49, the following standards shall apply to plugging and abandonment:
 - (1) For any well, the primary method of plugging and abandonment shall be removal of the well's casing and other components of the well including the screen, grout, bentonite seal, filter pack, concrete slab, protective casing, guard posts, and native soil in immediate contact

with the grout and subsequent installation of cement-bentonite grout from the bottom of the resulting borehole to the ground surface using the tremie method;

(2) In areas where all or part of the well's casing and other components of the well cannot be plugged and abandoned with the procedure stated in the previous paragraph, the well shall be plugged and abandoned by installation of cement-bentonite grout mixture inside the well's casing from the bottom of the well to the ground surface provided the annular seal is demonstrated to be adequately sealed and the following items are submitted:

a) Prior to plugging the well, supporting documentation is provided that demonstrates that removal of all or part of the well's casing and other components of the well will be detrimental to the environment; and/or

b) Certification and supporting documentation by a qualified groundwater professional that shows that removal of the well's casing was attempted, and that continued attempts to remove all or part of the well's casing, and other components of the well would have been detrimental to the environment.

(3) After plugging and abandoning a well, all surface features of the well including, but not limited to, the concrete slab, guard posts, protective casing, shall be dismantled and disposed of in an environmentally-sound manner, and the surface restored to its original condition.

G. Drinking Water Protection

There are no water supply wells in the vicinity of the landfill. Fort Hood receives its principal water supply from the City of Gatesville which is from Lake Belton on the Leon River. North Fort Hood and surrounding communities also rely upon the City of Gatesville for their water supply which is water from Belton Lake. The Travis Peak occurs at an estimated depth in excess of 400-feet at the site. Separating the Travis Peak Formation from the bottom of trenches at the proposed landfill, is an estimated thickness of 400-feet of primary strata including the Paluxy and Glen Rose Formations. Although the Paluxy is a major aquifer in other areas, its occurrence at the site is at its southern margin of deposition where its thickness feathers to 7-feet. It occurs as a remnant isolated by erosion and not hydraulically connected to the water producing portion of the formation. The Glen Rose Formation comprises the remainder of the strata overlying the Travis Peak. It is a dense, finely crystalline, chalky, and variably argillaceous limestone.

H. Surface Water Protection

1. Discharge Considerations

Applicant will not discharge pollutants into waters of the State or United States. Applicant will not discharge dredged/fill material into the waters of the United States. Applicant will not have nonpoint source pollution of waters of the United States. The proposed site is well outside the limits of any surface waters. The proposed construction (excavated for area fill) will limit the possibility of discharge to surface waters.

2. Surface Drainage Controls

The entire landfill is divided into 5 drainage areas A, B, C, D, and E with each drainage area divided into subdrainage areas. These drainage systems were designed to carry the flows of each subdrainage area from a 12-foot ditch to an 18-foot drainage ditch within an existing system. For the modified site, the amounts of runoff generated off-site and directed to the site are minimized. Mainly the drainage inside the limits of the landfill remains with the same configuration, with no flow being transferred from one system to another. All design calculations are based on the 25-year, 24-hour rainfall intensity for Coryell County, Texas, using the Texas Department of Transportation rational method. Velocity reducer devices (detailed on Sheet 13 of 16, Sequence No. C12) were included in the design, where slopes exceeded the maximum standards. No significant changes in runoff are expected. All systems and drainage areas are shown on the drainage area map of the Drawings, Part 3. Design calculations and runoff calculations are included in Part 3.

3. Contaminate Surface Water Handling

All contaminated surface water will be stored temporarily with small earthen berms in areas protected by the liner. A detail of the small earthen berms and how they will be used in the temporary storage areas is shown on Sheet 13 of 16, Sequence No. C12. A vacuum truck shall be utilized to remove contaminated surface water. The contaminated surface water will not be spread by irrigation during compaction of wastes. It will not be spread on top of the completed landfill. Provision has been made with the Bell County W.C.I.D. No. 1 Wastewater Treatment Plant for acceptance of the contaminated surface water and leachate removed from the landfill, pending analytical results to be done once to be accomplished when contaminated water and/or leachate is produced. A letter of from the W.C.I.D. No. 1 indicating conditional acceptance is included in Appendix II.

4. Protection from the 100-Year Flood

The Federal Emergency Management Agency, Flood Insurance Rate Map¹, indicates the project site to lie outside the 100-year flood. The nearest encroachment of the 100-year flood is within a drainage pathway, tributary to Clear Creek, approximately 1,000-feet east of the northeast corner of the project site.

I. Protection of Endangered Species

There are no endangered or threatened species in the vicinity of the proposed landfill. The basis for this conclusion is found within the report entitled "Ecological Baseline Report, Fort Hood, Texas," dated July 25, 1979, prepared for Fort Hood by Espey Huston and Associates. This report is incorporated within the Environmental Impact Statement for Fort Hood.

The United States Fish and Wildlife Service was requested to provide an opinion on the impact of the operation of the proposed sanitary landfill site on endangered or threatened species. Their letter states that there are no threatened or endangered species existing at the proposed site, and that it is their opinion that the proposed project will not impact any of these species.

J. Control of Methane

The site conditions and operation of the landfill should not create problems regarding methane generation and migration. It is anticipated that the methane generated at the site from the waste will flow vertically through the methane recovery and removal system and disperse into the atmosphere. Methane generated below the liner by natural means, may collect under the dome like structure of the liner. Operational procedures for the control of methane are expected to be required during the post-closure period. A methane gas recovery and venting system has been designed to remove gases from below the FML cap after final closure. A contingency methane venting system is included in Part 3, if methane becomes a problem.

Methane Gas Monitoring Plan

The entire landfill shall be subject to air-monitoring requirements. The U.S. Army, Fort Hood, included this comprehensive air-monitoring plan designed to limit methane gas to less than the lower-explosive limits at the landfill boundary, and to 25 percent of the lower-explosive limits in the control building.

¹ Flood Insurance Rate Map, Community Panel Number 480768 0306B, Federal Emergency Management Agency, September 30, 1981.

1. The type and frequency of monitoring proposed was based on the following factors:
 - a. soil conditions - clay with low permeability;
 - b. hydrogeologic conditions surrounding the facility - low water table with slow movement of water between sections;
 - c. hydraulic conditions surrounding the facility - sheet flow from units into drainage ditches which empty into natural drainage systems; and
 - d. the location of facility structures and property boundaries with accountability for prevailing winds - one small portable control building at the entrance to landfill and one portable building at the south fence line on Turkey Run Road, and a Maintenance building on the east border with surrounding range land owned by the U.S. Government. The nearest housing area is 2,400-feet southeast of the landfill. The prevailing wind is from the south (26.6%) and south-southeast (12.1%).²
2. Initial air monitoring of the landfill was done April 1, 1994. Four methane probe wells were drilled to a depth of 7-feet at the locations given in Part 3 and shown on the drawings. Installed in each well was a 1/2-inch PVC screen pipe to total depth with a screw cap on bottom and top, with 1/4-inch hose barb protruding from top cap for sampling purposes. Each well was gravel packed from bottom of hole to 3.5-feet to bring gravel just above perforated screen height. Then 3-inches of native soil was placed in hole for an isolation layer between gravel and bentonite. Finally, a 9-inch bentonite isolation layer was installed and hydrated. Native soil was placed to surface level and a protection box was placed around the top of the 1/2-inch sampler.
3. After the installation of the wells, each probe was tested for methane gas. The methane gas content was found to be zero (0) in each probe with a barometric pressure of 30.01.
4. Subsequent methane gas monitoring wells shall be drilled to the closest permanent low seasonal water table, or the depth of refuse within 1,000-feet of the probe. Upon completion of the wells, Fort Hood shall supply the TNRCC with the survey of the well's locations; natural ground elevations and top of riser at each well; depth of each well with screen length; and boring logs and well reports.

² Gray Army Airfield

5. Air monitoring will consist of taking readings every quarter from a hand-held instrument at the existing probe locations around the perimeter of the landfill and in the existing buildings. In addition, all buildings within 1,000-foot perimeter of the facility shall be equipped with permanent, continuous monitors. Fort Hood shall contract the operation of the air monitoring of the facility including the requirements for data collection. The air-monitoring instrument shall be calibrated according to the directions accompanying the instrument. Testing of the instrument with a sample that contains a known organic vapor compound, will be performed prior to any sampling to ensure it is operating properly. Parameters to be analyzed for are methane, oxygen, and carbon dioxide. Readings will initially be taken from locations upwind to obtain a baseline of background air quality. Operators shall then proceed to obtain readings from the downwind side of the landfill. Comparisons of upwind readings with downwind readings shall be done to determine any differences and if there is a potentially harmful or hazardous condition existing. The spacing of the readings shall be every 150-feet, unless specified otherwise by the Executive Director. The operator shall record name, date, time, and weather conditions obtained from Fort Hood 's Robert Gray Army Airfield (wind direction, wind velocity, temperature, sunny/cloudy/partly cloudy, and barometric pressure) as a heading for tabulating the readings. This table shall be established to record the monitoring locations and accompanying instrument readings. A Methane Gas Monitoring Well Sampling Form will also be used to perform the gas monitoring (See Part 3). If needed, to verify problem areas, each location shall then be spotted on a map with the instrument readings positioned next to locations. A migration pattern may be obtained and drawn on the map with concentrations decreasing away from the source. More readings may be necessary to determine lateral extent of readings. Other information will be supplied as requested by the Executive Director.

6. After final closure of Area 1, methane gas production and settlement within the unit is expected to begin. As part of the routine maintenance and inspection schedule, Area 1, and subsequently Area 2 through 6 shall be monitored monthly for any cracks or cavities that would allow gases to escape. The area(s) shall be monitored before a spark-producing device or machine enters the unit for purposes of performing a task involving the use of these spark-producing devices or machines. A permit procedure shall be administered to prevent inadvertent entry and activity, and/or operation of these spark producing devices or machines into those areas of passive gas venting. A list of illegal practices and ignition sources shall be developed that will include, as a minimum, the following: flares; smoking or flames; firearms; heliarc welding; or torch cutting equipment. This list will be posted on a sign at the entrance, along with the requirements for permits in passive gas venting areas and amended as needed to inform all personnel entering the landfill.

7. If methane gas levels exceeding the limits specified in this section are detected, Fort Hood shall:

- a) immediately take all necessary steps to ensure protection of human health and notify the Executive Director;
- b) within seven days of detection, submit a report to the Executive Director that provides the methane gas levels detected and a description of the steps taken to protect human health and the environment; and
- c) within 30 days of detection, submit a remediation plan for the methane gas releases to the Executive Director. The plan shall describe the nature and extent of the problems and the proposed remedy and shall include an implementation schedule. The plan shall be implemented within 60 days of detection.

The U.S. Army shall notify the Executive Director when strong odors occur at facility boundaries. Records of inspections and surveys shall be maintained at the facility. Odors shall be controlled by the best means practicable.

K. Soils Data

1. Geology Report

a. Physiography

Fort Hood Military Reservation is in southern Coryell County and a portion of adjoining Bell County, in central Texas. This region lies within the Lampasas Cut Plain District of the Great Plains physiographic province.³ The Lampasas Cut Plain is a broad, elevated, dissected limestone plateau forming the western border of the outcrop of the Fredericksburg Group. Dissection of this plateau has produced interstream ridges and outliers of limestone supported by the Edwards, Comanche Peak, and Walnut Formations. Flat-lying remnants of the Edwards limestone form a few extensive uplands on the main divides. Local relief can reach as high as 400-feet. The Balcones fault zone borders the district on its eastern edge and roughly conforms to the boundary between the Great Plains and Gulf Coastal Plain physiographic provinces.

Drainage leaves the reservation primarily via Cowhouse Creek and Owl Creek, both tributary to the Leon River.

b. General Geology

The general geology of the Fort Hood reservation has been developed from published data; primarily the Geologic Atlas of Texas, Waco Sheet,⁴ and Stratigraphy of the Fredericksburg Division, South Central Texas,⁵ both published by the Bureau of Economic Geology. The generalized geologic map, Part 3, shows the subject area underlain by three Lower Cretaceous age formations. From oldest to youngest they are the Glen Rose, Paluxy, and Walnut Formations. The Glen Rose is composed primarily of limestone and marl with some shale and sandy shale. The limestone is often dense, finely crystalline, fossiliferous and gray to tan in color, with marl and chalky limestone common. The Glen Rose has an estimated thickness of 400-feet. The project site is situated at the southern edge of the Paluxy outcrop where it has feathered to a thickness of approximately 7-feet.⁶ The Paluxy was deposited in a marginal continental to very near-shore environment. The formation consists of fine to medium grained, gray-brown quartz sand and sandstone interbedded with gray carbonaceous clay shale. The Walnut formation unconformably overlies the Paluxy formation. Sediments of the Walnut are marine in origin, consisting primarily of gray to black calcareous clays, thin bedded limestones, thick chalky nodular limestone and shell aggregates. The formation is extremely fossiliferous. The Walnut in this area is well developed. Five distinct members have been mapped in Bell and southern Coryell Counties.⁷ Only the lower three members are represented at the project site. From oldest to youngest they are the Bull Creek, Bee Cave, and Cedar Park Members (see Geology Map and Geologic Atlas of Texas, Waco Sheet, Part 3).

Bedrock is mantled with a thin veneer of residual and colluvial overburden. Overburden of any appreciable thickness is confined to the major stream valleys.

⁴ Geologic Atlas of Texas, Waco Sheet, Bureau of Economic Geology, 1970.

⁵ Stratigraphy of the Fredericksburg Division, South Central Texas, Bureau of Economic Geology, 1964.

⁶ Depositional Systems in the Paluxy Formation, Bureau of Economic Geology, 1977.

⁷ Bureau of Economic Geology, loc. cit.

c. Structure

The Lampasas Cut Plain district is a dissected limestone plateau dominated by a southeast dipping monocline. Regional dip in this area is generally east-southeast at approximately 40-feet per mile, decreasing to the northwest to about 15-feet per mile. Abruptly terminating the monocline approximately 26 miles east of the site is the Balcones fault zone. This is a major structural feature where the strata have been downthrown several hundred feet. No faults have been mapped in the immediate area.

d. Groundwater

Fort Hood receives its principal water supply from Lake Belton on the Leon River. Groundwater wells in the North Fort Hood area have been abandoned due to the water being saline. The nearest known groundwater wells are in Gatesville approximately 20 miles north of the landfill. The primary aquifer system used for groundwater is comprised of the Hensell and Hosston Members of the Travis Peak Formation.⁸ The Travis Peak Formation is comprised of the basal, Lower Cretaceous sands and has an estimated thickness of 200-feet in the study area. The Travis Peak directly underlies the Glen Rose Formation and occurs at a depth in excess of 400-feet at the site. There are no known wells developed in the Travis Peak in the vicinity of the study area.

2. Site Conditions

a. General

Subsurface conditions at the subject 154-acre site have been investigated with 35 continuously sampled core borings (See Erase Drawings, Part 3; Appendix IV for locations). The specific site is situated upon a relatively flat surface ridge supported by the three lower members of the Walnut Formation. Individual members were not identified at the site. Continuous core borings to a depth of 50-feet and spaced on approximately 500-foot centers defined the stratigraphy of the site in detail. See Soil Conservation Service Soil Map and Soil Aerial Photo.

North Fort Hood Groundwater Study, U.S. Army Corps of Engineers, Fort Worth District, April 1983.

b. Overburden

Overburden at the site consists of a thin mantle of residual soil developed on primary material. Material within this soil mantle is a dark brown, stiff, high plasticity clay with large amounts of limestone gravel interspersed throughout. Overburden does not totally mantle bedrock at the site revealing outcrops of primary material along the steeper margins of the site. Laboratory test data classify the material as a clayey gravel or gravelly clay. Average thickness of this stratum is 1.7-feet.

c. Primary

Primary material at the site includes in descending order: Walnut, Paluxy, and Glen Rose Formations.

Aerially, the majority of the site is supported by the Walnut Formation (see Geologic Map, Part 3). For purposes of correlation at the site, the Walnut has been divided into two units based upon predominant lithology. The upper unit of the Walnut, occurring at the higher elevations, consists mainly of clay shales that have been highly weathered to the consistency of silty, low plasticity clays. These clays are calcareous, fossiliferous, and contain thin beds of a moderately hard to hard, well-cemented, fossiliferous tan limestone. The unit reaches a maximum thickness of 34-feet in the southern portion of the site and has been completely removed by erosion to the north.

The lower unit of the Walnut is predominantly limestone with interbedded thin seams of clay shale. The limestone is hard, moderately weathered to unweathered, fossiliferous, argillaceous to sandy in zones, and light-brown to gray in color. This unit averages 23-feet in thickness but has been completely eroded out in the northeast portion of the site.

The Paluxy Formation at the site is comprised of predominantly dark gray, soft clay shale, interbedded with seams and laminations of fine-grained, cross-bedded sand and weakly cemented sandstones. Frequently, the clay shales, sands, and sandstones are carbonaceous. Infrequently, the Paluxy is comprised of predominantly fine-grained sand and weakly cemented sandstone interbedded with clay shale seams. The lithology of the Formation varies laterally, typical of fluctuating near shore depositional environment. The average thickness across the proposed site is 7-feet.

The deepest lithologic unit sampled by core borings is the Glen Rose Formation. The Glen Rose is a moderately hard to hard, argillaceous, fossiliferous, limestone. It is poorly bedded, with some thin clay shale seams and stringers. The limestone is unweathered, non-jointed and ranges in color from greenish-gray to gray. The Formation was not fully penetrated by exploratory borings.

d. Weathering

Primary materials are weathered to an average depth between 20 to 25-feet. The upper unit of the Walnut exhibits weathering throughout. The clay shales of this unit have been reduced to a silty, low plasticity clay. It is generally yellow-brown in color, containing some thin beds of light brown, moderately-jointed limestone. This unit is easily excavated with standard equipment.

The lower unit of the Walnut is weathered to a lesser degree. Physical weathering is predominant in this unit and occurs in the form of iron oxide stained joints. The matrix of the altered rock has been stained to a yellow-brown or reddish-brown. This material is excavated with difficulty. Below the weathered interval, the limestone is gray, non-jointed, hard, and contains stringers of black clay shale. The practical depth of excavation by standard methods is determined by the depth of weathering, as is evidenced by previous excavation operations at the southern end of the site.

e. Groundwater

(1) General

Groundwater data have been obtained from temporarily cased exploratory borings serving as observation wells. Each boring was bailed dry of all drilling fluids, cased with 2-inch slotted PVC pipe and sealed at the surface with a bentonite plug to prevent entry of surface water into the boring. Table 3 is a history of water level readings at the subject site.

The proposed sanitary landfill is situated upon materials which are capable of various types and rates of groundwater movement. Weathered clay shales allow some movement along cracks and fractures, as well as along the margins of interbedded limestone seams. The loss of structure which occurs as the shale is reduced to a clay consistency and may restrict groundwater flow and velocity. Groundwater movement in the limestones is restricted, dependent primarily upon secondary permeability features such as joints, fractures, and occasional vugs. Sand units, through interstices between individual grains, tend to be better

conductors of groundwater. Restrictions to this movement develop through cementation. Due to different modes of groundwater movement in different material types, in situ permeability is addressed below with respect to each material type.

(2) In Situ Permeability

Thirty-two field permeability tests were conducted at various depths utilizing twelve cluster piezometers. Each cluster piezometer consists of three piezometer installed in the same boring at various depths and hydraulically isolated from each other. The tests were conducted in accordance with procedures outlined in Appendix IV. The intervals hydraulically isolated correspond to the lithologic units described above. Results of the field permeability tests are summarized as follows:

<u>Lithologic Unit</u>	<u>Median Coefficient of Permeability</u>
Upper Walnut Shale	1.4×10^{-5}
Lower Walnut Limestone	4.0×10^{-5}
Paluxy Formation	7.5×10^{-5}
Glen Rose Formation	3.0×10^{-5}

(3) Hydraulic Gradient

Because water in the saturated zone is contained under water table conditions, the potentiometric surface conforms somewhat to local relief. The highest potentiometric levels are found along the topographically high drainage divide bisecting the proposed site and oriented along a north-south axis. Flow lines constructed perpendicular to the potentiometric lines describe the path at different points, followed by water moving through the saturated section. The direction of groundwater flow is laterally away from the central localized drainage divide. To the west of the divide, the flow direction is northwest towards the drainage path of House Creek. To the east, the flow direction is northeast towards the drainage path of Clear Creek. The average hydraulic gradient has been calculated to be 320-feet per mile, although it varies considerably over the site.

L. Active Geological Faults

Major structural features in the vicinity of the project area associated with faulting include the Balcones Fault Zone and the Llano Uplift. The Balcones Fault Zone is a complex system of normal faults trending northeast-southwest through central Texas. The fault zone occurs approximately 25 miles east of the site where the Balcones escarpment, an abrupt topographic rise, marks the west margin of

the fault zone. The fault zone is approximately 5 miles wide at this point.⁹ The Llano uplift raises Precambrian and Paleozoic rocks to the surface in nine counties southwest of the project site. Generally, the central core of Precambrian rocks are surrounded by a margin of Paleozoic rocks. Faults in the Paleozoic strata are comprised of northeast trending block faulting, which occurred during Middle Pennsylvanian time.¹⁰ The outer margin of the faulting occurs approximately 33 miles southwest of the site at its nearest point.¹¹ Faulting associated with both structures is considered inactive and no faults closer to the site have been mapped. See Tectonic Map of Texas.

M. Closure Plan

1. The landfill site is separated into three phrases.
2. Based on anticipated solid waste generation rates for Fort Hood, this landfill will be closed in 2053.
3. Landfill areas will be filled vertically up to 3 to 4-feet below the finished contours shown on Permit Modification Design Drawings, Part 3. Eighteen inches of intermediate final cover will be placed over the final lift of solid waste and daily cover material at the end of each day, so that these elevations are reached. All areas that have received waste but will be inactive for longer than 180 days shall provide intermediate cover. This operation will continue laterally in each expansion of the Area until the topographically high elevation on the line has been reached, then the operation will proceed to the next successive topographically low elevation in that Area, to begin operations and deposit waste until the high point is reached again. Final cover will then be applied to the entire Area.
 - a. Final Cover of the entire area will be applied no later than 30 days after final grades are reached in each area of the landfill, and no later than 30 days after the final date of known receipt of solid waste in the area. This deadline may be extended by the Executive Director to a maximum of 60 days.
 - b. Standing water will be removed.

⁹

Geologic Atlas of Texas, Llano Sheet, Bureau of Economic Geology, 1981.

¹⁰

The Ouachita System, Flawn, P.T., Goldstein, A., King, P.B., and Weaver, C.E., 1961.

¹¹

Bureau of Economic Geology, loc. cit.

- c. The runoff diversion system will be maintained until final cover is installed and will be modified to prevent overflow of the landfill to adjoining areas.
- d. Insect and rodent inspection will be documented before installation of final cover, and extermination measures used, if required, as a result of inspection.
- e. A gas collection and venting system will be installed in each Area between the waste with intermediate final cover and the final cover.

(1) The gas collection and venting system will consist of perforated high-density polyethylene pipe in a 1-foot square trench surrounded by graded gravel. The gravel shall be sized to ensure it is larger than the one-half inch (1/2") perforations in the collection pipe.

(2) The perforated pipe will be connected to a solid, rise pipe and gas vent to complete a passive collection and removal system. Whatever synthetic membrane is used for the liner and cap system, the same type of material or comparable material should be used in the piping systems throughout the landfill (i.e., polyethylene pipe with polyethylene FML and polyethylene FML Cap).

(3) A geotextile or graded soil filter will surround the graded gravel.

- f. A final cover will be placed on top of the intermediate final cover which consists of 18 inches of compacted clay with a permeability of less than 1×10^{-5} cm/sec or be at least as impermeable as the intermediate final cover.
- g. A very low-density textured polyethylene, a minimum of 60-mils thickness, or comparable synthetic membrane, will be installed on top of the recompact clay layer. The polyethylene cap layer will be entrenched with the polyethylene line and will be welded together to provide a seal between the cap and liner.
- h. A minimum of 6-inches of topsoil shall be installed on top of the polyethylene. The depth of the topsoil should be increased if it is determined that 6-inches of topsoil is not sufficient to support sufficient grassroot growth.
- i. After closure inspection and approval, the U.S. Army will plant ground cover to prevent erosion and to return the landfill to a more natural appearance. Prior to the establishment of

grass, an application system (i.e., a mulching) designed to control erosion and to promote soil retention and grassroot growth, may be used on the final soil layer.

j. The finished surfaces of the select fill will be maintained and inspected for vermin until turfing is accomplished. All finished surfaces of select fill, embankments, ditches, and areas disturbed by construction operations will receive turfing. Earthwork and turfing will be in accordance with the drawings in Part 3.

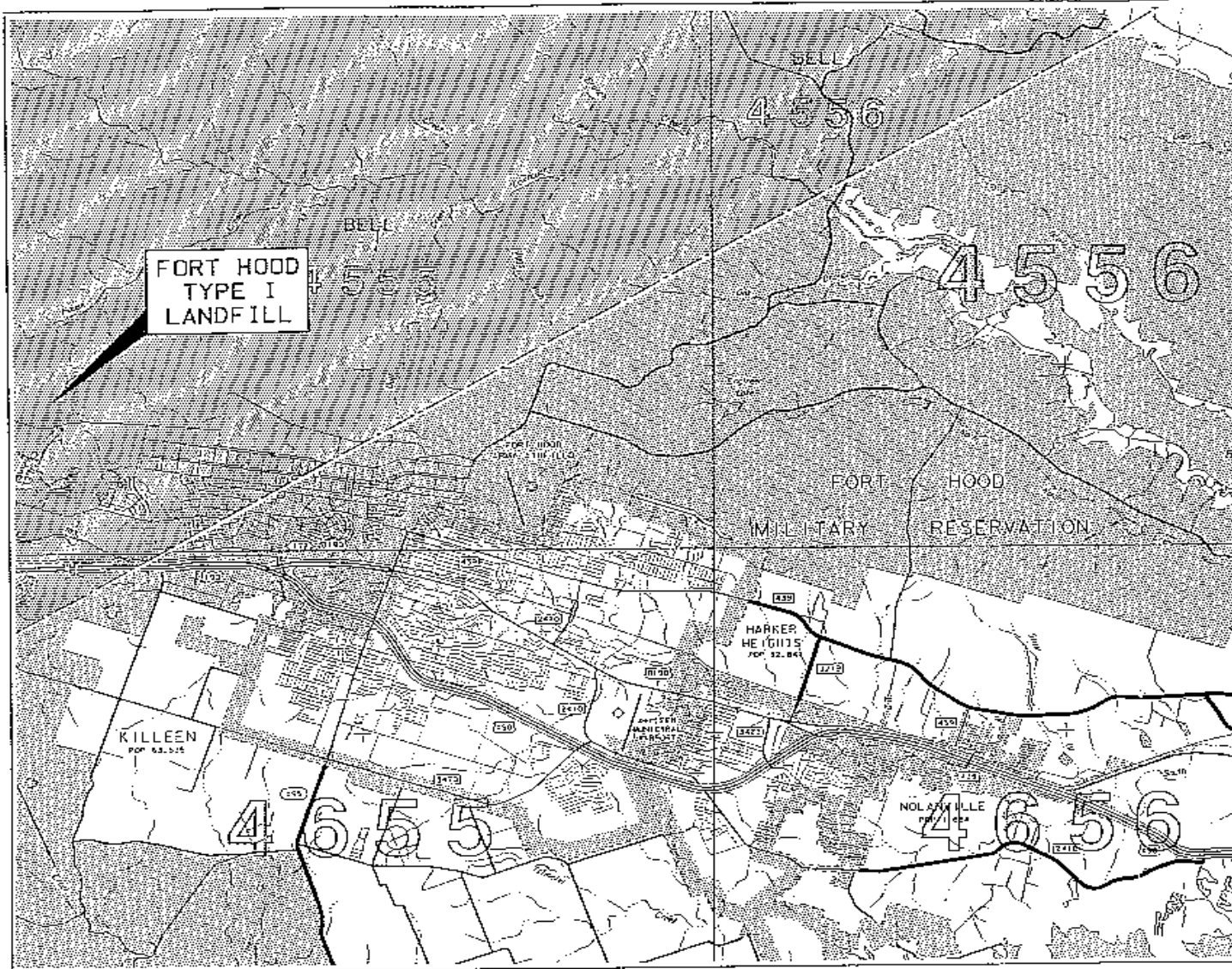
k. The U.S. Army will update the Coryell County mortgage and conveyance records, by entering the specific location of the landfill and specifying that the property was used for the disposal of solid waste.

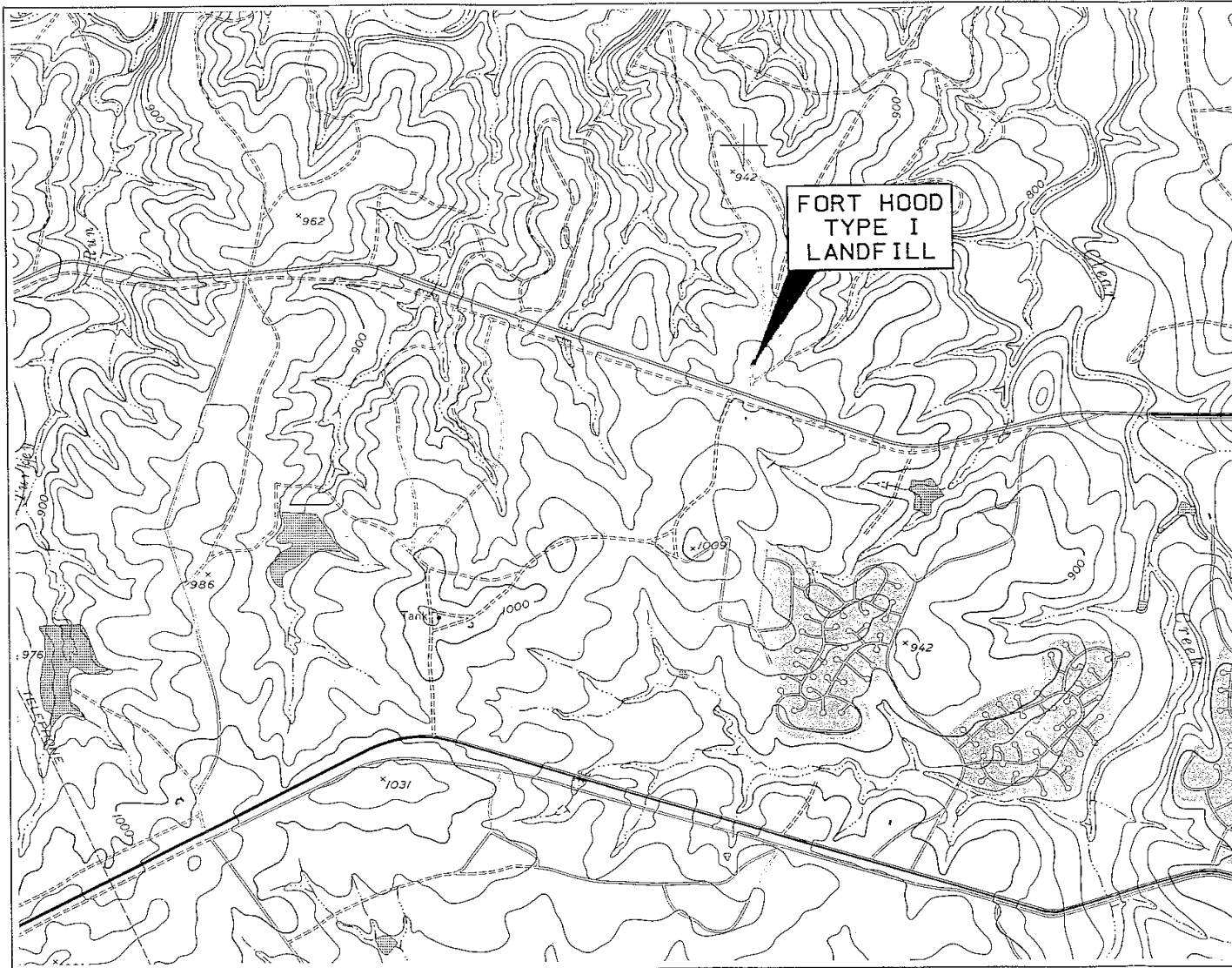
N. Post-Closure Care Maintenance Plan

1. After closure, a scheduled maintenance program will be established to include monthly inspections and repairs, as necessary, to maintain vigorous vegetation growth, refurbish surface soils damaged through erosion, shrinkage, subsidence, or animal burrowing, repair damaged fencing, and to upkeep monitoring wells. No development, human habitation, or recreational activities will occur on this site.
2. To assure public safety, the integrity of boundary fencing and access points will be maintained. Security forces and maintenance crews will monitor the site monthly. The Directorate of Public Works shall be responsible for overseeing the Post-Closure Care Maintenance. The address is U.S. Army, Headquarters III Corps and Fort Hood, Fort Hood, Texas 76544. The telephone number is (254) 287-2113. No habitable structures or public use areas (with the exception of roadways) will be constructed within 2,000-feet of the landfill. There is no plans in place to use the closed areas of the landfill.
3. Areas will remain in post-closure care for 30 years after closure of the landfill, unless it can be shown that the landfill is no longer a threat to human health and the environment and approved by the Executive Director for a shorter length of time.
 - a. Leachate collection and removal will continue until leachate is no longer generated or until the U.S. Army can demonstrate that the leachate no longer poses a threat to human health or the environment.
 - b. Gas collection and removal will continue after closure.
 - c. Air monitoring will continue until the U.S. Army can demonstrate that the gas no longer poses a threat to human health or the environment and is approved by the Executive Director.
 - d. According to the Groundwater Monitoring Plan, groundwater monitoring will continue after closure.
4. Ownership of the site will remain with the U.S. Government after closure.
5. Prior to abandonment of any monitoring well, the applicant will notify the Executive Director in writing of his intent. No well shall be abandoned until approval is obtained.

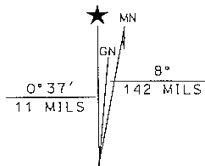
PART 3 - ATTACHMENTS

Fort Hood Military Installation Map

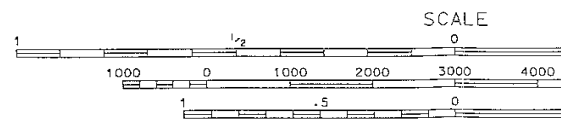




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UTM GRID AND 1978 MAGNETIC NORTH

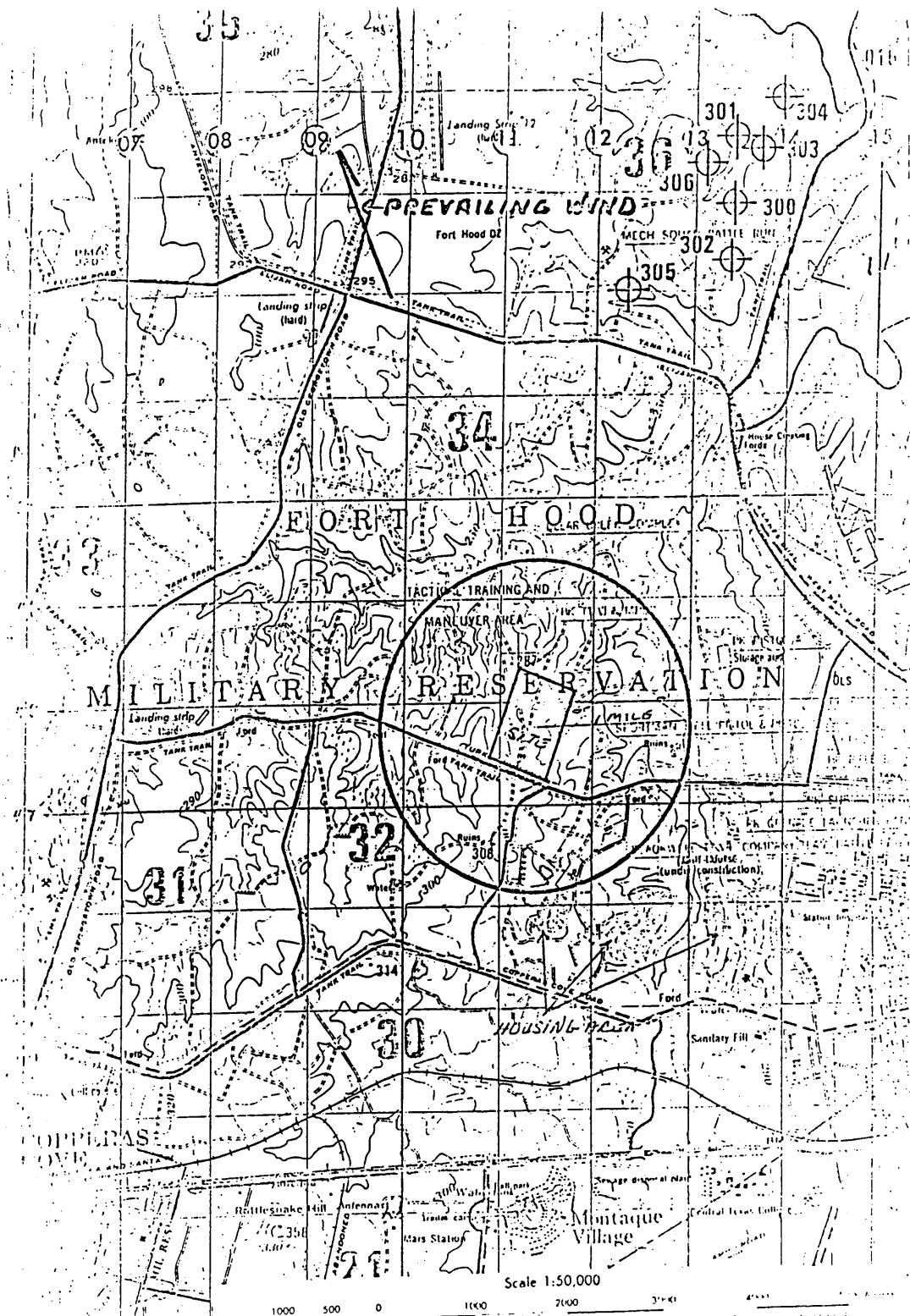


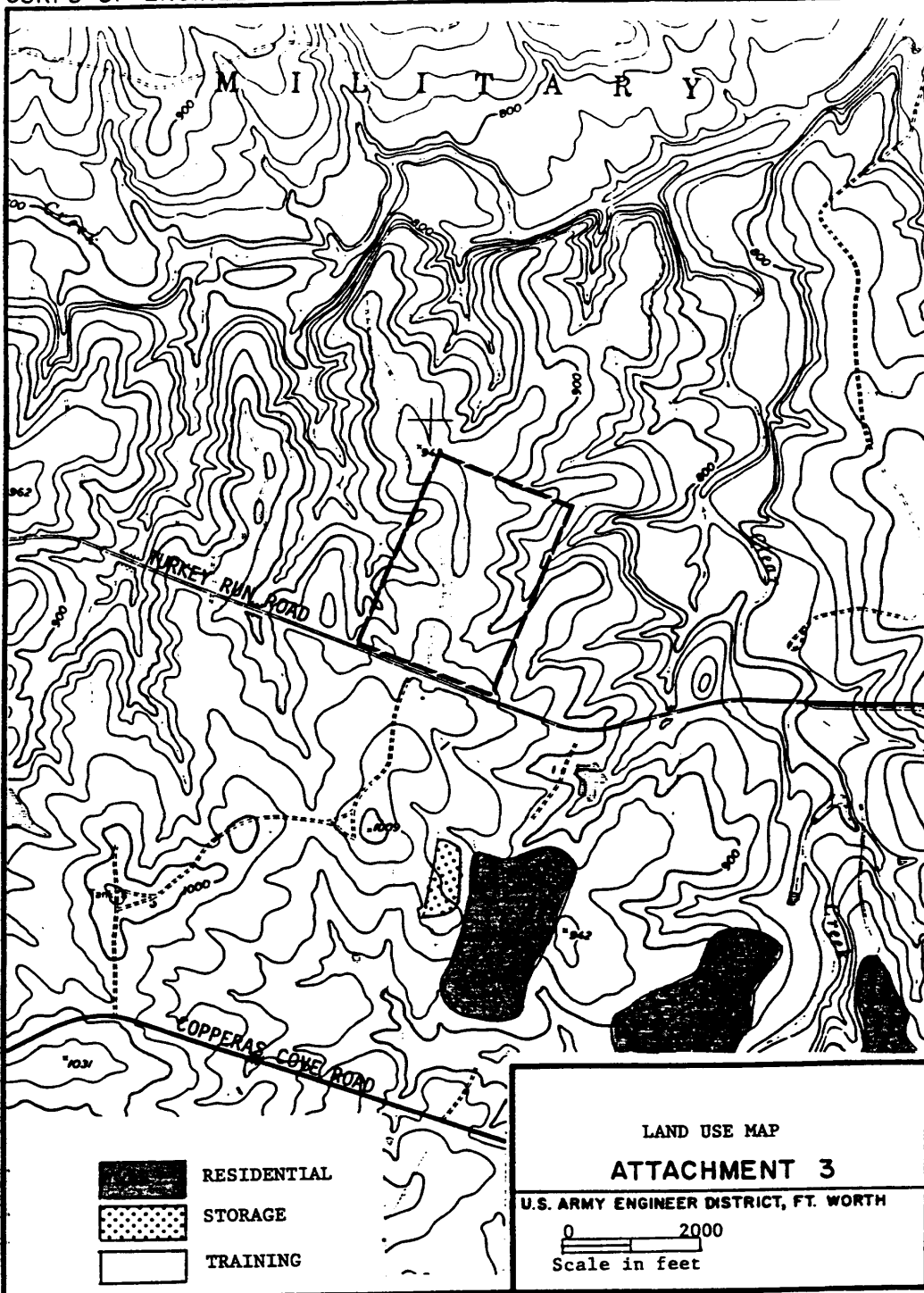
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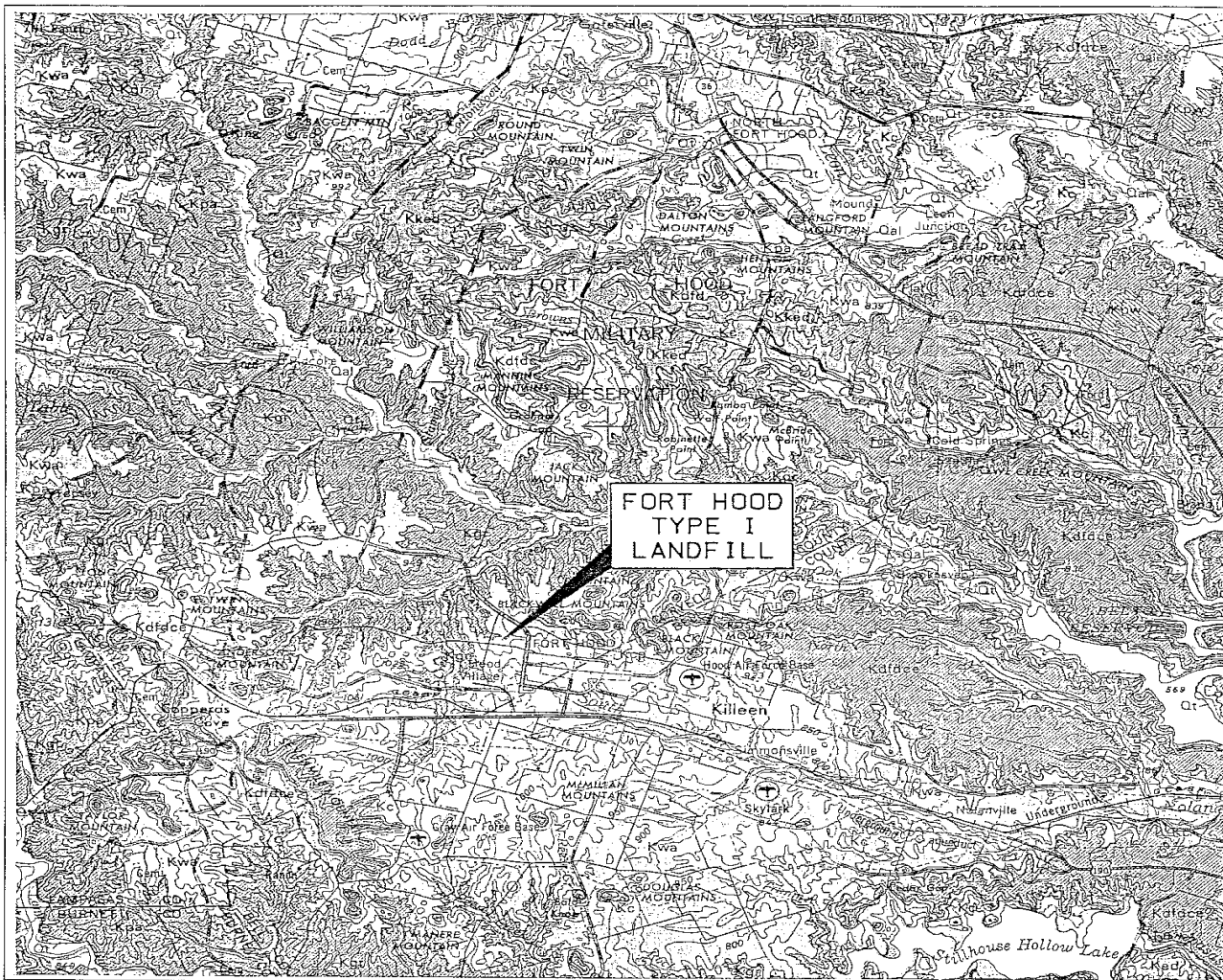
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INFORMATION PROVIDED WITH MAP

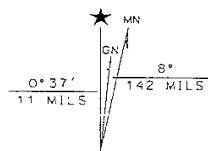
1. Map is Fort Hood Military Installation Map, Series V782S, Edition 5-DMA, and Scale 1:50,000, prepared in 1980. County map does not show Fort Hood area.
2. Site Boundaries are shown.
3. Prevailing Wind Direction: South-Southeast
4. Drainage Features are shown.
5. There are no water wells within 500 feet.
6. Housing Areas are shown.
7. No cemeteries within one mile.
8. No recreational areas within one mile of site.
9. Land Use. Maneuver Training Area for armor vehicles.
10. Political Areas. As shown.
11. Nearest Main Public Roadway. U.S. 190, as shown.
12. No public airports within four miles.
13. There are no easements on or adjacent to site.
14. Latitude 37° 09.30'. Longitude 97° 50.00'.







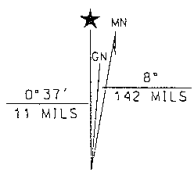
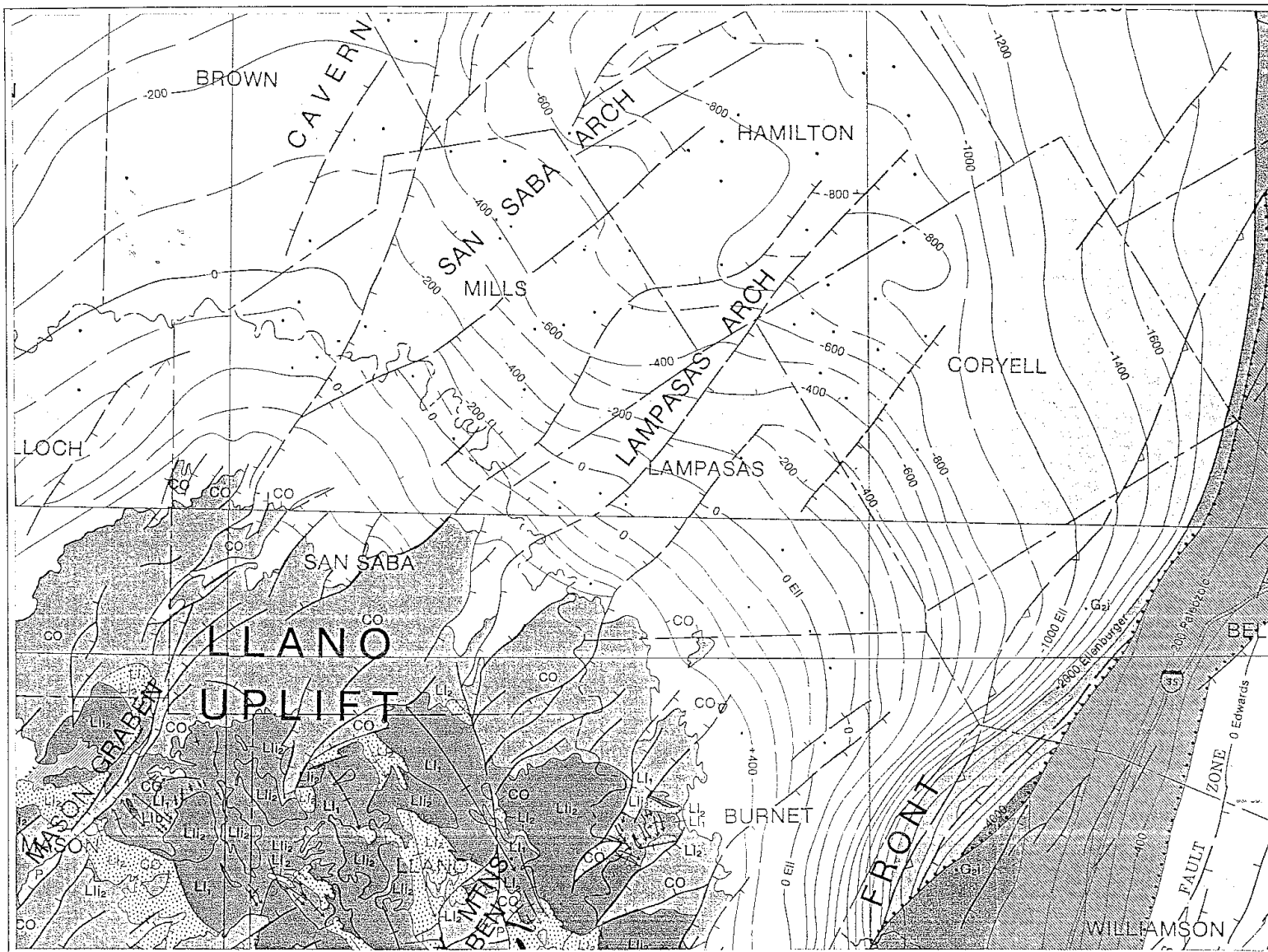
Mapped, edited, and published by the Army Map Service
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UTM GRID AND 1978 MAGNETIC NORTH

SCALE

CONTOUR IN
NATIONAL GEODETIC
GEOLOGIC ATLAS OF
LLOYD WILLIAM STEPHENSON
REPRINTED



SCALE 1:750,000

CONTOUR INTERVAL 200 METERS
WITH SUPPLEMENTAL 100-METER CONTOURS
DATUM IS MEAN SEA LEVEL

TECTONIC MAP OF TEXAS - 1990

PART 3 - ATTACHMENTS

Methane Gas Monitoring Wells and Results

Available at DPW.

PART 3 - ATTACHMENTS

Methane Gas Monitoring
Wells Sampling Form

METHANE GAS MONITORING WELL SAMPLING FORM

[illegible]

PART 3 - ATTACHMENTS
HELP Runs

Available at DPW

PART 3
Groundwater Monitoring Well
Sampling Form

PART 3
Flow Calculations

Available at DPW

SITE OPERATING PLAN

(A) Personnel and Equipment

(1) The applicant will require a minimum number of personnel on site during operating hours. The minimum will be one person at the gate, one working face monitor, one scale operator, and one landfill manager (with Class "A" certification). The Contracted operator will have other personnel on site as needed by his operation. The landfill manager will designate an alternate in his absence. One of these individuals will be on site at all times the landfill is open for operation.

(2) The operator will use equipment designed for landfill operations. Equipment will be able to excavate, spread, and compact cover material. The selection of equipment will be based on the ability of each machine to do the required job.

(3) The number of pieces of equipment on site will be adequate to dispose of the volume of waste received, and construct the soil liners and covers. The minimum equipment will be the following:

(a) Excavation and Hauling of soil material - Two 21 CY self-propelled scrapers.

(b) Compaction - Landfill Compactor (steel wheel) with assistance of Bulldozer.

(c) Spreading of waste and cover material - Landfill Compactor and/or Bulldozer.

(d) Construction of soil liner - Sheeps foot dirt compactor; Hyster steel wheel roller; Rhone Disc plow pulled by a bulldozer; motor graders; and water truck.

(4) Installation of the HDPE liner or other liner system will be contracted to the manufacturer.

(B) Security

(1) Access to the site will be limited to one route. Entrance to the site will be provided through lockable gates with the remainder of the active portion of the site enclosed by a minimum 7-foot high chain link fence or an equivalent metal fence. The gates will be kept locked to prevent the unsupervised dumping of solid waste, except during those times when operating personnel are present. A permanent control building exists near the landfill entrance (southeast corner) adjacent to the truck scale.

(2) The landfill will be open for receipt of waste during Fort Hood 's normal working hours (7:30 a.m. to 5:00 p.m., every day). Due to heavier usage and better weather, during certain months of the year, the landfill will remain open later to allow the operator to take advantage of all available daylight for disposal and/or construction purposes, as follows:

Jan, Feb, Nov, Dec: 5:00 p.m.

Mar, Apr, Sep, Oct: 6:30 p.m.

May, Jun, Jul, Aug: 8:00 p.m.

(3) Loads of waste entering the landfill will be screened at the gate, at the control building, and at the working face of the landfill area. This will involve a visual inspection of vehicles entering the site. Loads found to contain material that is not authorized for disposal in the landfill will be refused access and directed to the appropriate disposal or recycling facility.

(4) Traffic speed will be limited to 15 miles per hour inside the site to provide for safety and minimize dust problems.

(C) Sequence of Site Development

(1) The site development of the sanitary landfill will be divided into three phases, with each phase consisting of two landfill areas. Each phase will consist of approximately one-third of the entire permitted facility. Each phase will be operated until it reaches its design capacity. The landfill phases are identified as Phase I, Areas 1 and 2; Phase II, Areas 3 and 4; and Phase III, Areas 5 and 6. The design of the 154 acre site is in accordance with Subtitle D standards. The landfill was initially permitted in March 1991 with wastes deposited in the southeast corner of Phase I, into Area 1, Cell 1. Since that time, development has proceeded in a westerly direction into Cells 2, 3 and 4. Development is continuing in a westerly direction in Area 1, which contains an additional 8.46 acres for Cells 5 and 6. Area 2 contains the following three sections: a section comprising approximately the east two-thirds of Area 2, called the Special Waste Trench, for disposal of Class I industrial non-hazardous waste; a section of aerial fill above the Special Waste Trench for disposal of municipal solid waste; and a section comprising the west one-third of Area 2 for disposal of municipal solid waste. The municipal waste and Class I waste in Area 2 will be physically separated by a four-foot layer of compacted clay-rich soil and will never be co-mingled. The Special Waste Trench will be constructed and utilized simultaneously as Area 1 is being developed. After Area 1 is completed, development will continue north into the west section of Area 2, which contains 6.61 acres for municipal solid waste. When the Special Waste Trench reaches the designed grades (natural ground) and is closed, the section above it shall be utilized for disposal of municipal solid waste. After Area 2 is completed, development will continue north into Area 3, containing 20.21 acres. Area 4, with 20.18 acres will follow in succession, as will Area 5, with 20.19 acres; and Area 6, with 20.19 acres. This yields a total acreage of 105.92 acres. Solid waste placement rate averages 652 cubic yards per day at gross in-place density rates of approximately 700 to 900 pounds per cubic yard. The total landfill life of this facility is estimated to be 17 years. Final Closure and Post-Closure Care of the entire site will be according to the Subtitle D standards.

(2) Before opening the landfill to accept wastes, interior access roads, all weather roads, fences, diversion terraces and the liner system necessary for the first cell were complete. All groundwater monitoring wells were constructed prior to opening the landfill as initially permitted. Five additional groundwater monitoring wells were developed in September 1992 after the initial permit submittal. Well No. 9 was drilled in the middle of the proposed Area 6 of Phase 3. This well will be plugged and abandoned according to the Groundwater Monitoring Well Plugging Procedures in Section IV. F. of the Site Development Plan, or as directed by the Executive Director, to avoid providing a pathway from the refuse to the groundwater. Another monitoring well, No. 14, will be drilled in the southeast corner of the

landfill to replace No. 9 but primarily to provide an additional up-gradient well. All existing and proposed groundwater monitoring wells are indicated on the design drawings in Part 3. The Groundwater Sampling and Analysis Plan in Section IV. E. of the Site Development Plan describes the sampling and analysis of these wells.

(3) Disposal activities began in the southeast corner of the area designated as Phase One - Area 1 - Cell One on the Sectorized Fill Layout, initially permitted, and subsequently the Existing Contours drawing No. C3 in Part 3. The dimensions of each cell construction will be determined by several factors, including the landfill design, potential construction problems, etc. The actual working face will be limited to an area 100-feet by 100-feet.

(4) All solid waste accepted for disposal will be spread and compacted in approximately 2-foot layers along an inclined working face. At the end of each working day, the compacted wastes will be completely covered to protect personnel and the environment from exposures, prevent fires and scavenging, control disease vectors and windblown waste, and minimize odors, as follows: all exposed waste in the Special Waste Trench will be covered with a minimum of 6-inches of well-compacted soil, and all exposed waste in the active municipal cell will be covered with the ADC in accordance with the attached ADC Operating Plan. Personal protective equipment will be used by the ADC appliers if required in the Material Safety Data Sheets. An ADC quarterly status report shall be submitted to the Executive Director, describing the effectiveness of the alternative material, any problems that may have occurred, and corrective actions required as a result of such problems. If no problems occur within four consecutive quarters of use, status reports will no longer be submitted.

(D) Control of Dumping

(1) The packer trucks and other waste collection vehicles utilizing the site will normally be directed to deposit their load in a designated area at the base of the 100-foot by 100-foot working face, which will require backing into the area to unload.

(2) The landfill operator may, at his discretion, direct the refuse collection vehicles to deposit their loads at the top of the working face, when it is more convenient and effective for waste disposal operations.

(3) The landfill operator will maintain a guard at the gate, a monitor at the scale, and a monitor at the working face to inspect each load that is being deposited in the landfill. In addition to this inspection,

additional visual Quality Assurance/Quality Control (QA/QC) inspections will be performed on every fifth truck for each applicable waste stream. QA/QC results will be recorded and referenced by waste ticket number and maintained at the site operator's office for inspection. Only items authorized for disposal under the operating permit will be accepted at this facility subject to this plan. The primary evaluation criteria for the wastes intended for landfill disposal is to exclude hazardous (and PCB) waste, as defined in 40 CFR 261. All wastes subject to this plan will be evaluated by analytical testing and/or process knowledge. Any items found in the waste that are not authorized for disposal in the landfill will be removed by the transporter and directed to the appropriate destinations, as follows: Large items, such as white goods (refrigerators, washing machines, dryers, etc.) will be directed to the Fort Hood Recycle Center located on Fort Hood. Recyclable materials will be directed to the Fort Hood Recycle Center. Hazardous materials will be redirected to the hazardous material storage facility located on Fort Hood or other permitted hazardous waste treatment, storage and disposal facilities, as needed, on a case by case basis. The requirements for handling special wastes authorized for disposal in this facility are detailed in Section F. Other items not covered in this section will be handled on a case by case basis.

(4) Salvaging shall not be allowed to interfere with disposal of solid waste or create a public health nuisance. Salvaged materials shall be considered as potential recycled materials. Salvaged items shall be removed from the site often enough to prevent the items from becoming a nuisance, to prevent discharge of any pollutants from the area and to prevent an excessive accumulation of the material at the site. Special wastes received at the landfill shall not be salvaged. Pesticide, fungicide, rodenticide, and herbicide containers shall not be salvaged. Scavenging shall not be allowed on the landfill.

(E) Fire Prevention and Control

(1) The use of daily cover will contain any fire in a layer under construction from spreading to other layers. All equipment operators will keep a fire extinguisher on their machine at all times for small fires. A stockpile of earth shall be maintained within 2,500-feet of the working face for fire suppression also.

(2) If larger fires occur, the waste in the burning area will be spread out so that water or cover material can be applied to extinguish the fire. In such cases, the area of the fire will be excavated, and the liner examined and replace, if necessary.

(3) Should a collection truck arrive carrying burning waste, it will be routed as quickly as possible away from the working face to a safe area where its load will be dumped and the fire extinguished.

(4) The Fort Hood Fire Department is aware of the type of synthetic materials being used and has firefighting vehicles located 4 and 7 road miles from the landfill.

(F) Control of Special Wastes

- (1) A working face monitor will be on duty at all times to inspect each load of solid waste deposited.
- (2) The operator of the facility will maintain all required records at the site (or approved alternate location) for a minimum of 5 years. All transporter certificates as required by Texas Natural Resource Conservation Commission (TNRCC) will be collected and maintained. All records and certificates will be available for inspection by TNRCC representatives and authorized agents of local governments having jurisdiction over the site.
- (3) No hazardous or PCB waste will be accepted or disposed of in the permitted landfill.
- (4) The following special wastes will be accepted and handled as follows:
 - (a) Dead animals: Animal carcasses will be disposed of with other municipal solid waste and covered with 3 feet of other municipal solid waste or 2-feet of soil immediately upon disposal
 - (b) Water supply treatment plant sludge containing a minimum of 10 percent solids, which are not hauled in vacuum trucks, has been or is to be treated or processed and the treated/processed material has been analyzed by EPA Method 9095 (Paint Filter Test) and determined not to contain "free liquids": Incorporate with municipal solid waste.
 - (c) Stabilized sludge from domestic wastewater treatment plants, containing a minimum of 10 percent solids, which are not hazardous and are not hauled in vacuum trucks, and has been or is to be treated or processed and the treated/processed material has been analyzed and determined by EPA Method 9095 (Paint Filter Test) not to contain "free liquids": Quantities shall be limited to that which can be adequately handled at the site without creating odor problems, and shall be placed on the working face along with other solid waste and covered at the end of the day.
 - (d) Unstabilized sludge, which has passed through a primary and secondary digester, from domestic wastewater treatment plants, containing a minimum of 10 percent solids, which are non-hazardous and are not hauled in vacuum trucks, and has been or is to be treated or processed and the treated/processed material has been analyzed and determined by EPA Method 9095 (Paint Filter Test) not to contain "free liquids": It will be placed on the working face with other municipal solid waste and covered at the end of the day.

(e) Regulated Asbestos-Containing Material (RACM):

(1) Regulated Asbestos-Containing Material (RACM), as defined in 40 CFR 61, as amended, includes friable asbestos; Category I nonfriable ACM that has become friable; Category I nonfriable ACM that will be or has been subjected to sanding, grinding, cutting, or abrasion; and Category II nonfriable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of demolition or renovation operations.

(2) Delivery of RACM to the site will be coordinated with the site operator so that it can be properly handled and covered.

(3) RACM will only be accepted at the site wetted, and in tightly sealed and unruptured containers or bags, or will be wrapped with six-mil polyethylene, which have been approved by the TNRCC.

(4) The bags or containers holding the RACM shall be placed in areas designated for receipt of RACM with other municipal solid waste below natural grade level. Where it is not possible or practical to place the RACM below natural grade level, provisions shall be made to ensure that the RACM will not be subject to future exposure through erosion or weathering of the intermediate and/or final cover. RACM placed above natural grade shall be located such that when the unit is closed, the RACM is at least 20-feet from any final side slope of the unit and at least 10-feet below the final surface of the unit. The site operator will record the disposal location of the RACM, in accordance with the grid coordinates set forth on Design Drawings C5 - C11 and C13 - C15, in Part 3, which were surveyed by a Registered Professional Land Surveyor (RPLS).

(5) The bags or containers holding the RACM will be manually unloaded and carefully placed in their final disposal location. The bags or containers will be carefully covered immediately with a minimum of 12-inches of soil (clean earthen material) or 3-feet of non-ACM solid waste to prevent them from being ruptured.

(6) A contingency plan for handling accidental spills (i.e., ruptured bags or containers) has been developed and is referred to as the "Installation Spill Contingency Plan (ISCP)." The ISCP

is in the Environmental Preservation section contained in III Corps and FH REG 420-2, Environment and Natural Resources, May 1, 1993. In addition, there is a "Spill Prevention Control and Countermeasure Plan" in FH REG 200-10, February 15, 1989. These Fort Hood regulations contain procedures to be followed in incidents or immediate threat for a spill event along with a notification process to be followed. Copies of these publications will be made a part of the landfill operating record, with training of involved personnel to be completed. Copies of these documents are included in Appendix III.

(f) Nonfriable asbestos: Nonfriable asbestos waste will be placed with other municipal solid waste on the active working face and covered. Under no circumstances will any material containing nonfriable asbestos be placed on any surface or roadway which is subject to vehicular traffic or disposed of in any other manner which could crumble the material into a friable state.

(g) Empty containers which have been used for pesticides, herbicides, fungicides, or rodenticides: The containers will be triple rinsed prior to receipt at the site, rendered unusable prior to or upon receipt at the site, placed with other municipal solid waste, and covered the same day they are received. If triple-rinsing is not feasible or practical (i.e., paper bags, etc.), the containers will be placed on the working face and immediately covered with a minimum of 3-feet of municipal solid waste. Salvaging and/or scavenging of the containers will not be allowed.

(h) Municipal hazardous waste from Conditionally Exempt Small Quantity Generators (CESQG): A generator qualifies as a CESQG if it generates less than 100 kilograms (220 lbs) of municipal hazardous waste in a calendar month and accumulates less than 100 kilograms (220 lbs) of municipal hazardous waste on site. This waste will be placed with other municipal solid waste.

(i) Grease trap waste generated by food preparation facilities at Fort Hood:

(1) Grease trap waste transporters will use vacuum trucks for the collection of the grease trap waste. The liquid portion of grease trap waste will be decanted and disposed of in the sanitary sewer system.

(2) The remaining residue, which comprises approximately 50 percent of the total volume removed, will be transported to an approved facility to be solidified. The liquid residue will be mixed with dry soil or other absorbent material until solidified. Once the residue is solidified and has been tested in accordance with EPA Method 9095 (Paint Filter Test), and determined not to

contain "free liquids", it will be deposited on the working face along with other municipal solid waste and covered. Quantities will be limited to that which can be adequately disposed of at the site.

(j) Reserved.

(k) The Special Waste Trench within the permitted landfill is for the disposal of demolition debris which consists of family housing and other buildings and structures that contain lead-based paint in non-hazardous concentrations. Renovation/demolition debris with lead-based paint and asbestos-containing material with lead-based paint applied to it will be placed in the Special Waste Trench. The Special Waste Trench, (design included in Section C.3 of the Site Development Plan and in the design drawings in Part 3) will occupy the east two-thirds of Area 2. This special waste will be classified as stated below. Approval by the TNRCC before acceptance at the landfill is not required, as stated in the TNRCC letter dated November 9, 1994 (See Part 1). Only waste from within the Department of Army facility at Fort Hood will be accepted. Fort Hood 's Directorate of Public Works or a designated agent or department shall determine which shipments will be accepted at the site based on the results of the sampling and analysis. The sampling protocol used to characterize the demolition debris is the U. S. Army Environmental Hygiene Agency protocol entitled, "Sampling Protocol, Building Demolition Debris and Buildings Painted With Lead-Based Paint." A copy of the protocol which describes the sampling and testing procedures is included at the end of this revised Site Operating Plan. This protocol was reviewed by the TNRCC's Industrial and Hazardous Waste Division. (See TNRCC letter dated December 16, 1994 in Part 1). Debris waste shipments shall be analyzed according to the EPA Method 1311, Toxicity Characteristic Leaching Procedure (TCLP) as the method of extraction; and EPA Method 6010A (Inductively Coupled Plasma-Atomic Emission Spectroscopy) or EPA Method 7421 (Atomic Absorption, Furnace Technique) as the methods of analysis for lead. Only debris waste shipments found to contain levels of lead to be in the range of special waste or non-hazardous waste will be accepted. Specifically, the waste must be in the range between 5 and 1.5 milligrams/Liter [(mg/L), approximately equal to parts per million], noninclusive (i.e., <5.0 mg/L and >1.5 mg/L) to be included in the Special Waste Trench. All waste below and including 1.5 mg/L (1.5 mg/L) may be placed with the municipal solid waste. Hazardous waste (5.0 mg/L) shall not be accepted at this facility. Sampling, testing and documentation procedures shall be placed in the operating record to demonstrate compliance with these conditions. Other special wastes that would be allowed in the Special Waste Trench would be regulated and non-regulated asbestos-containing material that has lead-based paint applied on it. This would be contingent on the TCLP analysis results as described

above. Municipal solid waste will be placed above the Special Waste Trench after it has been covered with 4-feet of compacted clay and a leachate collection and removal system constructed. The location and size of the Special Waste trench is reflected on Sheet Nos. C13, C14, and C15 in Part 3. Each load that is delivered to the landfill for disposal in the Special Waste Trench will receive a visual Quality Assurance/Quality Control (QA/QC) inspection to verify contents and nature of the waste. In addition to visual QA/QC inspections, additional QA/QC inspections will be performed on every fifth truck per applicable waste stream. QA/QC results will be recorded and referenced by waste ticket number and maintained at the site operator's office for inspection. All information and documents pertaining to the special waste disposed of in the Special Waste Trench will be maintained by the operator for a period of no less than three years.

(G) Control of Wind-blown Material

- (1) The Alternative Daily Cover (ADC) will be installed over the compacted municipal waste in accordance with the attached ADC Operating Plan. A minimum of 6-inches of soil cover will be spread and compacted over the compacted Class I waste each day that Class I waste is received to minimize the amount of blowing litter.
- (2) A minimum 7-foot-high chain link fence or an equivalent metal fence around the perimeter of the active Phase of the landfill will be maintained at all times, to prevent wind-blown material from leaving the site.
- (3) Litter along fence lines will be policed daily.
- (4) Army troops patrol the routes to the landfill for 2 miles to recover items that have blown or fallen from vehicles hauling waste to the site.

(H) Vector Control

- (1) A properly designed and operated sanitary landfill will minimize the attraction of insects, rodents and birds to solid waste, but it is virtually impossible to eliminate these disease vectors.
- (2) The daily application of the ADC over the compacted municipal waste and 6-inches of soil cover over the Class I waste will prevent many problems associated with these pests.

(3) Use of pesticides for insect and rodent control will be used routinely at the landfill.

(4) These controls will be carried out under the supervision of the Fort Hood Staff Entomologist and with appropriate safety precautions to inhabitants of the area.

(I) Dewatering of Excavation

(1) When groundwater is encountered during the construction of the liner system, soil with a weight equivalent to 1-foot of compacted clay liner (125 pounds per cubic foot) for every 2-feet of static groundwater head will be used as ballast to support the liner.

(2) Any liner support required will be in addition to the liner system proposed for this site.

(3) Groundwater surface elevations are indicated on Potentiometric Surface and Monitoring Well Sections (See Appendix IV).

(J) Dust and Mud Control

To aid in dust and mud control, all-weather roads will be constructed from the access roads to the disposal areas as shown on the landfill drawings in Part 3. Any additional dust control required will be achieved with water trucks.

(K) Compaction, Intermediate Cover and Final Cover

(1) All solid waste will be compacted and covered at least daily.

(2) Compaction will consist of spreading and compacting the solid waste along the inclined working face to a depth of approximately 2-feet. Daily cover will consist of installing the Alternative Daily Cover (ADC) over the compacted municipal waste in accordance with the attached ADC Operating Plan and spreading and compacting a minimum of 6-inches of soil over the compacted Class I waste.

(3) After placement of the final lift of solid waste, intermediate cover (SC, CL, or CH soil) will be applied, as shown on the closure and cross-section drawings in Part 3, until the area or cell is completed. Depending on the area, final cover will consist of either compacted soil and a flexible membrane or a GCL and flexible membrane installed on top of the intermediate cover, as depicted in

the closure drawings in Part 3. The remainder of the final cover system will be installed as shown in the closure drawings in Part 3 and will be suitable for planting vegetation which will be placed over the area, compacted and sloped to prevent ponding of surface water. Final contours are shown on the closure drawings in Part 3.

(4) A cover application log will be maintained at the site and made available for inspection by TNRCC representatives. Each entry in the log will be certified by the signature of the landfill manager that the work was accomplished as stated in the log.

(a) For daily cover, this log will specify the date the daily cover was accomplished, how it was accomplished, and the last area covered.

(b) For intermediate cover, this log shall specify the area covered, the date cover was applied, how it was accomplished, and the thickness of the applied cover.

(c) For final cover, this log shall specify the area covered, the date the cover was applied, how it was accomplished and the thickness of each layer of the final cover system, including the methane recovery and venting system.

(L) Soil and Liner Quality Control

(1) The landfill manager will fully familiarize himself with and assure continuous compliance with the approved Soil and Liner Quality Control Plan (SLQCP), combined with the TNRCC's Liner Construction and Testing Handbook, included within this Part 4, as Section III. In the event there is an overlap of the requirements between the SLQCP and the TNRCC's Liner Construction and Testing Handbook, then the Liner Construction and Testing Handbook shall take precedence over the SLQCP. The operator shall notify the regional TNRCC office of their intent to begin construction of any liner prior to commencement of construction. All pertinent records and reports will be maintained at the site.

(2) The operator will submit, to the TNRCC, a Soil and Liner Evaluation Report (SLER) and a Flexible Membrane Liner Evaluation Report (FMLER) for each area prepared for the disposal of solid waste, in accordance with the construction methods and test procedures set forth in the SLQCP and the TNRCC's Liner Construction and Testing Handbook. The SLER and FMLER will be prepared in accordance with the SLQCP and the TNRCC's Liner Construction and Testing Handbook and submitted to the TNRCC at least 14 days prior to depositing any solid waste in that disposal area.

(M) Monitoring for Leachate and Methane

(1) Water quality monitoring wells are located around the perimeter of the landfill, except for monitoring well No. 9 which is located within the disposal area of Phase 3, Area 6. This well will be abandoned and another well (No. 14) will be drilled at the southeast corner of the landfill as an additional background groundwater monitoring well (See Phase 1 Closure, Drawing No. C5). These monitoring wells will provide information regarding the potential of a ground water contamination problem resulting from the operation of the sanitary landfill. A Groundwater Sampling and Analysis Plan is provided in the Site Development Plan, Attachment 11.

(2) The operator will take necessary precaution to prevent disturbances to or destruction of the monitoring wells. The operator will be responsible for any damage to the monitoring wells, including their destruction. Any groundwater monitoring wells that must be removed or abandoned will be done in accordance to the section entitled Plugging and Abandoning Wells in the Groundwater Protection Section (See Part 2).

(3) Any methane generated in the landfill above the HDPE liner or other approved liner system is expected to rise vertically, and vent into the atmosphere. A methane recovery and venting system will be constructed within the final cover system, but below the flexible membrane used for final cover. Following the installation of the final cover system, surface and subsurface monitoring for methane gas will be accomplished in accordance with the methane monitoring plan included in Part 2. In addition, a Methane Gas Monitoring Well Sampling Form will be used to accomplish the methane monitoring required.

(4) In addition to the permitted landfill site, all usable enclosed spaces (homes, offices, etc.) within 1000-feet of the landfill will be monitored continuously for methane gas concentration. If methane is detected, the contingency plan for methane gas will be implemented, which will require further testing.

(5) Landfill gas monitoring will consist of taking readings every quarter from a hand-held instrument at the existing probe or well locations around the perimeter of the landfill and in the existing buildings. In addition, all buildings within a 1,000-feet perimeter of the facility shall be equipped with permanent, continuous monitors. The gas monitoring instrument shall be calibrated according to the directions accompanying the instrument. Testing of the instrument with a sample that contains a known organic vapor compound, will be performed prior to any sampling to ensure it is operating properly.

Parameters to be analyzed for are methane, oxygen, and carbon dioxide. Readings will initially be taken from locations upwind to obtain a baseline of background air quality. Operators shall then proceed to obtain readings from the downwind side of the landfill. Comparisons of upwind readings with downwind readings shall be performed to determine any differences and if potentially harmful or hazardous condition exist. The spacing of the readings shall be every 150-feet, unless specified otherwise by the Executive Director. The operator shall record name, date, time and weather conditions obtained from Robert Gray Army Airfield (wind direction, wind velocity, temperature, sunny/cloudy/partly cloudy and barometric pressure) as a heading for tabulating the readings. This table shall be established to record the monitoring locations and accompanying instrument readings. A Methane Gas Monitoring Well Sampling Form will also be used to record the readings from the gas monitoring (See Part 3). If needed, to verify problem areas, each location shall then be identified on a map with the instrument readings positioned next to locations. A migration pattern may be obtained and drawn on the map with concentrations decreasing away from the source. More readings may be necessary to determine lateral extent of migration. Other information will be supplied as requested by the Executive Director.

(6) After final closure of Area 1, methane gas production and settlement within the unit is expected. As part of the routine maintenance and inspection schedule, Area 1, and subsequently Area 2 through 6 shall be monitored monthly for any cracks or cavities that would allow gases to escape. As part of the routine methane gas monitoring, gas monitoring probes and wells shall be inspected quarterly, including all piping or tubing, the concrete pads, and the bollards. Any defects discovered shall be corrected immediately. The area(s) shall be monitored before a spark-producing device or machine enters the unit for purposes of performing a task involving the use of these spark-producing devices or machines. A permit procedure shall be administered to prevent inadvertent entry and activity, and/or operation of these spark producing devices or machines into those areas of passive gas venting. A list of illegal practices and ignition sources shall be developed that will include, as a minimum, the following: flares; smoking or flames; firearms; heliarc welding; or torch cutting equipment. This list will be posted on a sign at the entrance, along with the requirements for permits in passive gas venting areas and amended as needed to inform all personnel entering the landfill.

(7) In the event that methane gas concentrations in excess of 25% LEL are detected in a gas monitoring probe or well, the monitoring frequency for that probe or well and two adjacent wells will be increased to monthly. In addition to increasing the sampling frequency for these wells, monthly monitoring of gas monitoring well pressure and temperature, barometric pressure and daily rainfall data will also be performed. If groundwater is found in any landfill gas monitoring probe or well,

measurements will be taken from the top static head. The increased sampling frequency and sampling parameters will continue until three consecutive months of methane gas concentrations of less than 25% LEL are obtained. At that time, monitoring will return to quarterly measurement as described above. Records of inspections and surveys shall be maintained at the facility.

(8) If methane gas levels exceeding the limits specified in this section are detected, Fort Hood shall:

- a) immediately take all necessary steps to ensure protection of human health and notify the Executive Director;
- b) within 7 days of detection, submit a report to the Executive Director that provides the methane gas levels detected and a description of the steps taken to protect human health and the environment; and
- c) within 30 days of detection, submit a remediation plan for the methane gas releases to the Executive Director. The plan shall describe the nature and extent of the problems and the proposed remedy and shall include an implementation schedule. The plan shall be implemented within 60 days of detection.

(9) The U.S. Army shall notify the Executive Director when strong odors occur at facility boundaries. Odors shall be controlled by the best means practicable.

(10) All future methane gas monitoring wells shall be drilled to the closest permanent low seasonal water table, or the depth of refuse within 1,000-feet of the well. Upon completion of the wells, Fort Hood shall supply the TNRCC with the survey of the wells locations; natural ground elevations and top of riser at each well; depth of each well with screen length; and boring logs and well reports of wells.

(11) All contaminated surface water will be contained temporarily with small earthen berms in areas protected by the liner. A detail of the small earthen berms and how they will be used in the temporary working areas is shown on Sheet 13 of 16, Sequence No. C12. A termination berm shall also be constructed at the limit of the lined area. This will include a sacrificial FML which will be removed when tie-in with the next cell occurs. This is to prevent contaminated water from being released, and to prevent run-on from the unlined areas. A vacuum truck shall be utilized to remove contaminated surface water. The contaminated surface water will not be recycled into the waste. Provision has been made with the Bell County W.C.I.D. No. 1 Wastewater Treatment Plant for discharge into the sanitary

sewer system of the contaminated surface water and leachate removed from the landfill. The letter from W.C.I.D. No. 1 indicating conditional approval of such discharges is included in Appendix II.

(12) Leachate Level Measurements and Monitoring Procedures

(a) At each of the sumps in the leachate collection system, extraction riser pipes will provide a means to extract leachate and to monitor leachate levels. Methods will be employed to maintain the leachate depth on the liner at less than 12-inches. In conjunction with manual pumping, leachate levels will be measured relative to the top of the sump elevation and may be measured by physical means such as a dipstick method or by water level indicator to determine the depth to leachate from the top of the riser. Using survey (elevation) documentation of the top of each sump and each riser, the depth of leachate may be calculated. A long measuring pole will be used to measure the level of the leachate on the liner. It shall be lowered down the 18-inch leachate extraction pipe to obtain a reading on the pole. The amount of leachate will then be calculated using the angle of the side slope and the distance on the pole.

(b) Initially, leachate levels will be monitored on a monthly basis for each leachate collection sump. If the depth of leachate during this monthly monitoring is determined to be greater than 12-inches, then the monitoring frequency will be increased to biweekly. If the depth of leachate during the bi-weekly monitoring is determined to be greater than 12-inches, the monitoring frequency will be increased to weekly monitoring. If the depth of leachate during weekly monitoring is determined to be greater than 12-inches, monitoring will be increased to semi-weekly basis. If the leachate depths for two consecutive monitoring events are less than 9-inches, the frequency of monitoring will be relaxed to the preceding monitoring frequency. For example, if two consecutive weekly monitoring readings indicate depths less than 9-inches, the monitoring frequency will be reduced to bi-weekly monitoring.

(c) At a minimum, monthly leachate depth monitoring will be performed for a period of six months. If after six months, the leachate depth has been consistently less than 9-inches, the frequency of leachate depth monitoring may be relaxed to every other month. If after six months, the leachate depth has not been consistently less than 9-inches, monthly leachate depth monitoring will continue.

(d) Leachate will be removed to maintain less than 12-inches of leachate head on the liner system. Whenever leachate is pumped, the system will be pumped to remove as much leachate as

possible, not simply to reduce the leachate depth to less than 12-inches. Leachate pumping will occur whenever leachate monitoring indicates a leachate depth of 12 inches or greater. Volumes of leachate removed from the leachate collection system along with the measured depths in the leachate collection system shall be recorded and inserted in the operating record.

(N) Signs and Site Rules

(1) The hours of operation will be posted on a sign at the landfill. The landfill will be open for receipt of waste during Fort Hood 's normal working hours (7:30 a.m. to 5:00 p.m., every day). Due to heavier usage and better weather, during certain months of the year, the landfill will remain open later to allow the operator to take advantage of all available daylight for disposal and/or construction purposes, as follows:

Jan, Feb, Nov, Dec: 5:00 p.m.

Mar, Apr, Sep, Oct: 6:30 p.m.

May, Jun, Jul, Aug: 8:00 p.m.

(2) This sign will list wastes which are not authorized for landfill disposal.

(3) Another sign will be posted at the landfill entrance listing illegal practices and ignition sources. The list of illegal practices or equipment shall include the following as a minimum: flares; smoking or flames; firearms; heliarc welding; or torch cutting equipment. The sign will also include the requirements for permits in passive gas venting areas and amended as needed to inform all personnel entering the landfill.

(4) All vehicles transporting waste will have the waste covered to prevent accidental loss of waste during transport. U. S. Army troops police the routes to the landfill to collect lost waste.

(O) Protection of On-Site Utilities and Easements

Utility lines servicing the landfill are a 6-inch water supply line and an 8-inch sanitary sewer line. The maintenance building is located at the southeast corner of the landfill and is depicted on the Site Map. Other equipment located north of the maintenance building includes a 10,000-gallon diesel above-ground storage tank with an electric dispensing pump; a 3,000-gallon above-ground gasoline storage tank with an electric dispensing pump; and a 500-gallon above-ground used oil storage tank. Each of

these tanks are within an earthen berm lined with 60-mil High Density Polyethylene (HDPE). A drawing depicting these three tanks is included in Part 3, for information purposes only. All used oil from the tank is removed and taken to the Fort Hood Recycle Center. An equipment wash area, comprised of a concrete pad with a sump drain connected to a sand/oil separator, is located east of the maintenance building. Fort Hood is required by U. S. Army regulation to have a contingency plan which would be implemented in the event of an accident, fire, explosion, spill or other emergency. This plan is entitled "Installation Spill Contingency Plan (ISCP)." It is the III Corps and Fort Hood REG 420-2, dated May 1, 1993. The "Spill Prevention Control and Countermeasure Plan (SPCCP)" provides a plan for prevention of oil, hazardous, and toxic substance spills and mitigation of the effect resulting from spills. This document is FH REG 200-10 and is dated February 15, 1989. Copies of these two documents are included as Appendix III and will be incorporated into operating procedures for landfill operations personnel.

(P) Wet Weather Operations

During wet weather, collection trucks will unload in areas designated by the operator.

(Q) Inspection and Maintenance

(1) Completed sections of the landfill will be graded to prevent ponding of surface water and promptly seeded with grasses native to the Fort Hood area. The selection of such grasses will be coordinated with the Fort Hood Agronomist.

(2) Completed sections of the landfill will be inspected and maintained by the operator. After closure, the post-closure maintenance program will be implemented, which includes monthly inspections and repairs, as necessary, to maintain vigorous vegetation growth, refurbish surface soils damaged through erosion, shrinkage, subsidence, or animal burrowing, repair damaged fencing and to maintain monitoring wells. No development (except road construction), human habitation, or recreational activities will occur on or within 2000 feet of the landfill. To assure public safety, the integrity of boundary fencing and access points will be maintained.

(3) Closed areas will remain in post-closure care for 30 years after final closure, unless it can be shown that the landfill is no longer a threat to human health and the environment and approved by the Executive Director for a shorter length of time.

(a) Leachate collection and removal will continue until leachate is no longer generated or until it can be demonstrated that the leachate no longer poses a threat to human health or the environment.

(b) Landfill gas collection and removal will continue after closure. Landfill gas monitoring will continue until it can demonstrate that the gas no longer poses a threat to human health or the environment.

(c) In accordance with the Groundwater Monitoring Plan, groundwater monitoring will continue after closure.

(4) Fort Hood will retain ownership of the site after closure.

(5) Prior to abandonment of any monitoring well, the Executive Director will be notified in writing. No well shall be abandoned until approval is obtained.

(R) Operator's Training

(1) Responses to environmental incidents requires careful preparation and prompt action to reduce the hazards. The health and safety of personnel and the general public must be protected. Training of the operators for the standard operation of the landfill is required for safe and orderly operations and to protect themselves and others from hazards. Refresher training will be required for the licensed landfill operators. Operators will also be trained in the proper personal protection and general safety required to work with municipal solid waste and the approved special wastes. Training in the proper wearing of the personal protective equipment and their limitations will be provided. Additional training will include planning for emergencies and the proper response in the event of a fire, explosion, spill or other emergency.

(2) Potential hazards must be recognized, and exposure risks evaluated whenever an incident occurs. A methodical preliminary site survey could yield enough information to benefit emergency response. These might include respirator and protective clothing selection or further definition of the hazard by qualitative assessment. Once the plan of action based on the initial survey is done, hazard evaluation is not over. No matter how passive the situation appears, the site is potentially very dynamic. To ensure a safe working environment, continuous or periodic monitoring of the hazards must be performed. Training in hazard recognition could prevent a potentially hazardous event.

(3) Fort Hood has a contingency plan which would be implemented in the event of an accident, fire, explosion, spill or other emergency. This plan is entitled "Installation Spill Contingency Plan (ISCP)." It is the III Corps and Fort Hood REG 420-2, dated May 1, 1993. The "Spill Prevention Control and Countermeasure Plan (SPCCP)" provides a plan for the prevention of oil, hazardous, and toxic substance spills and mitigation of the effect resulting from such spills. This document is Fort Hood REG 200-10 and is dated February 15, 1989. The ISCP and SPCCP will be incorporated into the landfill operating procedures. Personnel will be trained in the contents of these documents and the proper procedures to follow in the event of an emergency. Copies of these two documents are included as Appendix III.

(4) Personal Protective Equipment may be required in certain areas of the landfill. Air monitoring of personnel is recommended to determine what the exposure levels are for lead for each job in the Special Waste Trench. The proper respiratory protection would be determined from the results. A personal full-face air-purifying respirator with High Efficiency Particulate Air (HEPA) filters is required to minimize exposure to lead dust and asbestos fibers. In particular, an accidental breakage of the asbestos bags, and/or asbestos (transite) siding or panels in a non-friable state that has become friable (crushable by hand pressure) and that could have lead-based paint applied to it would require the use of a respirator of this type. Personnel should receive a physical examination and be properly fitted before wearing a respirator. The U. S. Occupational Safety and Health Administration (OSHA) has set limits based on Threshold Limit Values (TLVs), and other Federal and industry standards and are known as Permissible Exposure Limits (PELs). The PEL values are found in 29 CFR 1910.1000. PELs are legally enforceable standards and apply to all private industries and Federal agencies.

(5) A safety shower with eye wash is recommended for emergency situations. Other protection that may be required include personal protective clothing such as rubber gloves, tyvek suits, and rubber boots. Additional training should be implemented to instruct operators in the proper procedures to follow for decontamination in the event of an accident or spill to prevent furtherance of an emergency or prevent physical harm. The protective equipment and procedures to be implemented should be reviewed and revised as necessary.

PART 4, SECTION II
Quality Control Plan for
Grease Trap Waste Disposal

QUALITY CONTROL PLAN
FOR
GREASE TRAP WASTE DISPOSAL
AT
FORT HOOD SANITARY LANDFILL

1. Fort Hood contracts out the removal of grease trap waste from 47 locations throughout the installation. Each grease trap is cleaned and serviced once each month.
2. Approximately 15,000 gallons of grease trap is generated every month. This waste is composed of grease, putrescible material, sewer-clogging solids, liquids, and sediments from the bottom and side walls. Liquids shall be decanted and disposed of in the sanitary sewer system. The remaining residue, approximately 50% of the total volume removed, shall be transported in vacuum trucks to a contracted facility to be solidified before depositing in the Fort Hood Sanitary Landfill, where it shall be deposited upon the working face and covered with incoming refuse. The refuse shall be covered daily.
3. The transporter shall comply with sections 325.441-325.447 and 325.531 of the Municipal Solid Waste Management Regulations (MSWMR). The transporter shall be registered with the TNRCC; maintain a copy of the registration form, as annotated by the TNRCC with an assigned registration number, at their place of business and in every vehicle operated under the registration; and renew their registration with the TNRCC annually while their contract with Fort Hood is in effect.
4. The vacuum trucks used by the transporter shall be prominently marked to display the company's name and the department assigned registration number. The vacuum trucks shall be constructed, operated, and maintained to prevent loss of liquid or solid waste materials, prevent health nuisance and safety hazards to operating personnel and the public, and preclude odors and insect breeding.
5. The transporter shall initiate and maintain a record of each individual collection and deposit. Such record shall be in the form of a manifest or similar documentation approved by the TNRCC. The manifest shall include the name and registration number of the transporter; the location where the waste was generated and the date collected; the type and volume of waste collected and transported; the name of person(s) collecting, transporting, and depositing the waste; the date and location where the waste was deposited; the permit number, location, and operator of the facility where the waste was deposited; and the name and signature of the landfill operator acknowledging receipt of the waste and volume received.
6. The transporter shall provide a copy of each manifest to the generator and the landfill operator, while retaining a copy of each manifest for three years, making them available to the TNRCC upon request. The transporter shall submit to the TNRCC an annual summary of their activities up to December 31 of each year showing amounts and types of waste collected and the disposition of each such wastes. The annual report shall also include the total capacity of all vacuum trucks used during the reporting year and be submitted by March 1 of the year following the end of the reporting period. The report shall be prepared on a form approved by the TNRCC.
7. In the event of a spill of waste during the collection or transportation, the transporter shall take appropriate action to protect human health and the environment.

SLQCP OUTLINE

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III. CONSTRUCTION

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I. IV. QUALITY ASSURANCE/QUALITY CONTROL

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I. INTRODUCTION

A. GENERAL

This plan delineates the methods and procedures to be followed during liner system construction to ensure the protection of surface and groundwater quality beyond the boundary of the landfill. Also contained in the plan are requirements for Quality Assurance/Quality Control testing, to include guidance for Engineers/Geologists who prepare and sign the Soils and Liner Evaluation Report (SLER) and the Flexible Membrane Liner Evaluation Report (FMLER). This Soil and Liner Quality Control Plan will be used in conjunction with the Texas Natural Resource and Conservation Commission's (TNRCCs) Liner Construction and Testing Handbook. The two documents shall complement each other in that where one document does not address a concern the other does. The TNRCC shall have precedence in matters that are overlapping in nature, as it was published in accordance with 30 Texas Administrative Code, Chapter 330 the Municipal Solid Waste Regulations governing this site. The Liner Construction and Testing Handbook is located in Appendix IV.

B. DEFINITIONS

1. GEOTECHNICAL QUALITY CONTROL PROFESSIONAL (GQCP)

A professional engineer, registered in Texas, who possesses professional experience in geo-technical engineering and testing or a graduate geologist, who has a minimum of four years' experience in engineering geology and is experienced in geo-technical testing and its interpretations. For the purpose of this plan, the GQCP is the independent third-party engineer who is responsible for Quality Assurance/Quality Control and the preparation of the

SLER and FMLER.

2. QUALIFIED ENGINEERING TECHNICIAN

An engineering technician, who is NICET-certified in Geo-technical Engineering Technology at level 2 or higher or an engineering technician with a minimum of four years of directly related experience or a graduate engineer or geologist with one year of directly related experience. For the purpose of this plan, all qualified engineering technicians work at all times under the general supervision of the GQCP who signs the SLER and FMLER.

II. C. LINER SYSTEM DESCRIPTION

1. DESIGN

In situ soils at this facility do not meet Subtitle D groundwater protection requirements for an alternate liner. In lieu of a naturally occurring liner, a composite liner system shall be constructed. The constructed liner system shall consist of a two-foot compacted clay layer, a 60 mil Flexible Membrane Liner (FML), a geo-synthetic layer (a composite of geo-textile fabric and geo-net-type drainage material), a Leachate Collection System (LCS), a one-foot granular drainage layer, and a one-foot protective cover layer (see liner system detail). An alternate layer design, if approved by the Executive Director, would consist of a Geo-synthetic Clay Liner material (GCL), a 60 mil FML, a geo-synthetic layer, a LCS, a one foot granular drainage layer, and a one-foot protective cover layer. The GCL design will be submitted at a later date. For the purpose of this plan, the term "liner system" shall include the Leachate Collection System, with all associated piping, trenches, drainage media, sumps and other appurtenances.

2. DISCUSSION

This facility is designed to be an area fill operation. Lateral expansions are constructed in the form of cells, which are typically, 330 feet wide by 420 feet long. The width of expansions (cells), however, is not critical and may vary for a number of reasons, including rate of incoming waste, weather conditions, general geology, etc. The permanent side slopes of the excavation is designed on a 3 to 1 slope to ensure stability and constructability. For additional stability, a textured FML and a geo-textile fabric-drainage net composite is specified on the side slopes.

II. MATERIALS SUITABILITY

A. CLAY LAYER

1. Prior to clay layer construction, representative samples from each clay soil source to be used shall be tested in a geo-technical laboratory to ensure their suitability for lining purposes. The following set of tests shall be performed:

Test	Method	Suitability Criteria
Classification	ASTM D2487	Classified as CL or CH
Sieve Analysis (+40, +200, -200)	ASTM D422	30 + % (-200 sieve) 100% (1-1/2" screen)
Atterberg Limits Plasticity Index Liquid Limit	ASTM D4318	No less than 15 No less than 30
Coefficient of Permeability	Appendix VII of Corps Of Engineers Manual EM 1110-2-1096 or ASTM D5084-90	No greater than 1.0×10^{-7} Cm/sec

2. Soils from each source to be used which meet the Atterberg Limits requirements shall be compacted to determine the moisture-density relationship, including maximum dry density

and optimum moisture content using either the Standard Proctor (ASTM D698 – for Lightweight equipment) or the Modified Proctor (ASTM 1557 for heavy equipment). The selection of compaction test and permeability test shall be at the discretion of the GQCP, or their qualified engineering technician, based on their experience and knowledge of the actual compaction equipment and methods to be used.

B. FML LAYER

1. The Flexible Membrane Liner (FML) to be installed shall be High Density Polyethylene (HDPE) sheeting material manufactured from new, first quality products designed specifically for the purpose of leachate containment in landfills.

2. The manufacturer shall certify in writing that the FML material meets the physical property requirements for its intended application.

3. The FML material shall be extruded to produce a uniform sheet free of any defects that might affect its serviceability, such as holes, tears, nodules, blisters, undispersed raw materials, or foreign matter. Any such defect shall be repaired by extrusion welding in accordance with the manufacturer's recommendations.

4. The sheeting material shall be packaged as prefabricated panels in rolls with labels identifying the thickness, length, width, product information, and manufacturer's information.

5. The FML material shall be stored in its original unopened crates on pallets and protected from direct sunlight under a light-colored, heat reflective, opaque cover in a manner that provides a free flowing air space between the crate and the cover.

6. The manufacturer or his designated representative shall furnish panel layouts as required for the installation.

7. All FML material to be used shall meet or exceed (unless otherwise noted) the following minimum specifications, which meet or exceed the NSF Standard 54 Specifications for High Density Polyethylene (HDPE):

SHEET

Test	Method	Suitability Criteria
Thickness	ASTM D374/D751	60 mils
Tensile Strength (yield)	ASTM D638	2200 psi
Tensile Strength (break)	ASTM D638	3800 psi
Elongation (Yield)	ASTM D638	13%
Elongation (break)	ASTM D638	700%
Tear Resistance	ASTM D1004C	45 lbs
Puncture Resistance	FTMS 101C Method 2065	69 lbs
Friction Angle (for textured only)	GRI-GS7	29°

BASE RESIN

Test	Method	Suitability Criteria
Density	ASTM D792/1505	0.94 g/cm ³
Carbon Black Content	ASTM D1603	2 – 3 %
Carbon Black Dispersion	ASTM D3015	A1 or A2

8. Whenever a different FML manufacturer is to be used, samples of the new material shall be welded to samples of the current HDPE material and “compatibility” testing shall be performed. The following shall comprise this “compatibility” testing and shall be the minimum acceptable values:

Test	Method	Suitability Criteria
Shear Strength	ASTM D4437	2000 psi
Peel Adhesion (Fusion Weld)	ASTM D4437	1500 psi
Peel Adhesion (Extrusion Weld)	ASTM D4437	1300 psi

C. GEOSYNTHETIC LAYER

1. The geo-synthetic layer to be installed directly on the textured FML on the side slopes shall be a composite of a layer of HDPE drainage net material heat bonded between two layers of continuous filament, needle punched, polypropylene geo-textile fabric. This composite is specifically designed to transmit fluids in the plane of the sheet. The composite shall be chemically resistant within a pH range of 2 to 13 and shall be stabilized to resist degradation caused by prolonged exposure to ultraviolet (U.V) light. The geo-synthetic material shall meet the following minimum specifications:

Test	Method	Suitability Criteria
Specific Gravity	ASTM D1505	0.94 g/cm ³
U. V. Resistance	ASTM D4355	80% of strength retained
Permeability (of geo-textile)	ASTM D4491 (500 hours)	0.4 cm/sec
Transmissivity (of drainage net)	ASTM D4716 (@ 0.25 gradient)	10 gal/min/ft width
Ply Adhesion	ASTM DF904 (2" x 5" @ 2 ipm)	2.0 lbs/inch

2. The geo-synthetic material shall be uniformly rolled onto a cardboard core and wrapped in plastic to protect the material from moisture and damage. It shall be scored off the ground, out of weather, and in its original wrapping until installation. The rolls shall be tagged

externally to indicate the manufacturer, product type, product grade, lot number, and physical dimensions. Each roll shall also be certified by the manufacturer that it contains no broken needles.

D. LEACHATE COLLECTION SYSTEM (LCS) COMPONENTS

1. GENERAL

The HELP model was used to aid in the design of this leachate collection system. The LCS design was developed based on the worst case scenario, in which the greatest amount of leachate was generated from several different simulated situations (see LCS typical details).

2. All components of the leachate collection system shall be manufactured of materials which are chemically resistant to the leachate expected to be generated; of sufficient strength and thickness to prevent collapse under the pressures exerted by overlying wastes, waste cover materials, and by any equipment used at the landfill; and designed and operated to function throughout the scheduled closure and post-closure care period of the landfill.

3. The following are considered components of the LCS: SDR-35 Perforated Drain Pipe and fittings (6" Dia.), SDR-35 Solid Wall Drain Pipe and fittings (8" Dia.), 1 ½" crushed aggregate (with less than 15% Calcium Carbonate), Clay Sump, and 16 oz. Geo-textile fabric "wrap" around the perforated pipe.

4. The SDR-35 pipe was selected for its excellent chemical resistance, strength, wall thickness, and drainage characteristics. The drainage laterals and headers shall employ perforated pipe, whereas, the clean-outs shall employ solid wall pipe.

5. The specification for the crushed aggregate was developed based on drainage characteristics, chemical resistance, and the size of the holes (1/2") in the perforated pipe. The specified aggregate size (1-1/2") will preclude any material from entering (and clogging) the drainage laterals and/or headers.

6. The sump shall be constructed as a single unit to prevent loss/leakage of leachate before it can be removed.

7. The geo-textile fabric used to "wrap" the LCS drainage laterals and headers shall act as a filter to prevent clogging of the LCS system and shall be chemically resistant within a pH range of 2 to 13 and shall be stabilized to resist degradation caused by prolonged exposure to ultraviolet (U.V.) light. The geo-textile shall have small enough openings to retain fine-grained soil particles and have large enough openings to retain fine-grained soil particles and have large enough openings to allow leachate to pass through without a significant reduction in flow. The geo-textile fabric shall meet the following minimum specifications:

Test	Method	Suitability Criteria
Fabric Weight	ASTM D3776	16 oz/SY
Mullen Burst	ASTM D3786	450 psi
U.V. Resistance	ASTM D4355 (500 hours)	80% of strength retained
Trapezoidal Tear	ASTM D4533	140 lbs

Grab Tensile Strength	ASTM D4632	320 lbs
Grab Elongation	ASTM D4632	90%

E. DRAINAGE LAYER

1. The material to be used for the drainage layer is a course sand. There are no soils of this type on site; therefore, this material must be purchased and trucked to the site.
2. The sand to be used shall have a coefficient of permeability of 1.0×10^{-2} cm/sec or greater.

F. PROTECTIVE COVER LAYER

1. The protective cover layer shall be constructed from on site soils, classified as CL or CH (ASTM D2487). The material/layer shall have no particles larger than 3 inches in any dimension.
2. This layer shall serve as filter to prevent the high permeability drainage layer from clogging.

III. CONSTRUCTION

A. GENERAL

Soil liner construction and testing shall be conducted systematically and shall not exceed 60 working days unless adverse weather prevents timely completion.

B. EXCAVATION/GRADING

1. INITIAL PREPERATION

Areas of the landfill to be excavated for lateral expansions shall be cleared and grubbed of all trees, stumps, roots, brush, debris, fences, and other items which would interfere with construction operations.

2. DEPTH OF EXCAVATION

Each area shall then be excavated as shown on plans.

3. GRADING

The bottom of the excavation shall be graded to a two percent minimum slope toward the leachate collection system laterals and sump.

4. EXCAVATED SOILS

Topsoil shall be carefully removed and stockpiled for final closure. All other excavated soil shall be stockpiled in a separate location for liner suitability testing or use as daily cover.

5. PROTECTION OF EXCAVATION

Surface water shall be directed away from the excavation and construction sites by diversion ditches, dikes, etc. Any surface or groundwater accumulating in the excavation or construction site shall be pumped off the site as soon as possible.

6. TRENCHING FOR LCS

A rough trench for the leachate laterals and headers shall be constructed, as well as a rough sump. These rough excavations or the LCS shall be filled during clay layer construction to provide the extra thickness required to ensure a minimum of two foot of clay under the LCS trenches and sump.

7. PREPARATION FOR CLAY PLACEMENT

The finished, graded bottom of the excavation shall be rolled with a flat wheel roller prior to placement of the first lift of clay.

C. CLAY LAYER

1. GENERAL

The clay layer shall be constructed monolithically with the side slopes in parallel lifts. Based on the 3 to 1 side slope, no key is required at the toe of the slope.

2. CLOD SIZE REDUCTION

Prior to adding water, clod sizes shall be reduced by disking, pulverizing, or screening, as determined by the GQCP, or their qualified engineering technician. The GQCP, or their qualified engineering technician, shall determine which method best suits the soils present and when the clods have been sufficiently reduced for hydration. The maximum clod size of compacted clay layer shall be no greater than one inch in diameter. The clay shall also contain no rocks larger than one inch in diameter nor total more than 10 % by weight.

3. HYDRATING LINER SOIL

Clean, uncontaminated water shall be added and the soil mixed thoroughly, allowing sufficient time for the soil to hydrate completely.

4. COMPACTION EFFORT

The clay layer shall be compacted with a tamping-foot or prong-foot roller with minimum 6-inch pad or prong length. The lift thickness shall be controlled to achieve total penetration through the loose (8-to 9-inch) lift being compacted into the previously compacted lift, thereby ensuring adequate bonding between lifts. The top of the first three 6-inch lifts shall be roughened to a shallow depth and moistened prior to the placement of the next lift, to further ensure adequate bonding between lifts.

5. MARKERS

The limits of the constructed clay layer shall be clearly marked by discernible SLER markers, which shall be tied into the landfill grid system and maintained at all times until they are no longer required. The SLER markers shall not penetrate the constructed clay layer for any reason.

6. LIFT PROTECTION

The top surface of the completed clay layer shall be bladed or flat-wheel rolled smooth prior to final liner thickness verification and subsequent FML placement. Further, the top surface shall be proof rolled any time that construction is scheduled to be halted for more than 24 hours to help mitigate the effects of desiccation.

7. LINER TIE-IN

The leading 10 to 20 feet of liner edge shall not receive waste in order to facilitate tie-in with the next lateral expansion (the exact width shall be dictated by the GQCP, or their qualified engineering technician). The top foot of previously compacted clay shall be removed across the 10 to 20-foot width to create a ledge (stair) on which to construct the new liner edge (see tie-in detail).

8. HYDROSTATIC PRESSURE

No waste disposal area shall be constructed below the seasonal high groundwater table, as defined in the initial soil boring logs. If and when groundwater is encountered during excavation and is determined to be under hydrostatic pressure, the GQCP, or their qualified engineering technician, shall dictate the type, amount of, and procedures for ballasting. At a minimum, one foot of additional ballast shall be required for each two additional feet of hydrostatic head (the two-foot clay layer will itself offset four feet of hydrostatic pressure). The clay layer shall not be constructed in standing water.

9. PONDED WATER

No water shall be allowed to pond on the clay layer.

D. FML LAYER

1. GENERAL

The FML layer shall be constructed directly atop the two-foot clay layer as soon as possible after approval of the Soils and Liner Evaluation Report (SLER). Installation of the HDPE flexible membrane liner shall be performed in strict accordance with the manufacturer's recommendations, all applicable drawings, and this plan.

2. CERTIFICATION OF PROFICIENCY OF INSTALLER

The installer shall have successfully installed at least ten million square feet of the same FML during the last five years. At least one of the installations shall have been a liner application at a municipal solid waste disposal site.

3. CERTIFICATION OF PROFICIENCY OF INSTALLER'S PERSONNEL

Placement of the FML shall be performed by trained and experienced personnel, who are qualified and approved by the manufacturer to install this material, and under the supervision of a single field engineer, responsible for panel layout, seaming, repairing, and testing, who has supervised the installation of at least two million square feet of this FML.

4. ROLL INSPECTION

The installer's quality control inspector shall inspect each roll, panel, or section of the FML to verify that it meets the required specifications and to check material thickness with a micrometer.

5. PANEL LAYOUT AND OVERLAP

Only those panels that are to be placed and seamed in one day shall be unrolled. Panels shall be positioned with the overlap recommended by the manufacturer, but not less than two inches. The quality control inspector shall inspect and verify that the FML is placed with sufficient overlap.

6. PANEL DEPLOYMENT

The FML rolls/panels shall be deployed in a manner which minimizes rutting (less than 1 inch deep) in the clay layer surface.

7. PROTECTION OF THE FML

No support equipment shall be allowed on the FML. Personnel working on the FML shall not smoke, wear damaging shoes, or engage in any other activity likely to damage the FML.

8. ANCHOR TRENCHES

Anchor trenches shall be completed around the entire area to be lined (except for edges to be tied into on future expansions, which shall be anchored by sandbags). Excavated anchor trench soil shall be removed from the trenches. No loose soil shall underlie the FML in the anchor trenches. The anchor trenches shall be carefully backfilled and compacted to the specifications used for the clay layer, without damaging the FML (see anchor trench and termination details).

9. PREPARATION FOR FIELD SEAMING

Prior to field seaming, all wrinkles shall be walked out removed as much as possible. All foreign matter shall be removed from the area to be bonded.

10. FIELD SEAMING

Field seaming shall be performed by the methods approved by the manufacturer only (see fusion and extrusion weld details). As much as practical, field seaming shall start in the middle and work toward an open end in order to minimize cutting and patching of large wrinkles that become trapped. All completed seams shall be tightly bonded and sealed. Tack welds, if used, shall use heat only. No double-sided tape or

glue shall be permitted. No folds, wrinkles, or fish-mouths shall be allowed within the seamed area. In general, seams shall be oriented up and down, instead of across the slope. No horizontal seam shall be less than 5 feet from the toe of the slope. In corners, sumps, trenches, and other odd-shaped configurations, the number of field seams shall be minimized.

11. SEAMING/WELDING EQUIPMENT

Only welding systems which utilize the manufacture's recommended welding process shall be used for bonding the FML. The welding equipment used shall be capable of continuously monitoring and controlling the temperatures in the zone of contact where the machine is actually fusing the FML so as to ensure changes in environmental conditions shall not affect the integrity of the weld.

12. REPAIRS

Where wrinkles or folds occur, the FML shall be cut and overlapped, and an extrusion weld shall be applied. Where destructive samples are taken, or seams fail, a patch shall be installed, which extends at least six inches beyond the edge of the defect, with all corners of patches rounded to a three-inch radius.

During wrinkle or fold repairs, the adjacent FML may not be required to meet the two-inch minimum overlap if approved by the installer's quality control inspector.

13. SEAMING CONDITIONS

No seaming shall be attempted above 104 degrees Fahrenheit air temperature or in high wind. If seaming is to be performed below 41 degrees Fahrenheit ambient air temperature, preheating of the FML may be required.

14. UNSEAMED EDGES

At the end of each day or installation segment, all be anchored with sandbags or other approved devices. Staples, U-shaped rods, or other penetrating anchors shall not be used to secure the FML.

15. CERTIFICATION OF INSTALLATION

Upon completion of the FML installation, the installer shall certify in writing that the FML has been installed in accordance with the manufacturer's recommendations, all applicable drawings, and this plan.

16. WARRANTY

The manufacturer of the FML shall warrant, in writing, the material on a pro-rated (straight-line) basis against manufacturing defects and degradation of the material for a period of twenty (20) years from the date of final acceptance following installation. The manufacturer shall replace and reinstall any material which fails within the warranty period. In addition to the materials warranty, the installer shall warrant the installation of the FML against defects or failure for a period of one (1) year. The

installation warranty shall commence from the date of final acceptance of the installed FML and shall include the services of qualified technicians and all materials required due to defects in installation and/or workmanship.

E. LEACHATE COLLECTION SYSTEM

1. GENERAL

The leachate collection system shall be constructed in accordance with all applicable permit drawings, details, etc. (see LCS details).

2. TRENCHES AND SUMPS

After approval of the SLER for the clay layer, the trench for the LCS laterals and headers, as well as the sump, shall be carefully excavated. The edges of the trench and sump shall be rounded to prevent sharp folds or bends in the FML material. The depth of the trench and sump may vary, but the thickness of clay liner below the trench and sump shall always be two feet or greater.

3. GEOTEXTILE "WRAP"

After placement, seaming, and testing of the FML, the geo-textile material shall be placed in the LCS trenches. The width of the geo-textile shall be sufficient to completely envelope the LCS laterals and headers after aggregate backfill.

4. PIPE AND SUMP INSTALLATION

The SDR-35 perforated pipe shall then be installed in the lateral and header trenches and connected together. The sump shall then be carefully excavated and leveled. All connections shall be in accordance with standard plumbing practice.

5. TRENCH BACKFILL

After pipe installation, the trenches shall be carefully backfilled with the 1-1/2" crushed aggregate around the drainage pipes. Upon completion of the backfill operation, the exposed edges (flaps) of the geo-textile fabric shall be "wrapped" each way over the pipes and backfill, enveloping the drainage piping system.

6. CLEANOUTS

After lateral and header pipe installation and backfill, the SDR-35 solid wall clean-out pipe shall be installed. See LCS Details for clean-out. The pipe shall be secured in place with sand bags or some other means until placement of the protective cover layer.

E. GEOSYNTHETIC LAYER

1. GENERAL

Installation of the geo-synthetic composite layer on the side slopes shall be performed in strict accordance with the manufacture's recommendations, all applicable drawings, and this plan.

2. CERTIFICATION OF PROFICIENCY OF INSTALLER'S PERSONNEL

Placement of the geo-synthetic composite shall be performed by trained and experienced personnel, including supervisors, foremen, and skilled laborers, who are qualified and approved by the manufacturer to install this material.

3. SEAMING

The geo-synthetic composite shall be seamed by overlapping the panels at least six inches and tying them together with plastic ties every few feet along the seam. The plastic ties shall tie the drainage nets in both panels together, not just the geo-textile fabric. No wire ties shall be used. The layer of geo-synthetic composite in place shall be free of holes, tears, wrinkles, creases, and other imperfections.

4. REPAIRS

Any required repairs on damaged composite shall be made in accordance with the manufacturer's recommendations.

5. EDGES

All edges of the geo-synthetic composite shall be anchored as shown on the anchor trench detail, termination detail, or floor tie-in detail.

6. CERTIFICATION OF INSTALLATION

Upon completion of the installation of the geo-synthetic composite, the installer shall certify in writing that the geo-synthetic layer has been installed in accordance with the manufacturer's recommendations, all applicable drawings, and this plan.

E. DRAINAGE LAYER

1. General

The installation of the granular drainage material (sand) shall be performed under the close supervision of the GQCP, or their qualified engineering technician.

2. INSTALLATION PROCEDURE

Using a light-weight dozer, a "pad" shall be constructed approximately three feet thick initially. When the pad is stabilized, "fingers" shall be constructed across the floor to minimize wrinkles in the FML caused by pushing the sand outward. The dozer shall continue pushing the sand and increasing the pad and finger dimensions. As the pad and finger dimensions increase, the thickness gradually decreases. When the total required amount of sand has been brought in, the thickness shall be checked by survey methods and adjusted until it is uniform across the floor.

3. ADDITIONAL THICKNESS

The area directly over the LCS trenches shall receive a two foot thick layer of sand. Also, the protective cover layer shall be constructed with “chimneys” to allow leachate to pass through the protective cover layer into the drainage layer and into the leachate collection system (see LCS details).

4. TERMINATION AND TIE-IN

The sand drainage layer shall be terminated and tied-in as shown on the LCS details.

5. SUMP BACKFILL

The space around the prefabricated concrete sump shall be backfilled with sand to the same elevation as the sand layer in the floor.

E. PROTECTIVE COVER LAYER

1. GENERAL

A layer of uniform native soil, free of rocks, vegetation, and debris that may damage the liner system, shall be installed directly on top of the sand drainage layer on the floor and on top of the geo-synthetic layer on the side slopes to act as a protective cover layer and filter to prevent clogging of the drainage media. All protective cover layers shall be installed as a single lift, regardless of thickness.

2. FLOOR

A one-foot, uncompacted protective cover layer shall be placed on the sand drainage layer immediately after verification of drainage layer thickness, upon approval of the GQCP, or their qualified engineering technician (see termination detail). Protective cover material shall not be placed over the additional thickness of the drainage layer over the LCS laterals and headers or over the chimneys through protective cover layer (see LCS details)

3. SIDE SLOPES

A two-foot, uncompacted protective cover layer shall be placed on the geo-synthetic composite layer immediately after placement of the one-foot protective cover layer on the floor (see tie-in detail). Placement of the single two-foot lift shall begin at the toe of the slope and progress up the slope to minimize loading on the FML.

IV. QUALITY ASSURANCE/QUALITY CONTROL

A. GENERAL

1. OBJECTIVE

The objective of Quality Assurance/Quality Control (QA/QC) during liner system construction is to ensure that each material component is structurally sound and that all construction procedures are correctly and properly followed so that the quality

of the surface and groundwater is absolutely protected. This objective shall be met by using a GQCP whose credentials have already been submitted to and approved by the Commission, as well as requiring GQCP, or their qualified engineering technician, to be on site at all times during liner system construction.

2. ACCEPTABLE CRITERIA

The entire lateral expansion (cell construction) project shall be in accordance with all applicable drawings, plans, etc., and under the direct supervision of the GQCP, or their qualified engineering technician. The GQCP, or their qualified engineering technician, shall ensure that the QC/QA objective is met and shall have the authority and responsibility for making any required field decisions or judgements concerning liner system construction.

3. TESTING AND REPORTING

The GQCP shall be responsible for all Quality Control/Quality Assurance testing (both field and lab) and reporting (both SLER and FMLER). Specific procedures and requirements for QC/QA testing and reporting follow.

A. EXCAVATION/GRADING

1. EXCAVATION

The GQCP, or their qualified engineering technician, shall observe all excavation and trenching operations to ensure that areas are not over-excavated. The GQCP shall certify that the depth of all excavations and trenches are in accordance with this plan and the Site Development Plan (SDP).

2. GRADING

The GQCP, or their qualified engineering technician, shall be responsible for verifying all grades and lines using instrument survey methods and shall certify that the floor and LCS lateral trenches are on a two percent minimum slope toward the sump. The GQCP, or their qualified engineering technician, shall approve the graded excavation and the bladed/rolled surface for clay placement.

A. CLAY LAYER

1. GENERAL

The GQCP, or their qualified engineering technician, shall closely monitor the plasticity of the soils used for liner construction and may require additional soil suitability testing if he feels at any time that the initial soil suitability testing was inadequate for all the clay soils being used for lining.

2. TESTING

The GQCP, or their qualified engineering technician, shall perform the following Quality Assurance tests:

Test	Method	Frequency	Criteria
Field Density	ASTM D2922	Per 8,000 SF per 6" Lift (at least 3 per 6" lift)	95 % of Max. Dry Density at Opt. Moisture or wetter
Steve Analysis*	ASTM D422	Per 100,000 SF/6" Lift 100% (1 ½" Screen)	30 + % (-200 sieve)
Atterberg Limits* Plasticity Index	ASTM D4318	Per 100,000 SF/6" Lift	No less than 15 No less than 30
Coefficient of Permeability	Appendix VII Corps of Engineers Manual EM1110-2-1906 or ASTM D5084-90	Per 100,000 SF/6" Lift	No greater than 1.0 X 10 ⁻⁷ cm/sec

- At least one test per 6" lift

1. THICKNESS VERIFICATION

Thickness shall be verified by instrument survey methods at a minimum of one verification point per 5,000 SF (at least two points).

2. FAILED TESTS

Sections of clay liner that fail the density/moisture criteria shall be reworked and retested until they pass. The procedure used to accurately define the failing section/area shall be determined by the GQCP.

3. TEST HOLES

All holes created for testing purposes shall be backfilled with bentonite.

A. FML LAYER

1. TEST SEAMS

Each day prior to commencing field seaming, test seams shall be made on fragment pieces of the FML to verify that seaming conditions are adequate. Such test seams shall be made at the beginning of each seaming period and at least once every four hours for each seaming apparatus used that day. Also, each seamer shall make at least one test seam each day. The GQCP, or their qualified engineering technician, shall observe and oversee all test seam procedures and field testing. The test seam shall be at least three feet long by one foot wide. Six adjoining one inch wide specimens shall be die-cut from the test seam sample. These specimens shall be tested in the field for shear and peel (three each), and must not fail in the seam. If additional test specimens fail at the seam, the seaming method used shall be investigated to determine the cause of the failures. Once the problem is identified and resolved, a new set of specimens shall be tested for verification.

2. CONFIRMANCE TESTING

Conformance testing samples shall be taken at the rate of one per resin lot and at least one per 100,000 SF. Sample size shall be three feet by the roll width. The GQCP shall perform conformance testing of the FML material as follows:

Test	Method	Suitability Criteria
Thickness	ASTM D751	60 mil
Tensile Strength (Yield)	ASTM D638	2200 psi
Tensile Strength (Break)	ASTM D638	3800 psi
Elongation (Yield)	ASTM D638	13 %
Elongation (Break)	ASTM D638	600 %

1. QA TESTING OF PANELS

The acceptance criteria for QA testing is the same as for its suitability and is delineated herein. The GQCP, or their qualified engineering technician, shall perform QA testing on the FML material as follows:

Test	Method	Frequency
Thickness	ASTM D374	5' increments along leading and trailing edges of each geomembrane panel
Tensile	ASTM D638	Per 100,000' & every resin lot
Density	ASTM D792/1505	Per 100,000' & every resin lot
CBlack Content	ASTM D1603	Per 100,000' & every resin lot
CBlack Dispersion	ASTM D3015	Per 100,000' & every resin lot
Puncture Resistance	FTMS 101C(2065)	Per 100,000' & every resin lot

1. QA TESTING OF SEAMS – NONDESTRUCTIVE

Nondestructive seam testing shall be performed to verify the continuity of seams. The installer shall test the entire length of all double weld (fusion) seams using the manufacturer's recommended air pressure test procedure. The installer shall test all extrusion welds using the manufacturer's recommended vacuum box test procedure. The GQCP, or their qualified engineering technician, shall ensure that precautions are taken to prevent damage to the FML by the testing equipment. All testing information and results shall be reduced by the GQCP, or their qualified engineering technician, for reporting purposes.

2. QA TESTING OF SEAMS – DESTRUCTIVE

Destructive seam testing shall be performed to evaluate the strength of seams. The location of destructive samples shall be determined by the GQCP, or their qualified engineering technician, and the installer's quality control inspector. The number and

size of destructive samples to be taken shall also be determined by the GQCP, or their qualified engineering technician, and the installer's quality control inspector and may vary for field testing, lab testing, and archive storage. At each location, a minimum of one sample for field testing (peel), two samples for lab testing (peel and shear), and two samples for archive storage shall be taken. The GQCP shall order the following tests on the lab destructive samples:

Test	Method	Frequency	Criteria
Shear Strength	ASTM D4437	Per 500 LF of seam per welder	2000 psi
Peel Adhesion (Fusion Weld)	ASTM D4437	Per 500 LF of seam per welder	1500 psi
Peel Adhesion (extrusion weld)	ASTM D4437	Per 500 LF of seam per welder	1300 psi

1. VISUAL INSPECTION

The GQCP, or their qualified engineering technician, shall visually inspect 100 % of the FML surface and certify that it is free of visible defects.

A. LEACHATE COLLECTION SYSTEM

1. GENERAL

The GQCP, or their qualified engineering technician, shall observe all phases of the leachate collection system construction and shall verify that all LCS components being installed comply with the specifications.

2. EXCAVATION AND TRENCHING

Prior to installation of the SDR-35 pipe and the prefabricated concrete sump, the GQCP, or their qualified engineering technician, shall approve the trench and sump excavations by verifying minimum slopes and maximum depths.

3. LCS LAYOUT AND INSTALLATION

The GQCP, or their qualified engineering technician, shall ensure that spacing and installation of the LCS components is in accordance with all applicable details and drawings and that only standard plumbing practices are used for all pipe connections. The GQCP, or their qualified engineering technician, shall intensely monitor installation of all LCS components to ensure FML integrity is maintained.

4. BACKFILLING

The GQCP, or their qualified engineering technician, shall oversee the backfilling and "wrapping" of the LCS laterals, headers, and sumps, ensuring that care is taken to prevent damage to and minimize movement of the LCS components.

5. LCS FLUSHING

The GQCP, or their qualified engineering technician, shall as a functional check of the leachate collection system, flush out all laterals and headers with uncontaminated water, verifying that all piping drains properly into the sump. At the conclusion of the flush test, all water in the sump shall be pumped out.

A. GEOSYNTHETIC LAYER

The GQCP, or their qualified engineering technician, shall observe all field seaming of the geo-synthetic composite material and shall verify that all seaming was performed in accordance with the manufacturer's recommended procedures and that no wire ties were used for the seams.

B. DRAINAGE LAYER

1. GENERAL

The GQCP, or their qualified engineering technician, shall closely monitor the placement of the sand drainage layer to ensure that the FML layer is not damaged.

2. TESTING

The GQCP, or their qualified engineering technician, shall perform the following Quality Assurance tests:

Test	Method	Frequency	Criteria
Coefficient of Permeability*	Appendix VII Corps of Engineers Manual EM1110-2- 1906 or ASTM D5084-90	Per 1000,000 SF	No less than 1.0×10^{-2} cm/sec

*At least one test

1. THICKNESS VERIFICATION

Thickness shall be verified by instrument survey methods at a minimum of one verification point per 5,000 SF (at least two points). The sand drainage layer shall be at least one foot thick at all locations. The thickness above the LCS laterals and headers shall be two foot (see LCS details).

A. PROTECTIVE COVER LAYER

1. GENERAL

The GQCP, or their qualified engineering technician, shall closely monitor the placement of the protective cover layer to ensure that the sand drainage layer is not disturbed.

2. CHIMNEYS

The GQCP, or their qualified engineering technician, shall inspect and verify that all chimneys have been constructed in accordance with all applicable drawings and that their dimensions and spacing are correct.

3. THICKNESS VERIFICATION

Thickness shall be verified by instrument survey methods at a minimum of one verification point per 5,000 SF (at least two points). The protective cover layer shall be at least one foot thick at all locations on the floor. The protective cover layer shall be at least two foot thick at all locations on the side slopes.

A. REPORTING

1. SLER

All excavations and construction, as well as field and lab sampling and testing of components of the composite liner system shall be under the direct supervision of the GQCP, or their qualified engineering technician. The GQCP, or their qualified engineering technician, shall collect all test results from the excavation and construction of the clay layer and prepare a Soils and Liner Evaluation Report (SLER) for each new lateral expansion. Construction above the clay layer shall not commence until the SLER is accepted and approved by the Commission, unless conditions require more expedient progress, in which case, all construction above the clay layer shall be at the risk of the owner. The GQCP shall use TNRCC Form No. GWP-1 for SLER preparation. The GQCP shall include all maps, sketches, drawings, calculations, test results, and any other information required by the Commission in the SLER submission.

2. FMLER

The GQCP, or their qualified engineering technician, shall collect all test results for all construction above the clay layer and prepare a Flexible Membrane Liner Evaluation Report (FMLER) for each new lateral expansion. No waste shall be deposited in a new lateral expansion without approval of the Commission. The GQCP shall use TNRCC Form No. GWP-2 for FMLER preparation. The GQCP shall include all maps, sketches, drawings, calculations, test results, and any other information required by the omission in the FMLER submission. Each FMLER prepared by the GQCP shall include:

- A. Names of each seamer for start-up testing and afternoon break, including time and tip temperatures for each seaming apparatus used each day.
- B. Manufacturer's QC documents.
- C. Field micrometer measurements in one foot increments along the leading and trailing edges of geo-membrane panel.

- D. Nondestructive tests, including start and stop times, before and after pressure readings (for double track testing) for all seams.
- E. Roll shipment/receipt and Storage/handling information.
- F. All pertinent drawings, including the Sectorized Fill Layout Plan, the location of cell with SLER/FMLER markers, previous filled and active areas, panel placement including patches, repairs, caps, etc., and destructive tests and repairs.
- G. Verification of drainage layer and protective cover layer thickness.
- H. Report must include the qualitative results of Destructive tests for both field and laboratory values of Shear and Peel (both tracks when possible for double track and record location of failures).
- I. Report must discuss the full-time quality assurance by the GQCP, or their qualified engineering technician, nondestructive testing equipment, including testing procedures and methods of evaluation and passing criteria, subgrade acceptance, anchor trench preparation and backfilling, wrinkles/fishmouths, and normal manufacturing creases, deployment of panels, seam preparation/orientation, weather and air temperature/sheet temperature limits, continuous 100 % nondestructive seam testing; equipment placed on the geomembrane; repairs, including preparation and procedure, patch size and shape, failed test seams and testing either side of failed seam; placement of cover material up slope for sidewalls; thickness and sieve analysis requirements of drainage and protective cover material; field panel identification/placement; specification for air pressure testing/vacuum testing; 100 % visual inspection of geomembrane for defects, damage, etc.; handling and storage of geomembrane materials; quality control/assurance certificates from the manufacturer, including the testing required and frequency of testing; and daily start-up and additional testing for each seaming machine.